Chapter 11

Composite kernels and the odd two-body kernels

Chapter 10 wrote the dilatation component D_1 of a local operator as four times the logarithmic pair kernels, a one-body term and a composite term, and the translation component P_1 as a one-body term and a composite term, the composite graphs being those in which $S^{(1)}(\Delta)$ or $S^{(1)}(P_b)$ sits on the cycle as a line from a ghost leg to a gauge leg (Theorem **??**, Proposition **??**); it evaluated the composite term on the planar scalar sector, and on a general operator the locality of the composite term is a statement to be proved: on the letters $A|_V$, λ, c the renormalised composite two-point functional contains $\log|p|^2$ of the momentum (Remark **??**). This chapter shows that Q maps the composite graphs of a cocycle to cycles with one special edge and gives in outline the proof that the composite term is then a sum over the pairs of letters of a local kernel, the composite pair kernel; it computes these kernels as simplex integrals and evaluates them on the parts with the fewest fields of twist two, of the $\mathfrak{su}(1|1)$ sector, of the fermion and gluon $\mathfrak{sl}(2)$ sectors and of two mixed families. On all of them the composite pair kernels satisfy the two identities that Corollary **??** requires, the translation identity along the null direction of the derivatives, and along every direction on twist two for $S' \leq 4$ and on the letters without derivatives; for a one-loop quantisation through arity two along (Δ, P_b) the arity-two equation then fixes the three sector coefficients of $\hbar F_1^{(1)}(\Delta)$ to $-\frac{1}{3}$ units in the normalisation of Theorem 11.14: D_1 is four times the map of Theorem **??**, with spectrum twice the one-loop anomalous dimensions, and P_1 vanishes on cohomology in the translation-covariant scheme. The last section sets up the two-body kernels of Q_1, S_1 and K_1 in the same scheme: their graphs and output bounds, the relations of S_1 to Q_1 and of K_1 to P_1, their renormalisation freedom and the comparison with the one-loop differential of the holomorphic twist.

The unit is $\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2(X,Y)}$, equal to $-\frac{g^2 N}{32\pi^2}$ for $\mathfrak{g} = \mathfrak{su}(N)$ (Remark **??**). Edges carry $P(\varepsilon, T)$ of Definition **??**; the special edge is $K_\varepsilon - K_0 = -\int_0^\varepsilon (H \otimes 1) K_t \, dt$, the factor $e^{-\varepsilon|p|^2} - 1$ times the components of K_0 (Proposition **??**). Every component below is independent of the endpoint $T \in (0, \infty]$ of the transfer (Theorem **??** (ii)) and is evaluated at $T = \infty$, where the graph sums converge at generic momenta (Proposition **??**); along K_b it is taken at a finite T (Definition **??**). On arity zero $s = Q + \{I, -\}$ (Theorem **??**). Letters, the pair weight, the pair degree and the Koszul signs are those of Definition **??** and (**??**). The field-counting functional is $2\int(A, A^*) - 4\int(\lambda^*, \lambda) - \int(c^*, c)$, whose bracket multiplies the coordinate functionals of c, A, λ, A^*, λ^*, c^* by 1, 2, 4, -2, -4, -1 (Lemma **??**), and the map of Theorem **??** is the coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](O)$ minus the unit times the bracket of O with it (Section 10.7). Along Δ, $v = 2E$ and $v.$ is the operator of Lemma **??**, so that $\{S^{(1)}(\Delta), M\} = v.M$; along P_b, $v.$ is ∂_b on every slot (Proposition **??**); for $x \in \{\Delta, P_b\}$, $v_x.$ is the operator of the vector field of x (Definition **??**). The two-component form of Definition **??** of every main formula is in Remark 11.60.

11.1 The composite term as a sum over pairs

Definition 11.1 (Square kernel and composite kernels)**.** For $0 < \varepsilon < T \le \infty$ and $P = P(\varepsilon, T)$, the square kernel is $\big((P\star)^2 \otimes 1\big)K_0$, and the composite kernel along $x \in \{\Delta, P_b\}$ is

$$\big(P \star v_x. \otimes 1 + 1 \otimes P \star v_x.\big)P,$$

the two legs of $S^{(1)}(x)$ joined, in both orders, to two edges P (Definition **??**, Theorem **??**). The sector parts of $S^{(1)}(P_b)$ are

$$\int \big(\partial_b(A|_{W^\perp_{\mathbb{R}}}), A^*|_{W^\perp_{\mathbb{R}}}\big), \qquad -\int (\partial_b\lambda, \lambda^*), \qquad \int \big(\partial_b(A|_V), A^*|_V\big) - \int (\partial_b c, c^*);$$

their sum is $S^{(1)}(P_b)$, and each sector part of $S^{(1)}(\Delta)$ of Section 10.8 is $2\int x^\mu$ times the density of the sector part of $S^{(1)}(P_{e_\mu})$ of the same sector, summed over μ, plus a translation-invariant functional, with e_μ an orthonormal basis of V and x^μ the coordinates.

Lemma 11.2 (Structure of the composite kernels)**.**

- *(i) The composite kernel along x is $2\big(P \star v_x.\, P \star \otimes 1\big)K_0 = (v_x. \otimes 1 - 1 \otimes v_x.)$ applied to the square kernel.*
- *(ii) $(v_x. \otimes 1 + 1 \otimes v_x.)$ applied to the square kernel is $-4\big(\varepsilon\partial_\varepsilon + T\partial_T\big)$ of the square kernel for $x = \Delta$ and 0 for $x = P_b$. Hence the composite kernel along Δ is $2(v. \otimes 1)$ of the square kernel plus $4(\varepsilon\partial_\varepsilon + T\partial_T)$ of it, and the composite kernel along P_b is $2(\partial_b \otimes 1)$ of the square kernel.*
- *(iii) The square kernel and the composite kernels have components in $c \otimes A|_V$ and $A|_V \otimes c$ alone: as operators, $(P\star)^2$ and $P \star v_x.\, P\star$ map the slot of $A^*|_V$ to the slot of c and the slot of c^* to the slot of $A|_V$, and vanish on the other slots.*
- *(iv) In a graph the composite kernel is the vertex $S^{(1)}(x)$ with both legs on edges P. Its term $\int (v_x.A|_V, A^*|_V)$ contributes, with the leg $A|_V$ on the component $A \otimes A$ of P and the leg $A^*|_V$ on the component $A^*|_V \otimes c$. Along Δ this internal vertex is*

  $$\int \big((2E+2)A|_V, A^*|_V\big) = 2\int x^\mu \big(\partial_\mu(A|_V), A^*|_V\big) + 2\int \big(A|_V, A^*|_V\big),$$

  *along P_b it is $\int (\partial_b(A|_V), A^*|_V)$. The components of P lie in $A \otimes A$, $\lambda \otimes \lambda$, $c \otimes A^*|_V$ and $A^*|_V \otimes c$ (Proposition **??**), so the terms of $S^{(1)}(x)$ in the sectors of $A|_{W^\perp_{\mathbb{R}}}$, λ and c, whose antifield legs are $A^*|_{W^\perp_{\mathbb{R}}}$, λ^*, c^*, enter a cycle through the special edge.*
- *(v) $(Q \otimes 1 + 1 \otimes Q)$ annihilates the composite kernel; the operator ∂ of the composite kernel is odd and anticommutes with Q.*
- *(vi) At $T = \infty$ the component of the square kernel is $(Q^{GF})^2$ on the slot of $A^*|_V$, which is $2d^*$ with values in the slot of c (Proposition **??** (ii)), times $(-,-)^{-1}$ and*

  $$\int_\varepsilon^\infty \int_\varepsilon^\infty e^{-(s'+s'')|p|^2}\, ds'\, ds'' = e^{-2\varepsilon|p|^2}\,|p|^{-4}:$$

  *a doubled edge with two Schwinger times $s', s'' \in [\varepsilon, \infty)$, of momentum degree -3, which obeys the weight rule of Proposition **??** for the slots c (weight 0) and $A|_V$ (weight 1).*

Proof. (i) The first equality is Proposition **??** (i) along Δ; along P_b the same computation holds, ∂_b preserving the pairing and commuting with Q^{GF} and H (Proposition **??**). For the second, $2P\star\, v_x.\, P\star = v_x.(P\star)^2 + (P\star)^2 v_x.$, along Δ because $[v., P\star]$ is Q^{GF} times a function of H and commutes with $P\star$

(proof of Proposition **??** (i)), along P_b because ∂_b commutes with $P\star$. For the even operator $X = (P\star)^2$, $(X\, v_x.\otimes 1)K_0 = (X\otimes 1)(v_x.\otimes 1)K_0 = -(1\otimes v_x.)(X\otimes 1)K_0$ by Lemma **??**, $v_x.$ being even and pairing preserving.

(ii) Since $(v_x.\otimes 1 + 1\otimes v_x.)K_0 = 0$, the left side is $([v_x., X]\otimes 1)K_0$. Along Δ, $X = (Q^{GF})^2 g(H)^2$ with $g(H) = \int_\varepsilon^T e^{-tH}dt$, $[v., Q^{GF}] = -4Q^{GF}$ and $[v., f(H)] = -4Hf'(H)$ (Proposition **??**). From $Hg'(H) = Te^{-TH} - \varepsilon e^{-\varepsilon H} - g = (\varepsilon\partial_\varepsilon + T\partial_T)g - g$,

$$[v., X] = -8X - 4(Q^{GF})^2 H\partial_H\big(g^2\big) = -8X - 4\big((\varepsilon\partial_\varepsilon + T\partial_T)X - 2X\big) = -4\big(\varepsilon\partial_\varepsilon + T\partial_T\big)X.$$

Along P_b, $[\partial_b, X] = 0$. Insert into (i).

(iii) $(Q^{GF})^2$ vanishes on the slots of c, A, λ, λ^*; on A^* and c^* it is $2d^*(A^*|_V)$ in the slot of c and $-2dc^*$ in the slot of $A|_V$ (Proposition **??** (ii)); $v_x.$ preserves the slots, and the components $A^*\otimes A$ and $c^*\otimes c$ of K_0 go to $c\otimes A|_V$ and $A|_V\otimes c$.

(iv) Applied to $A^*\otimes A$, the kernel $(P\star\, v_x.\, P\star\otimes 1)K_0$ is the component $A^*\to A$ of Q^{GF}, then $v_x.$ on the slot of A, then the component $A\to c$ of Q^{GF}: the vertex $\int(v_x.A|_V, A^*|_V)$ between an $A\otimes A$ and an $A^*|_V\otimes c$ component, the factor 2 counting the two orientations. On $A|_V$, $L_{2E}A|_V = 2x^\mu\partial_\mu(A|_V) + 2A|_V$ (Definition **??**).

(v) Proposition **??** (iii) along Δ; along P_b, $QP\star + P\star Q = e^{-\varepsilon H} - e^{-TH}$ commutes with ∂_b, which commutes with Q, so $Q(P\star\,\partial_b\, P\star) = (P\star\,\partial_b\, P\star)Q$. The identity $[\partial_\phi, Q] = \partial_{(Q\otimes 1+1\otimes Q)\phi}$ of Lemma **??** gives the anticommutation; the composite kernel has degree -1.

(vi) Proposition **??** (iv) and $\big(\int_\varepsilon^\infty e^{-s|p|^2}ds\big)^2 = e^{-2\varepsilon|p|^2}|p|^{-4}$. □

Definition 11.3 (Composite graphs and the arity-one counterterm of an operator)**.** Let M be a local functional, δ a square-zero parameter of degree $-|M|$, $x\in\{\Delta, P_b\}$ and $P = P(\varepsilon,\infty)$. The composite graphs of M along x are the δ-linear part of

$$\tfrac{\hbar}{2}\,\partial_{\text{composite kernel along } x}\, W(P, I+\delta M)_{\text{tree}}:$$

the composite kernel closes two legs of a tree of vertices I and one vertex M. The cycle so formed either passes through two letters X_a, X_b of M, a composite cycle through the pair (X_a, X_b) with trees of I on its outputs, or passes through vertices of I alone and is joined to M by a tree through one letter, a composite loop of I on a leg of M. The arity-one counterterm of M along x is $\hbar\,\mathrm{Sing}_\varepsilon$ of the amputated composite cycles through the pairs, summed over the pairs with the Koszul signs of (**??**). The singular part of the composite loops of I is $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ (Proposition **??** (i)); its quadratic part along Δ is the unit times $\log\varepsilon$ times $Q\big(\int(\iota_x A|_V, c^*) - \frac12\int(\rho(x)\lambda^*, \lambda^*)\big)$, with no term in ε^{-1} (Theorem **??** (i)).

Proposition 11.4 (The image of the composite graphs under Q)**.** *For every local functional M, $x\in\{\Delta, P_b\}$ and $\varepsilon > 0$, at $T = \infty$,*

$$Q\big(\textit{composite graphs of } M \textit{ along } x\big) = -\tfrac{\hbar}{2}\,\partial_{\textit{composite kernel along } x}\Big[\textit{tree-level flow, } \delta\textit{-linear, with one vertex } sM \textit{ or one}$$

*the special edge joining a vertex I to a letter of M, from $\{I, M\}_\varepsilon - \{I, M\}$, or two vertices I, from $\frac12(\{I, I\}_\varepsilon - \{I, I\})$. Since $\Delta_\varepsilon - \Delta_0 = -\partial_{K_\varepsilon - K_0}$ (Definition **??**, Proposition **??**), $\{f, g\}_\varepsilon - \{f, g\}$ is minus f and g joined by one edge $K_\varepsilon - K_0$; for $sM = 0$ the right side is $\frac{\hbar}{2}\,\partial_{\textit{composite kernel along } x}$ of the trees with exactly one edge $K_\varepsilon - K_0$.*

Proof. By Lemma 11.2 (v), Q anticommutes with ∂ of the composite kernel. By Proposition **??** at order $\hbar^0$, $Q\,W(P, J)_{\text{tree}}$ is the tree-level flow of $QJ + \frac12\{J, J\}_\varepsilon$; the term $\hbar\Delta_\varepsilon J$ is of order $\hbar$, and its composite closure has two cycles. For $J = I + \delta M$ the classical master equation $QI + \frac12\{I, I\} = 0$ (Proposition **??**) leaves $\frac12(\{I, I\}_\varepsilon - \{I, I\})$ in the δ-independent part and $sM + (\{I, M\}_\varepsilon - \{I, M\})$ in the δ-linear part. □

Definition 11.5 (Special-edge composite cycles and the composite pair kernel)**.** For two letters X_a, X_b of any slots and $x \in \{\Delta, P_b\}$, the special-edge composite cycles of (X_a, X_b) along x are

$$\tfrac{\hbar}{2}\,\partial_{\text{composite kernel along } x}\,\partial_{K_\varepsilon - K_0}\big(\text{amputated chains from } X_a \text{ to } X_b\big)\,:$$

the chains of Definition **??** with $k \geq 1$ vertices of I in which one edge e_c is the composite kernel along x, one other edge e_σ is the special edge, with the components of K_0 (each field slot with its antifield slot), which may end at a letter of any slot, and every remaining edge is $P(\varepsilon, \infty)$. By Proposition 11.4 they are minus $\frac{\hbar}{2}\,\partial$ of the composite kernel applied to the chains with one bracket-defect vertex. A chain is read with the signs of Lemma **??** along the chain from X_a, times

$$(-1)^{n(<\sigma)+n(<c)+[c<\sigma]},$$

where $n(<j)$ is the number of odd outputs at the vertices before the edge e_j, and $[c < \sigma]$ is 1 if e_c precedes e_σ and 0 otherwise; the outputs are left open as in Definition **??**. The composite pair kernel of (X_a, X_b) along x is the coefficient of ε^0 of its special-edge composite cycles.

By Proposition 11.4, for $sM = 0$ the image under Q of the composite cycles through the pairs and of the composite loops on the legs is the sum of the special-edge composite cycles with trees of I on their outputs, together with the composite closures in which the special edge lies on a tree; Theorem 11.14 identifies the composite term of a cocycle with the sum over its pairs of letters of the composite pair kernels.

Proposition 11.6 (Reachability and census)**.**

- *(i) The composite kernel joins a slot c to a slot $A|_V$. The letters c and $A|_V$ reach it directly; the letters $A|_{W_{\mathbb{R}}^\perp}$, λ and $A^*|_V$ reach it through one vertex, $A|_V(A|_{W_{\mathbb{R}}^\perp})^2$ entered at $A|_{W_{\mathbb{R}}^\perp}$, $\lambda|_{S_+\otimes W}\lambda|_{S_-\otimes W^*}A|_V$ entered at λ, $([A|_V, c], A^*|_V)$ entered at c, each left at $A|_V$; the letters $A^*|_{W_{\mathbb{R}}^\perp}$, λ^*, c^*, on which P and the composite kernel have no component, reach it through the special edge, to a slot $A|_{W_{\mathbb{R}}^\perp}$, λ, c.*

- *(ii) A chain of Definition 11.5 exists for* 45 *of the* 55 *unordered pairs of the ten slots; the other ten are the pairs of two letters among $A^*|_{W_{\mathbb{R}}^\perp}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$, c^*, which would need two special edges. Every chain conserves the degree, the fermion parity and the centre $\mathbb{Z}/4$ of $SL(4,\mathbb{C})$, and its outputs carry the pair degree; the composite kernel has degree -1 and the special edge $+1$.*

- *(iii) Every chain has $k \geq 1$ vertices. A chain with $k = 1$ has the composite kernel at one letter and the special edge at the other, so $k = 1$ occurs exactly for the pairs with a letter $A|_V$ and for the pairs $(c, A^*|_V)$, $(c, A^*|_{W_{\mathbb{R}}^\perp})$, (c, λ^*), (c, c^*); the other pairs, the field pairs $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$, $(A|_{W_{\mathbb{R}}^\perp}, \lambda)$, (λ, λ) among them, start at $k = 2$.*

- *(iv) At the pair weight without derivatives the leg sets of the ten pairs of field letters are those of Table 11.1; the numbers of leg sets with $n \leq 3$ derivatives on the two letters, for the* 21 *pairs of the active slots of Lemma* **??***, are those of Table 11.1.*

Proof. (i) Lemma 11.2 (iii), Lemma **??** and the vertices of Definition **??**. (ii), (iv) Enumeration of the chains with one composite kernel and one special edge, the vertices of I and the components of P and K_0, up to six vertices and output weight four above the pair weight; every chain is checked for degree, parity and the centre. A pair of two letters among $A^*|_{W_{\mathbb{R}}^\perp}$, λ^*, c^* needs an edge with a component on each letter; K_0 alone has one, and a chain has one special edge. (iii) A chain with $k = 0$ would join the two letters by the composite kernel and the special edge at once; the rest is read off from the enumeration. □

pair	pair weight	leg sets of the special-edge composite cycles, fewest vertices in brackets; †: position part along Δ and part along P_b
$(A\|_V, A\|_V)$	2	$A\|_V$ (1), $A\|_V^2$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^2$ (2), $A\|_V^{3\dagger}$ (2), $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^{2\dagger}$ (2), $A^*\|_V c^\dagger$ (2), $\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (2)
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	2	$A\|_{W_{\mathbb{R}}^\perp}$ (1), $A\|_V A\|_{W_{\mathbb{R}}^\perp}$ (2), $A\|_V^2 A\|_{W_{\mathbb{R}}^\perp}{}^\dagger$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^{3\dagger}$ (2), $A^*\|_{W_{\mathbb{R}}^\perp} c^\dagger$ (2), $(\lambda\|_{S_-\otimes W^*})^{2\dagger}$ (2), $(\lambda\|_{S_+\otimes W})^{2\dagger}$ (2)
$(A\|_V, \lambda\|_{S_+\otimes W})$	$\frac{5}{2}$	$\lambda\|_{S_+\otimes W}$ (1), $A\|_V \lambda\|_{S_+\otimes W}$ (2), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}$ (2), $c\lambda^*\|_{S_-\otimes W}$ (2), $A\|_V^2\lambda\|_{S_+\otimes W}{}^\dagger$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}{}^\dagger$ (3), $A\|_V c\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_+\otimes W}{}^\dagger$ (2), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	$\frac{5}{2}$	$\lambda\|_{S_-\otimes W^*}$ (1), $A\|_V \lambda\|_{S_-\otimes W^*}$ (2), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}$ (2), $c\lambda^*\|_{S_+\otimes W^*}$ (2), $A\|_V^2\lambda\|_{S_-\otimes W^*}{}^\dagger$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $A\|_V c\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_-\otimes W^*}{}^\dagger$ (2), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3)
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	2	$(A\|_{W_{\mathbb{R}}^\perp})^2$ (2), $A\|_V(A\|_{W_{\mathbb{R}}^\perp})^{2\dagger}$ (2)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	$\frac{5}{2}$	$A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_+\otimes W}{}^\dagger$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_-\otimes W^*}{}^\dagger$ (3), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	$\frac{5}{2}$	$A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}$ (2), $A\|_V A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}{}^\dagger$ (2), $(A\|_{W_{\mathbb{R}}^\perp})^2\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $A\|_{W_{\mathbb{R}}^\perp}c\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	3	$(\lambda\|_{S_+\otimes W})^2$ (2), $A\|_V(\lambda\|_{S_+\otimes W})^{2\dagger}$ (3), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $c\lambda\|_{S_+\otimes W}\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	3	$\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}$ (2), $A\|_V \lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $A\|_{W_{\mathbb{R}}^\perp}(\lambda\|_{S_-\otimes W^*})^{2\dagger}$ (3), $A\|_{W_{\mathbb{R}}^\perp}(\lambda\|_{S_+\otimes W})^{2\dagger}$ (3), $c\lambda\|_{S_-\otimes W^*}\lambda^*\|_{S_-\otimes W}{}^\dagger$ (3), $c\lambda\|_{S_+\otimes W}\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	3	$(\lambda\|_{S_-\otimes W^*})^2$ (2), $A\|_V(\lambda\|_{S_-\otimes W^*})^{2\dagger}$ (3), $A\|_{W_{\mathbb{R}}^\perp}\lambda\|_{S_-\otimes W^*}\lambda\|_{S_+\otimes W}{}^\dagger$ (3), $c\lambda\|_{S_-\otimes W^*}\lambda^*\|_{S_+\otimes W^*}{}^\dagger$ (3)

The leg sets of the special-edge composite cycles through the ten pairs of field letters at the pair weight without derivatives. The leg sets without † enter the translation-invariant part along Δ as well.

pair	$n=0$	$n=1$	$n=2$	$n=3$
$(A\|_V, A\|_V)$	7 (3)	18 (7)	34 (18)	67 (34)
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	7 (2)	16 (7)	32 (16)	59 (32)
$(A\|_V, \lambda\|_{S_+\otimes W})$	9 (4)	20 (9)	42 (20)	79 (42)
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	9 (4)	20 (9)	42 (20)	79 (42)
$(A\|_V, c)$	4 (2)	8 (4)	19 (8)	35 (19)
$(A\|_V, A^*\|_V)$	15 (6)	35 (15)	73 (35)	142 (73)
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	2 (1)	7 (2)	16 (7)	38 (16)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	4 (1)	12 (4)	30 (12)	62 (30)
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	4 (1)	12 (4)	30 (12)	62 (30)
$(A\|_{W_{\mathbb{R}}^\perp}, c)$	2 (1)	7 (2)	16 (7)	32 (16)
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	10 (2)	26 (10)	64 (26)	127 (64)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	4 (1)	12 (4)	28 (12)	63 (28)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	6 (1)	15 (6)	36 (15)	75 (36)
$(\lambda\|_{S_+\otimes W}, c)$	4 (1)	9 (4)	20 (9)	42 (20)
$(\lambda\|_{S_+\otimes W}, A^*\|_V)$	17 (5)	42 (17)	88 (42)	173 (88)
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	4 (1)	12 (4)	28 (12)	63 (28)
$(\lambda\|_{S_-\otimes W^*}, c)$	4 (1)	9 (4)	20 (9)	42 (20)
$(\lambda\|_{S_-\otimes W^*}, A^*\|_V)$	17 (5)	42 (17)	88 (42)	173 (88)
(c, c)	2 (1)	4 (2)	8 (4)	19 (8)
$(c, A^*\|_V)$	18 (7)	34 (18)	67 (34)	121 (67)
$(A^*\|_V, A^*\|_V)$	18 (5)	52 (18)	114 (52)	235 (114)

The number of leg sets of the special-edge composite cycles entering the part along P_b and, in brackets, the translation-invariant part along Δ, for n derivatives on the two letters.

Proposition 11.7 (Output bound). • *(i) A chain of Definition 11.5 with outputs of total Weyl weight w has superficial degree $w_{X_a} + w_{X_b} - w$ before the insertion factor of $S^{(1)}(x)$; the factor ∂_b adds one, and the translation-invariant part along Δ adds none.*

- *(ii) The composite pair kernel along P_b is a translation-invariant local functional of Weyl weight $w_{X_a} + w_{X_b} + 1$; the composite pair kernel along Δ is local of Weyl weight $w_{X_a} + w_{X_b}$, the position counted at weight -1, at most linear in the position. Both have the pair degree and the parity $|X_a| + |X_b|$.*

- *(iii) Chains with $w \le w_{X_a} + w_{X_b} + 1$ contribute to the kernel along P_b and to the part along Δ linear in the position, chains with $w \le w_{X_a} + w_{X_b}$ to the translation-invariant part along Δ; a chain above these bounds has a vanishing limit as $\varepsilon \to 0$.*

- *(iv) The number of output fields is at most $w_{X_a} + w_{X_b} + 1$ plus the pair degree for both kernels; the coefficients of ε^{-m} of the special-edge composite cycles have Weyl weight $w_{X_a} + w_{X_b} - 2m$ along Δ and $w_{X_a} + w_{X_b} + 1 - 2m$ along P_b.*

The same bounds hold for the composite cycles through the pair and for the arity-one counterterm, of degree one less than the pair degree: the coefficient of $\log\varepsilon$ *of the arity-one counterterm of* X_aX_b *has Weyl weight* $w_{X_a}+w_{X_b}$ *along* Δ *and* $w_{X_a}+w_{X_b}+1$ *along* P_b*.*

Proof. (i) Every edge obeys the weight rule w_1+w_2-4: P (Proposition **??**), K_0 (a field and its antifield have the weights w and $4-w$, momentum degree 0) and the composite kernel (Lemma 11.2 (vi)); the count of Lemma **??** applies. (ii) The coefficient of ε^0 is a polynomial in the output momenta of degree equal to the superficial degree (Theorem 11.11); along Δ the position enters once, through the factor $2x^\mu$ of Lemma 11.2 (iv). Degree and parity: Proposition 11.6 (ii). (iii) A polynomial of negative degree vanishes, and so does the limit of a chain of negative degree with a special edge (Theorem 11.11). (iv) Weyl weight plus degree is at least 1 for every output (Proposition **??**), so the number of outputs is at most w plus the pair degree. The statements on the arity-one counterterm follow in the same way from Lemma **??**. □

11.2 Closed forms and locality

Definition 11.8 (Position part and translation-invariant part)**.** A local functional is in canonical form if it is written $\int\omega\,D$ with $\omega\in\Omega^4_c(\mathbb{R}^4)$ and D a polynomial in letters, every derivative on a field. A map M of local functionals, linear in ω, of the form

$$M\Big(\int\omega\,D\Big)=\int\omega\,\big(x^\mu M_\mu(D)+M_0(D)\big),$$

with M_μ, M_0 maps of densities at one point and x^μ the coordinates of V in an orthonormal basis, has the position part $b^\mu M_\mu$ along $b\in V$ and the translation-invariant part M_0. A local functional $G=\int\big(x^\mu g_\mu+g_0\big)$ at most linear in the position has the position part $\int b^\mu g_\mu$ along b and the translation-invariant part $\int g_0$. On plane waves, with the momenta of the fields independent and ω absorbing the total momentum, a graph is $\alpha\,\delta(\cdot)+\beta^\mu\partial_\mu\delta(\cdot)$, and M_0 corresponds to α, M_μ to $i\beta^\mu$.

Lemma 11.9 (Translations and the canonical form)**.** *M is well defined, independently of integrations by parts against ω, exactly when every M_μ commutes with the derivatives and $[M_0,\partial_\mu]=M_\mu$. Then $[M,\{S^{(1)}(P_b),-\}]$ is the map $\int\omega\,D\mapsto\int\omega\,b^\mu M_\mu(D)$. For $M=v.=\{S^{(1)}(\Delta),-\}$ the position part is $2\partial_b$ and the translation-invariant part is the multiplication by $2w_D$; $\{S^{(1)}(P_b),-\}$ has position part 0.*

Proof. $M(\int\omega\,\partial_\mu D)$ and $M(-\int(\partial_\mu\omega)D)=\int\omega\big(M_\mu(D)+x^\nu\partial_\mu M_\nu(D)+\partial_\mu M_0(D)\big)$ agree for all ω exactly under the two conditions. Then $M(\int\omega\,\partial_bD)-\{S^{(1)}(P_b),M(\int\omega\,D)\}=\int\omega\,\big(M_0(\partial_bD)-\partial_bM_0(D)\big)=\int\omega\,b^\mu M_\mu(D)$, since $\{S^{(1)}(P_b),-\}$ acts by ∂_b on the fields and leaves the position and ω untouched. Lemma **??** (i) gives $v.\int\omega\,D=\int\omega\,2(x\cdot\partial+w_D)D$. □

Proposition 11.10 (Position decomposition)**.**

- *(i) For every pair of letters, the position part along b of the composite pair kernel along Δ is twice the composite pair kernel along P_b. For every local functional the position part of its arity-one counterterm along Δ is twice its arity-one counterterm along P_b; on cocycles the position part of the composite term along Δ is twice the composite term along P_b; and the position part of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(\Delta)$ along b is $2I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(P_b)$. In particular the quadratic part of the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(P_b)$ is $\frac12$ unit times $Q\big(\int(\iota_bA|_V,c^*)-\frac12\int(\rho(b)\lambda^*,\lambda^*)\big)$, and the quadratic part of the arity-one obstruction along P_b vanishes.*

- *(ii) The translation-invariant part of the composite pair kernel along Δ is computed with the internal vertex $\int\big(2(x-x_O)^\mu\partial_\mu(A|_V)+2A|_V,A^*|_V\big)$, x_O the point of the density, and pair by pair its commutator with $\{S^{(1)}(P_b),-\}$ is twice the composite pair kernel along P_b.*

Proof. (i) By Lemma 11.2 (iv) the internal vertex along Δ is $2\int x^\mu s_\mu + \int s_0$, with $\int s_\mu$ the internal vertex along P_{e_μ} and $s_0 = 2(A|_V, A^*|_V)$; insert $x^\mu = x_O^\mu + (x - x_O)^\mu$. On plane waves the factor x^μ is $-i\partial/\partial r^\mu$ of $e^{i(r,x)}$, with the momentum r injected at the internal vertex and absorbed by ω; its action on the momentum delta function is the position part, $-i$ times the graph with $\int s_\mu$ at $r = 0$, which is $2 \cdot (-i)$ times the graph along P_{e_μ}; its action on the integrand is translation invariant. The same holds graph by graph for the composite cycles, their singular parts, the counterterm trees and the composite loops of I. Theorem **??** (i) gives the quadratic part along Δ, which is Q of $\int x^\mu$ times a density with Q mapping letters to letters; Theorem **??** (ii) gives a quadratic obstruction along Δ without explicit position. (ii) The first statement is the decomposition of (i); the second is Lemma 11.9 applied to the composite pair kernel along Δ, whose position part is given by (i). □

Theorem 11.11 (Special-edge composite cycles as simplex integrals)**.** *For a chain of Definition 11.5 with k vertices of I and edges $e_0, \dots, e_k$, e_0 at X_a, let e_c be the composite kernel and e_σ, $\sigma \neq c$, the special edge. The cycle has $k+2$ Schwinger times: $t_j \in [\varepsilon, \infty)$ for $j \notin \{c, \sigma\}$, $t_\sigma \in [0, \varepsilon]$, and $t'_c, t''_c \in [\varepsilon, \infty)$ for the two segments of the composite kernel (Lemma 11.2 (vi)), which meet at the internal vertex of $S^{(1)}(x)$. The edge e_j carries $l + q_j$ as in Definition **??**, both segments carry $l + q_c$, and along Δ a momentum r is injected at the internal vertex and carried by the second segment and by $e_{c+1}, \dots, e_k$ to X_b and ω. The numerator is the product of the vertex functions, the edge factors of Definition **??**, the special-edge factor $-|l + q_\sigma|^2$ times the fibre factor of K_0, the segment factors $2(-,-)^{-1}$ times the inverse metric and $-i(l + q_c)(-,-)^{-1}$, and the insertion: $i(l + q_c)_b$ along P_b; 2 for $2\int(A|_V, A^*|_V)$ and $2i(l + q_c)_\mu$ for $2\int x^\mu(\partial_\mu(A|_V), A^*|_V)$ along Δ. With the simplex of dimension $k+1$ of the coordinates $\alpha = (\alpha_j)_{j\neq c} \cup (\alpha'_c, \alpha''_c)$, its centre, momentum form and m-th simplex integrand as in Definition **??** with $k+1$ in place of k, at generic output momenta and $T = \infty$,*

$$
\begin{aligned}
16\pi^2\big(\textit{coefficient of } \varepsilon^{-m}\big) &= \frac{1}{m}\int_{\alpha_\sigma \ \textit{smallest}} \Big[\big(\min_{f\neq\sigma}\alpha_f\big)^m - \alpha_\sigma^m\Big]\big(m\textit{-th simplex integrand}\big)\, d\alpha \qquad (m \geq 1),\\
16\pi^2\big(\textit{coefficient of } \varepsilon^{0}\big) &= \int_{\alpha_\sigma \ \textit{smallest}} \big(0\textit{-th simplex integrand}\big)\, \log\frac{\min_{f\neq\sigma}\alpha_f}{\alpha_\sigma}\, d\alpha,
\end{aligned}
\tag{11.1}
$$

the coefficient of $\log\varepsilon$ is 0, and the remainder tends to 0. Each coefficient is a polynomial in the output momenta and in r, of degree the superficial degree minus $2m$, and the limit vanishes for negative superficial degree. The coefficients of ε^0 are integrals of polynomials times logarithms of ratios of affine functions over rational subsimplices; for $k = 1$ they lie in $\mathbb{Q} + \mathbb{Q}\log 2 + \mathbb{Q}\log 3$ (Proposition 11.13).

Proof. The domain $\{t_\sigma \leq \varepsilon,\ t_f \geq \varepsilon\ (f \neq \sigma)\}$ is the difference of the domains with the cutoff shapes $1/\min_{f\neq\sigma}\alpha_f$ and $1/\min_f \alpha_f$, which differ on the region where α_σ is the smallest coordinate. Proposition **??** applies to each with $k+2$ edges, the Schwinger measure being $\tau^{k+1}d\tau\, d\alpha$, and the difference is taken as in Lemma **??**: the two terms in $\log\varepsilon$ cancel, and the finite parts differ by $-\int F\log(\alpha_\sigma/\min_{f\neq\sigma}\alpha_f)$ over that region, F the 0-th simplex integrand. The region $\tau \geq 1$ converges at $T = \infty$: the composite kernel is of order $|p|^{-3}$ at small momentum (Proposition **??** (iv)) and the other edges carry distinct generic momenta (proof of Theorem **??** (ii)), so the integrand is integrable and the measure of $\{\alpha_\sigma \leq \varepsilon/\tau\}$ is of order ε; the injected momentum enters polynomially at each order in r. The degree statement is Proposition 11.7 (i). □

Proposition 11.12 (Generating functions in the point splitting)**.** *With the letters at $x + y_a$ and $x + y_b$, $y = y_b - y_a$, the m-th simplex integrand for the numerator $(\textit{chain numerator})\, e^{i(y,l)}$ is the chain numerator with l replaced by $-i\partial_y$, applied to*

$$
(-1)^{m+k}e^{i(y,\textit{centre})}\sum_{i\geq 0}\frac{(|y|^2/4)^{i+m+k}(\textit{momentum form})^i}{i!\,(i+m+k)!} = (-1)^{m+k}e^{i(y,\textit{centre})}\Big(\frac{|y|}{2\sqrt{\textit{momentum form}}}\Big)^{m+k} I_{m+k}\big(|y|\sqrt{n}
$$

and, summed over the chains with the signs of Definition 11.5,

$$\sum_{\alpha,\beta}\frac{y_a^\alpha y_b^\beta}{\alpha!\,\beta!}\big(\textit{composite pair kernel of } (\partial^\alpha\varphi_a,\partial^\beta\varphi_b) \textit{ along } x\big)=\frac{\hbar}{16\pi^2}\,e^{i(y_b,q_k)}\int_{\alpha_\sigma\ \textit{smallest}}\big(0\textit{-th simplex integrand}\big)\log\frac{\min}{}$$

with q_k including r along Δ; the coefficients of ε^{-m} have the same generating function with the weights of (11.1)*. In position space iq_j is the derivative of the product of the outputs at x.*

Proof. As Proposition **??**, with $k+2$ edges: the power τ^{-1-m} of the Gaussian-integrated integrand times τ^{k+1} fixes the index shift $m+k$, and Theorem 11.11 replaces the weight $(\min_e \alpha_e)^m$ by the weights of the region where α_σ is the smallest. □

Proposition 11.13 (One vertex in closed form)**.** *Let $k=1$, $\beta:=\alpha_\sigma$, and let γ' and $\gamma''=1-\beta-\gamma'$ be the coordinates of the two segments. At $r=0$ the centre $-\beta q_\sigma-(1-\beta)q_c$ and the momentum form $\beta(1-\beta)|q_c-q_\sigma|^2$ depend on β alone, the region where β is the smallest coordinate requires $\beta\le\frac13$, and*

$$16\pi^2\big(\textit{composite pair kernel}\big)=\int_0^{1/3}\big(0\textit{-th simplex integrand}\big)(\beta)\Big[(1-\beta)\log\frac{1-\beta}{2\beta}-(1-3\beta)\Big]d\beta,$$

$$16\pi^2\big(\textit{coefficient of }\varepsilon^{-m}\big)=\int_0^{1/3}\big(m\textit{-th simplex integrand}\big)(\beta)\,\frac{2}{m}\Big[\frac{\big(\frac{1-\beta}{2}\big)^{m+1}-\beta^{m+1}}{m+1}-\beta^m\Big(\frac{1-\beta}{2}-\beta\Big)\Big]d\beta.$$

The part linear in r carries one factor γ'' in the simplex integrand, and $\int\gamma''(\cdots)\,d\gamma'=\frac{1-\beta}{2}\int(\cdots)\,d\gamma'$ by the symmetry $\gamma'\leftrightarrow\gamma''$ of the region and of the weights. The coefficients are integrals of $\beta^a(1-\beta)^b$ times these weights over $[0,\frac13]$, in $\mathbb{Q}+\mathbb{Q}\log2+\mathbb{Q}\log3$.

Proof. On the region, $\min_{f\neq\sigma}\alpha_f=\min(\gamma',\gamma'')$ and $\gamma'\in[\beta,1-2\beta]$. By the symmetry about $\gamma'=\frac{1-\beta}{2}$,

$$\int\log\frac{\min(\gamma',\gamma'')}{\beta}\,d\gamma'=2\int_\beta^{(1-\beta)/2}\log\frac{\gamma'}{\beta}\,d\gamma'=(1-\beta)\log\frac{1-\beta}{2\beta}-(1-3\beta),$$

and $\frac1m\int\big(\min(\gamma',\gamma'')^m-\beta^m\big)d\gamma'$ is the second weight. At $r=0$ both segments carry $l+q_c$, so the Gaussian data depend on β alone; along Δ the derivative in r of $\sum_j t_j|l+q_j|^2$ is $2t_c''(l+q_c)\cdot$, linear in γ''. The integrals $\int_0^{1/3}\beta^a(1-\beta)^b\log\frac{1-\beta}{2\beta}\,d\beta$ are rational combinations of 1, $\log2$, $\log3$, the endpoint $\beta=\frac13$ giving $\log\frac23$. □

Theorem 11.14 (The composite term is two-body and local)**.** *Let $O=\int\omega\,X_1\cdots X_n$ be a local functional with $sO=0$, $x\in\{\Delta,P_b\}$, $T=\infty$, generic momenta, and $\hbar F_1^{(0)}$ as in (i) of Definition **??**.*

- *(i) The special-edge composite cycles of (X_a,X_b) along x are $\sum_{m\ge1}\varepsilon^{-m}$ times local coefficients, plus the composite pair kernel, plus a remainder tending to 0; they contain no $\log\varepsilon$.*
- *(ii) s annihilates the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, and $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ is $\log\varepsilon$ times that coefficient in every sector.*
- *(iii) (Arity-one BRST covariance.) s of the coefficient of $\log\varepsilon$ in the arity-one counterterm of O along x is minus the bracket of O with the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, and the sum over the pairs, with the Koszul signs of* (**??**)*, of the coefficients of ε^{-m} of the special-edge composite cycles is s of the coefficient of ε^{-m} in the arity-one counterterm of O.*

- *(iv) The composite term of O along Δ (Definition **??**) and along P_b (Proposition **??**) is*

$$\big(\textit{composite term of } O \textit{ along } x\big) = \sum_{a<b} (-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)} \int \omega\, X_1 \cdots X_{a-1} \big(\textit{composite pair kernel of } (X_a, X \tag{11.2}$$

  a local two-body functional with no one-body term.

- *(v) Hence the composite terms of the pairs and of the legs of Theorem **??**, summed, equal the sum over the pairs of the composite pair kernels along Δ: the nonlocal parts of the brackets of the letters $A|_V$, λ, c with the renormalised composite two-point functional, $\log|p|^2$ of the leg momentum and $(p,a)(p,b)|p|^{-2}$ on $A|_V$ (Remark **??**), cancel against those of the images under Q of the renormalised composite cycles through the pairs containing these letters.*

- *(vi) For $sO \neq 0$ the right side of* (11.2) *acquires the renormalised composite graphs of sO, minus $\frac{\hbar}{2}\,\partial$ of the composite kernel applied to the trees with one vertex sO; these contain the nonlocal bracket with the renormalised composite two-point functional on the letters of sO.*

Outline. By Proposition 11.4, Q of the composite graphs of O is $\frac{\hbar}{2}\,\partial$ of the composite kernel applied to the trees with one edge $K_\varepsilon - K_0$. Sort them by the position of the special edge relative to the cycle closed by the composite kernel: (A) on a cycle through two letters, the special-edge composite cycles with trees on the outputs; (B) on a tree attached to a cycle through two letters, at an output or at a spectator letter, the factor $e^{-\varepsilon|p|^2}-1$ at a generic momentum times the unrenormalised composite cycle; (C) on a composite loop of I attached to O by a tree: by Proposition 11.4 for $M=0$, read on the amputated loops, where the collapse of a tree edge at a leg of the loop by $QP\star + P\star Q = e^{-\varepsilon H}$ gives the bracket with I, minus the amputated loops with one special edge is s of the amputated composite loops of I, that is, s of the renormalised composite functional along x plus $s\,I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, and (C) is its tree insertion into O; (D) on the tree joining a composite loop of I to O, the factor $e^{-\varepsilon|p|^2}-1$ times the unrenormalised loop. The counterterm trees, with the vertices $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ and the arity-one counterterm of O, have by Proposition **??** and Lemma **??** the images under Q

$$\begin{aligned} Q\big(-\text{tree}(I^{CT,(1)},O)\big) &= -\text{tree}\big(sI^{CT,(1)},O\big) - \{I^{CT,(1)},O\} + (\text{special edges}),\\ Q\big(-\text{tree}(\text{arity-one counterterm of } O)\big) &= -\text{tree}\big(s(\text{arity-one counterterm of } O)\big) + (\text{special edges}), \end{aligned}$$

with $I^{CT,(1)}$ short for $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, an odd vertex, and the special edges joining the counterterm vertex to a vertex I. These special edges cancel the singular parts of (B) and (D), whose sums are $e^{-\varepsilon|p|^2}-1$ times a renormalised cycle or loop, which have limits, and tend to 0.

(ii) By (C), s of the amputated composite loops of I is minus the loops with one special edge, which contain no $\log\varepsilon$ (Theorem 11.11). The coefficient of ε^{-m} of $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ is a colour-singlet local functional of degree -1 and Weyl weight $4-2m$ along Δ, the position at weight -1 and at most linear in it, and $5-2m$ along P_b; a colour singlet has at least two adjoint fields, and degree -1 requires an antifield of degree -1, of weight at least $\frac{5}{2}$, and a field of degree 0, of weight at least 1, or c^* and c, of weights 4 and 0: the field weight is at least $\frac{7}{2}$, and at least $\frac{5}{2}$ with one position, above $4-2m\leq 2$ and $5-2m\leq 3$.

(iii), (iv) The loops (C) give the tree insertion of s of the renormalised composite functional, and by (ii) $-\text{tree}(sI^{CT,(1)},O)$ vanishes; s of the renormalised composite functional is the composite part (Remark **??**, Proposition **??** (ii)), whose tree insertion is subtracted in the definition of the composite term. What remains is (A) minus the trees of s of the arity-one counterterm of O minus $\{I^{CT,(1)},O\}$. The left side has a limit, the composite graphs minus their counterterm trees converging (Proposition **??** (i)), and (A) contains no $\log\varepsilon$ by (i): the coefficient of $\log\varepsilon$ gives (iii), and so do the coefficients of ε^{-m}. At order ε^0 the trees on the outputs of the special-edge composite cycles and of the poles of

the arity-one counterterm are the same trees with edges $P(\varepsilon,\infty)$, so after the cancellation of the poles their difference tends to the trees of the composite pair kernels with edges $P(0,\infty)$, and the injectivity of the tree-level flow gives (11.2). Statement (i) is Theorem 11.11 for each chain.

(v) Definition **??** writes the same limit as the sum of the composite terms of the pairs and of the legs (proof of Theorem **??**), so the two sums agree on cocycles, and the right side is local. The mechanism: Q of a renormalised composite cycle through (X_a, X_b) produces, through $QP\star + P\star Q = e^{-\varepsilon H}$ on the edge at X_a, the composite cycles of the letters of $\{I, X_a\}$, and $sO = 0$ trades them for the composite loops on the legs, whose renormalised value is the renormalised composite two-point functional. (vi) For $sO \neq 0$, Proposition 11.4 keeps the trees with one vertex sO, whose composite closures are the composite graphs of sO of both kinds. □

Remark 11.15 (What the locality uses and what it fixes)**.** The argument uses $sO = 0$ exactly, the classical master equation, the anticommutation of Q with ∂ of the composite kernel, which rests on $[v_x., Q] = 0$ and $QP\star + P\star Q = e^{-\varepsilon H}$ (Lemma 11.2 (v)), the absence of $\log\varepsilon$ in cycles with one special edge, a property of the cutoff on every edge (Theorem 11.11), the absence of a term ε^{-1} in the composite loops of I (Theorem 11.14 (ii)), and generic momenta at $T = \infty$ for the convergence of the cycles and of the composite loops, which are of order $|p|^{-3}$ (Proposition **??** (iv)), or any finite T (Theorem **??** (ii)); (i) of Definition **??** enters through the terms with QW of the proof of Theorem **??**, which are outside the composite term. For a functional of the fields $A|_V$, $A|_{W_{\mathbb{R}}^{\perp}}$, λ alone, $sO = 0$ is the gauge invariance of its density. Statement (iv) fixes the one-body part of the composite term: a term t units times $v.X_a$ on every letter, added to the right side of (11.2), would replace each sector coefficient θ by $\theta + t$ in Sections 11.3–11.5 and leave every spectrum unchanged; the value $\theta = -\frac{1}{3}$ of Proposition 11.29 is the value in the normalisation of (11.2). The proof of Theorem 11.14 is an outline: steps (C) and (D), and the cancellation of (B) and (D) against the special edges of the counterterm trees, are stated for the classes of graphs. Statement (iv) is the input of Corollary 11.16, Propositions 11.17 and 11.31 and Sections 11.3–11.5; its consequences are evaluated on the bilinears, where the single chain of Example 11.18 reproduces the value of the route of the special edges, on the planar scalar sector, and through the spectra of Table 11.5, which a common one-body term leaves unchanged.

Corollary 11.16 (Local two-body form of D_1 and P_1)**.** *Under the hypotheses of Definition* **??***, and of the same definition with P_b in place of Δ, for $O = \int \omega\, X_1 \cdots X_n$, with the Koszul signs of* (**??**)*,*

$$D_1(O) = \sum_{a<b} \pm \int \omega \cdots \Big[4\big(\textit{coefficient of } \log\varepsilon \textit{ in } \partial_\delta I^{CT}[\varepsilon](X_aX_b)\big) + \big(\textit{composite pair kernel of } (X_a,X_b) \textit{ along } \Delta\big)\Big] \cdots$$
$$+ \sum_a \int \omega \cdots \Big[-4\,\textit{units}\cdot(\textit{weight of } X_a)\, X_a - \big\{\hbar F_1^{(1)}(\Delta) - 4\,\textit{units}\cdot(\textit{field-counting functional}), X_a\big\}\Big] \cdots,$$
$$P_1(O) = \sum_{a<b} \pm \int \omega \cdots \big(\textit{composite pair kernel of } (X_a,X_b) \textit{ along } P_b\big) \cdots - \sum_a \int \omega \cdots \big\{\hbar F_1^{(1)}(P_b), X_a\big\} \cdots,$$

every term local; D_1 minus four times the map of Theorem **??** *is the sum over the pairs of the composite pair kernels along Δ minus the bracket of O with $\hbar F_1^{(1)}(\Delta) - 4$ units times the field-counting functional.*

Proof. Insert (11.2) into Theorem **??** and Proposition **??**; four times the coefficient of $\log\varepsilon$ is two-body by Theorem **??** (iii), and the brackets with the field-counting functional, with $\hbar F_1^{(1)}(\Delta)$ and with $\hbar F_1^{(1)}(P_b)$ are derivations. □

Proposition 11.17 (The route of the special edges)**.** *For O as in Corollary 11.16, the limit as $\varepsilon \to 0$ of the cycles through the pairs of letters of O with one special edge and one vertex $S^{(1)}(\Delta)$ anywhere on the cycle, as the composite kernel or with one leg on the special edge, minus the trees of s of the arity-one counterterm of O, the graphs of Lemma* **??** *used in Theorem* **??***, is the sum over the pairs of four times the coefficient of $\log\varepsilon$ of the pair kernel plus the composite pair kernel along Δ.*

Proof. Both routes compute the local representative $D_1(O)$, which is unique (Theorem **??** (i)). In the route of the special edges the one-body terms are the trees with $-\{\hbar F_1^{(1)}(\Delta), O\}$, the arity-one defect of the action joined to O vanishing by (**??**) (proof of Theorem **??**); in Corollary 11.16 the one-body terms are -4 units times the weights minus the bracket with $\hbar F_1^{(1)}(\Delta) - 4$ units times the field-counting functional, which is $-\{\hbar F_1^{(1)}(\Delta), -\}$. The two-body sums therefore agree. □

Example 11.18 (The scalar bilinears)**.** For $O = \int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$, on the part with two legs $A|_{W_{\mathbb{R}}^\perp}$, Theorem **??** and Remark **??** give the value 4 (coefficient of $\log\varepsilon$ in $\partial_\delta I^{CT}[\varepsilon](O)) - \frac{4}{3}\,O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ units of the route of the special edges, so by Proposition 11.17

$$\big(\text{composite pair kernel of } (A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp}) \text{ along } \Delta\big)\Big|_{\text{outputs } A|_{W_{\mathbb{R}}^\perp} A|_{W_{\mathbb{R}}^\perp}} = \Big[-\tfrac{1}{3}\,v.O - \tfrac{4}{3}\,O\Big] \text{ units},$$

for the traceless and the trace parts alike. It is the single chain of Table 11.1,

$$A|_{W_{\mathbb{R}}^\perp} \xrightarrow{K_\varepsilon - K_0} \big([A|_{W_{\mathbb{R}}^\perp}, c], A^*|_{W_{\mathbb{R}}^\perp}\big) \xrightarrow{\text{composite kernel}} A|_V (A|_{W_{\mathbb{R}}^\perp})^2 \xrightarrow{P} A|_{W_{\mathbb{R}}^\perp}.$$

Its position part along b is $-\frac{2}{3}$ units times ∂_b, so the composite pair kernel along P_b is $-\frac{1}{3}$ units times $\{S^{(1)}(P_b), O\}$, in agreement with Theorem **??**, the composite triangle and the composite two-point functional along P_b giving $-2 + \frac{5}{3}$; its translation-invariant part is $-\frac{2}{3}\cdot 2\,O - \frac{4}{3}\,O = -\frac{8}{3}\,O$ units.

11.3 The free directions and the arity-two equation

Definition 11.19 (Finite counterterms along Δ and P_b)**.** Write the solutions of the arity-one equations along Δ and along P_b as

$$\begin{aligned}
\hbar F_1^{(1)}(\Delta) = {} & 4\,\text{units}\cdot(\text{field-counting functional}) + \Big[-\tfrac{2}{3}\int (A, A^*) + \tfrac{1}{3}\int(\lambda^*,\lambda) + \tfrac{4}{3}\int (c^*, c)\Big]\,\text{units} \\
& + \theta\,\text{units}\int \big(2E(A|_{W_{\mathbb{R}}^\perp}) + 2A|_{W_{\mathbb{R}}^\perp}, A^*|_{W_{\mathbb{R}}^\perp}\big) - \theta_\lambda\,\text{units}\int \big(2E(\lambda) + 3\lambda, \lambda^*\big) \\
& + \theta_V\,\text{units}\Big[\int \big(2E(A|_V) + 2A|_V, A^*|_V\big) - \int \big(2E(c), c^*\big)\Big] + \text{units}\cdot Q\Big(\kappa_V \int (\iota_x A|_V, c^*) + \kappa_\lambda \int (\rho(x)\lambda^*, \lambda \\
\hbar F_1^{(1)}(P_b) = {} & \theta_P\,\text{units}\int \big(\partial_b(A|_{W_{\mathbb{R}}^\perp}), A^*|_{W_{\mathbb{R}}^\perp}\big) - \theta_{P,\lambda}\,\text{units}\int (\partial_b\lambda, \lambda^*) + \theta_{P,V}\,\text{units}\Big[\int \big(\partial_b(A|_V), A^*|_V\big) - \int (\partial_b c, c^*)\Big] \\
& + Q\big(\text{a combination of } \textstyle\int(\iota_b A|_V, c^*) \text{ and } \int(\rho(b)\lambda^*, \lambda^*)\big) + (\text{cubic and higher}),
\end{aligned}$$

with numbers $\theta, \theta_\lambda, \theta_V, \kappa_V, \kappa_\lambda, \theta_P, \theta_{P,\lambda}, \theta_{P,V}$; θ is the coefficient of (**??**) and θ_P that of Theorem **??**. The three terms with θ, θ_λ, θ_V are the sector parts of $S^{(1)}(\Delta)$ in the sectors of $A|_{W_{\mathbb{R}}^\perp}$, of λ and of $(A|_V, c)$ times these numbers, and those with θ_P, $\theta_{P,\lambda}$, $\theta_{P,V}$ the sector parts of $S^{(1)}(P_b)$ (Definition 11.1). The scheme is translation covariant if the position part along b of $\hbar F_1^{(1)}(\Delta)$ is $2\hbar F_1^{(1)}(P_b)$; then $\theta_P = \theta$, $\theta_{P,\lambda} = \theta_\lambda$, $\theta_{P,V} = \theta_V$.

Proposition 11.20 (Completeness of the quadratic directions)**.** *Let $\mathcal{F}_{-1}$ and $\mathcal{F}_{-2}$ be the quadratic local functionals of degree -1 and -2, of Weyl weight 4 with the position at weight -1, at most linear in the position and invariant under* $\mathrm{Spin}(4)\times\mathrm{Spin}(6)$*. On the parity-even subspaces of Theorem* **??** *(iii), invariant under the reflection of one direction of V together with one direction of $W_{\mathbb{R}}^\perp$,*

$$\frac{\ker\big(Q|_{\mathcal{F}_{-1}}\big)}{Q\,\mathcal{F}_{-2}} = \mathrm{span}\big\{\textit{the three sector parts of } S^{(1)}(\Delta)\big\}, \qquad Q\,\mathcal{F}_{-2} = \mathrm{span}\Big\{Q\int(\iota_x A|_V, c^*),\ Q\int(\rho(x)\lambda^*, \lambda^*)\Big\}.$$

Without the reflection the quotient has one further class, $\int(\lambda^*, \rho(e_1e_2e_3e_4)\lambda)$, with $e_1,\dots,e_4$ an oriented orthonormal basis of V.

Proof. By Schur's lemma, $\mathcal{F}_{-1}$ is spanned by $\int(A^*|_V, D\,A|_V)$ with the structures D built from 1, $x\cdot\partial$, $x_\mu\partial_\nu$, $x_\nu\partial_\mu$ on the vector index and $\epsilon_{\mu\nu\kappa\tau}x^\kappa\partial^\tau$; by $\int(A^*|_{W_{\mathbb{R}}^\perp}, D\,A|_{W_{\mathbb{R}}^\perp})$ with $D\in\{1, x\cdot\partial\}$; by $\int(\lambda^*, D\lambda)$ with $D\in\{1,\ x\cdot\partial,\ \rho(x)\rho(\partial)\}$ times $\{1,\rho(e_1e_2e_3e_4)\}$; and by $\int(c^*, Dc)$ with $D\in\{1, x\cdot\partial\}$. $\mathcal{F}_{-2}$ is spanned by $\int x^\mu(A_\mu, c^*)$, $\int(\lambda^*, \rho(x)\lambda^*)$ and $\int(\lambda^*, \rho(e_1e_2e_3e_4)\rho(x)\lambda^*)$. The structures with ϵ or with one factor $\rho(e_1e_2e_3e_4)$ change sign under the reflection. With Q of Proposition **??**, the ranks on the plane-wave normal forms, computed exactly, are, in the sectors of $A|_{W_{\mathbb{R}}^\perp}$, of λ and of $(A|_V, c)$: $\dim\mathcal{F}_{-1} = 2,3,6$, $\dim\ker Q = 1,2,2$, $\dim Q\mathcal{F}_{-2} = 0,1,1$ for the reflection-invariant subspaces. The kernel is spanned by the sector part of $A|_{W_{\mathbb{R}}^\perp}$; by the sector part of λ and $Q\int(\rho(x)\lambda^*, \lambda^*)$; by the sector part of $(A|_V, c)$ and $Q\int(\iota_x A|_V, c^*)$. Without the reflection, $\dim\mathcal{F}_{-1} = 15$, the ϵ-structure is not Q-closed, $\int(\lambda^*, \rho(e_1e_2e_3e_4)\lambda)$ is closed and not exact, and $\int(\lambda^*, \rho(e_1e_2e_3e_4)\rho(x)\lambda^*) = 0$. □

Proposition 11.21 (What the arity-one equations fix)**.**

- *(i) Along Δ the quadratic part of* (**??**) *fixes the translation-invariant part $\big[-\frac{2}{3}\int(A,A^*)+\frac{1}{3}\int(\lambda^*,\lambda)+\frac{4}{3}\int(c^*,c)\big]$ units (Theorem **??** (iii)) and leaves θ, θ_λ, θ_V, κ_V, κ_λ free. Along P_b the quadratic part of the arity-one obstruction vanishes (Proposition 11.10 (i)), so the quadratic part of $\hbar F_1^{(1)}(P_b)$ is Q-closed and every coefficient of Definition 11.19 along P_b is free; the translation-covariant choice, $\hbar F_1^{(1)}(P_b)$ equal to half the position part along b of $\hbar F_1^{(1)}(\Delta)$, solves the equation along P_b at every order in the fields.*

- *(ii) Two solutions along Δ differ by an s-closed local functional of degree -1 and Weyl weight 4. The common shift of θ, θ_λ, θ_V by t adds t units times $S^{(1)}(\Delta)$, which is s-closed (Lemma **??** (iii)); the directions $Q\int(\iota_x A|_V, c^*)$ and $Q\int(\rho(x)\lambda^*,\lambda^*)$ complete to $s\int(\iota_x A|_V, c^*)$ and $s\int(\rho(x)\lambda^*,\lambda^*)$, and an s-exact direction sG contributes $s\{G,O\}$ to $D_1(O)$ on cocycles, zero on cohomology.*

- *(iii) Since $\{I, S^{(1)}(\Delta)\} = 0$ while the brackets of I with the sector parts are nonzero, the cubic part of* (**??**) *depends on the sector coefficients through $(\theta-\theta_V)$ times the bracket of the cubic part of I with the sector part of $A|_{W_{\mathbb{R}}^\perp}$ plus $(\theta_\lambda-\theta_V)$ times its bracket with the sector part of λ. It is solvable exactly when this sum is congruent to the cubic part of the composite part minus the bracket of the cubic part of I with the translation-invariant part, modulo the images under Q of the cubic local functionals of degree -1. If the classes of the two brackets modulo the images under Q of the cubic local functionals of degree -1, Weyl weight 4 and degree at most one in the position are linearly independent, the cubic equation fixes $\theta-\theta_V$ and $\theta_\lambda-\theta_V$.*

Proof. (i) Theorem **??** (iii) along Δ, Proposition 11.10 (i) along P_b. For the covariant choice: s maps letters to letters and carries no position, so it commutes with taking the position part, and the position part of the arity-one obstruction along Δ is twice the obstruction along P_b by Proposition 11.10 (i), the term 4 units times s of the field-counting functional having no position part. (ii) Theorem **??** (iii) restricted to arity one along Δ, and Lemma **??**: $\{sG,O\} = -s\{G,O\}$ for $sO=0$ and G of degree -2. (iii) s of the sector terms is the sum of θ, θ_λ, θ_V units times the brackets of I with the sector parts, whose sum over the three sectors vanishes; at cubic order s of the solution is Q of its cubic part plus the bracket of the cubic part of I with its quadratic part. □

Theorem 11.22 (The position part of D_1 is exact)**.** *Assume a one-loop quantisation through arity two along Δ and P_b, that is,* (**??**) *modulo $\hbar^2$ in its components of arity at most two along these generators, and local representatives through arity two along (Δ, P_b) on a space $\mathfrak{F}$ of local functionals stable under Λ, with P_1 translation covariant, $P_1(\int\omega\,D) = \int\omega\,P_1(D)$ for a map P_1 of densities commuting with the derivatives; this holds when $\hbar F_1^{(1)}(P_b)$ is translation invariant. Then for every $O=\int\omega\,D\in\mathfrak{F}$ with $sO=0$*

$$\big(\textit{position part of } D_1 \textit{ along } b\big)(O) = \big[D_1, \{S^{(1)}(P_b), -\}\big](O) = \pm\, s\Big(\hbar L_1(O)^{(2)}(\Delta, P_b)\Big)\,:$$

the position part of D_1 is s-exact on cocycles, with the arity-two component of O as primitive.

Proof. By Corollary **??** (ii), $[\Lambda^{(1)}(\Delta),\Lambda^{(1)}(P_b)]+\Lambda^{(1)}([\Delta,P_b])$ is a graded commutator of $\Lambda^{(0)}$ with an operator built from $\Lambda^{(2)}(\Delta,P_b)$. With $[\Delta,P_b]=-2P_b$ (Remark **??**) and $S^{(2)}(\Delta,P_b)=0$ (Definition **??**), the order-$\hbar^0$ part of that operator vanishes and the order-$\hbar$ part of the identity reads

$$\big[D_1,\{S^{(1)}(P_b),-\}\big]+\big[v.,P_1\big]-2P_1=\big[s,H_1\big],$$

with H_1 built from $\hbar L_1^{(2)}(\Delta,P_b)$. For translation-covariant P_1, $v.(P_1O)=\int\omega\,2(x\cdot\partial+w_D+1)P_1(D)$ and $P_1(v.O)=\int\omega\,\big(2x^\mu P_1(\partial_\mu D)+2w_D P_1(D)\big)$, so $[v.,P_1]=2P_1$. Lemma 11.9 identifies $[D_1,\{S^{(1)}(P_b),-\}]$ with the position part, and on a cocycle $[s,H_1]O=s(H_1O)$. □

Lemma 11.23 (Position part of a bracket)**.** *For a local functional $G=\int(x^\mu g_\mu+g_0)$, the position part along b of $\{G,-\}$ is $\{\int b^\mu g_\mu,-\}$.*

Proof. The bracket contracts a slot of g_μ at y with a letter $\partial^\alpha\varphi$ of D at x through K_0, which gives $\partial_x^\alpha(x^\mu\cdots)$; the term with no derivative on x^μ is x^μ times the bracket with $\int g_\mu$, and the others carry no position. □

Corollary 11.24 (The composite term along P_b and the translation component)**.** *Under the hypotheses of Theorem 11.22 and of Definition **??**, on every cocycle $O\in\mathfrak{F}$:*

- *(i) twice the composite term of O along P_b minus the bracket of O with the position part along b of $\hbar F_1^{(1)}(\Delta)$ is s-exact; this position part is*

  2θ *units* $\int\big(\partial_b(A|_{W_\mathbb{R}^\perp}),A^*|_{W_\mathbb{R}^\perp}\big)-2\theta_\lambda$ *units* $\int(\partial_b\lambda,\lambda^*)+2\theta_V$ *units*$\Big[\int\big(\partial_b(A|_V),A^*|_V\big)-\int(\partial_b c,c^*)\Big]+$*units*$\cdot Q\Big(\kappa_V$

  plus the position parts of the higher terms;

- *(ii) if $\theta=\theta_\lambda=\theta_V$, the composite term along P_b induces θ units times $\{S^{(1)}(P_b),-\}$ on cohomology on the parts with the fewest fields of the cocycles of $\mathfrak{F}$, on which $\hbar F_1^{(1)}(\Delta)$ acts through its quadratic part, and on every family in which $\{S^{(1)}(P_b),O\}$ is not exact for some cocycle the composite term along P_b determines θ;*

- *(iii) in the translation-covariant scheme the position part of D_1 is $2P_1$, and the arity-two equation along (Δ,P_b) is equivalent to $P_1\cong 0$ on cohomology: $P_1(O)=\pm\frac12\,s\big(\hbar L_1(O)^{(2)}(\Delta,P_b)\big)$;*

- *(iv) if $\theta=\theta_\lambda=\theta_V$ and $\theta_P=\theta_{P,\lambda}=\theta_{P,V}$, then P_1 induces $(\theta-\theta_P)$ units times $\{S^{(1)}(P_b),-\}$ on cohomology, and the finite part of the operators $\hbar\psi'=\frac{a}{2}\{S^{(1)}(\Delta),-\}$ (Definition **??**; the letter ψ of Section 11.5 is a different object), $a=(\theta-\theta_P)$ units, of Proposition **??** gives $P_1\cong 0$ on every family at once, leaves D_1, M_1, R_1 unchanged and changes the components along K_b, Σ and Σ^* by a, $-\frac{a}{2}$ and $\frac{a}{2}$ times the classical transformation. If the differences of the coefficients along P_b and along Δ depend on the sector, a multiple of $\{S^{(1)}(\Delta),-\}$ removes P_1 on the families with letters of one sector, and the translation-covariant scheme removes it on every family.*

Proof. (i) By Corollary 11.16 and Proposition 11.10 (i), the position part of D_1 is twice the sum over the pairs of the composite pair kernels along P_b minus the bracket with the position part of $\hbar F_1^{(1)}(\Delta)$, by Lemma 11.23; the coefficients of $\log\varepsilon$ and the bracket with the field-counting functional commute with the translations. Theorem 11.22 gives exactness, and the position part of each sector part of $S^{(1)}(\Delta)$ is twice the sector part of $S^{(1)}(P_b)$ (Definition 11.1). (ii) For equal coefficients the θ-terms are 2θ units times $S^{(1)}(P_b)$, which is s-closed, and the other directions are s-exact after completion, with $\{sG,O\}=-s\{G,O\}$. (iii) In the covariant scheme the position part of $\hbar F_1^{(1)}(\Delta)$ is $2\hbar F_1^{(1)}(P_b)$, and P_1 is the sum of the composite pair kernels along P_b minus the bracket with $\hbar F_1^{(1)}(P_b)$ (Corollary 11.16).

(iv) By (ii), $P_1 \cong \theta$ units times $\{S^{(1)}(P_b), -\}$ minus θ_P units times the same; Proposition **??** with $\lambda_{P_b} = -2$. For sector-dependent differences P_1 induces the sum over the sectors of the differences times the brackets with the sector parts of $S^{(1)}(P_b)$, plus the completions, which is a multiple of $\{S^{(1)}(P_b), -\}$ on the cocycles with letters of one sector. □

11.4 The dilatation identities in kernel form

Corollary **??** assumes that on cohomology the composite term minus the bracket with the finite counterterm beyond the field-counting functional induces $-\frac{1}{3}$ units times $\{S^{(1)}(\Delta), -\}$. By Corollary 11.16 this is a statement about the composite pair kernels, which splits into a statement along P_b and a statement about the translation-invariant part.

Lemma 11.25 (One-body terms)**.** *For $\theta = \theta_\lambda = \theta_V$, $\kappa_V = \kappa_\lambda = 0$ and the quadratic part of $\hbar F_1^{(1)}(\Delta)$ of Definition 11.19, the translation-invariant part of $-\{\hbar F_1^{(1)}(\Delta) - 4\,\text{units}\cdot(\text{field-counting functional}), X\}$ on a field letter $X = \partial^\alpha\varphi$ is $(c_\varphi - 2\theta\, w_X)$ units times X, with $c_\varphi = \frac{2}{3}$ for $A|_V$ and $A|_{W_{\mathbb{R}}^\perp}$, $\frac{1}{3}$ for λ and $\frac{4}{3}$ for c, and its position part along b is -2θ units times $\partial_b X$. At $\theta = -\frac{1}{3}$ the translation-invariant part is $\frac{2}{3}(2 + |\alpha|)$ units times X for every field slot and for c.*

Proof. $\{\int(A, A^*), X\} = X$, $\{\int(\lambda^*, \lambda), \lambda\} = -\lambda$ and $\{\int(c^*, c), c\} = -c$ (Lemma **??**), and the bracket of the sector part with a letter of its sector is $v.X$, with translation-invariant part $2w_X X$ and position part $2\partial_b X$ (Lemma 11.9). At $\theta = -\frac{1}{3}$: $\frac{2}{3} + \frac{2}{3}(1 + |\alpha|)$, $\frac{1}{3} + \frac{2}{3}(\frac{3}{2} + |\alpha|)$ and $\frac{4}{3} + \frac{2}{3}|\alpha|$ are all $\frac{2}{3}(2 + |\alpha|)$. □

Definition 11.26 (The identities in kernel form)**.** Let $\mathfrak{F}$ be a family of gauge-invariant cocycles of field letters, preserved by D_1 and contained in a space of local functionals stable under Λ, satisfying (ii) and (iii) of Definition **??**. The letter count is the derivation that multiplies a monomial $\int \omega\, X_1 \cdots X_n$ of letters $X_a = \partial^{\alpha_a}\varphi_a$, $\varphi_a \in \{A|_V, A|_{W_{\mathbb{R}}^\perp}, \lambda\}$, by $\sum_a (2 + |\alpha_a|)$; it is defined on canonical forms, and its commutator with $\{S^{(1)}(P_b), -\}$ is $\{S^{(1)}(P_b), -\}$. The three identities on $\mathfrak{F}$ are:

- (a) the cubic part of (**??**) forces $\theta = \theta_\lambda = \theta_V$;
- (b) the sum over the pairs, with the Koszul signs of (**??**), of the composite pair kernels along P_b induces $-\frac{1}{3}$ units times $\{S^{(1)}(P_b), -\}$ on the cohomology of $\mathfrak{F}$;
- (c) the sum over the pairs of the translation-invariant parts of the composite pair kernels along Δ induces $-\frac{2}{3}$ units times the letter count on the cohomology of $\mathfrak{F}$.

Remark 11.27 (Dependences among the identities)**.** (1) By Proposition 11.10 (ii), pair by pair the commutator of the translation-invariant part of the composite pair kernel along Δ with $\{S^{(1)}(P_b), -\}$ is twice the composite pair kernel along P_b, and the commutator of the letter count with $\{S^{(1)}(P_b), -\}$ is $\{S^{(1)}(P_b), -\}$; hence (c) on $\mathfrak{F}$ and on $\{S^{(1)}(P_b), \mathfrak{F}\}$ gives (b) along b, and the content of (b) beyond (c) lies in the directions b with $\{S^{(1)}(P_b), \mathfrak{F}\}$ outside $\mathfrak{F}$. (2) Identity (a) follows from (b) on three families with letters of one sector each, under the hypotheses of Theorem 11.22 (Proposition 11.29); the cubic arity-one equation is then a consistency condition. (3) Identities (b) and (c) are identities between graph sums at the regulator of Definition **??**; the numbers $-\frac{1}{3}$ and $-\frac{2}{3}$ are finite parts of special-edge cycles, which depend on the cutoff shape (Remark **??** (d), Proposition **??**). The statement independent of the scheme is the spectrum of D_1 at the coefficients fixed by the arity-two equation (Remark **??**).

Theorem 11.28 (Consequences of the identities)**.** *Assume (a), (b) and (c) of Definition 11.26 on $\mathfrak{F}$.*

- *(i) On the parts with the fewest fields of the cocycles of $\mathfrak{F}$, on which $\hbar F_1^{(1)}(\Delta)$ acts through its quadratic part, D_1 minus four times the map of Theorem **??** induces $-(\frac{1}{3} + \theta)$ units times $\{S^{(1)}(\Delta), -\}$ on cohomology: the assumption of Corollary **??**, for the solution with sector coefficient θ.*

- *(ii) Under the hypotheses of Theorem 11.22, and if $\{S^{(1)}(P_b), O\}$ is not exact for some cocycle $O \in \mathfrak{F}$, $\theta = -\frac{1}{3}$, and D_1 induces four times the map of Theorem* **??***; its spectrum on $\mathfrak{F}$ in a fixed classical Weyl weight is twice that of the one-loop dilatation coefficients, -8 units times the eigenvalues of $\sum_l H_{l,l+1}$ of [B03c, (1.1)–(1.3)] on the planar families of Corollary* **??** *and of Table 11.5.*

- *(iii) Under the hypotheses of (ii), in the translation-covariant scheme $P_1 \cong 0$ on $\mathfrak{F}$; otherwise Corollary 11.24 (iv) applies.*

Conversely, if D_1 minus four times the map induces $-(\frac{1}{3}+\theta)$ units times $\{S^{(1)}(\Delta), -\}$ for equal sector coefficients, then (b) and (c) hold on $\mathfrak{F}$.

Proof. (i) By Corollary 11.16, D_1 minus four times the map is the sum of the composite pair kernels along Δ minus the bracket with $\hbar F_1^{(1)}(\Delta) - 4$ units times the field-counting functional. Its position part along b is, by (b) and Lemma 11.25, $2 \cdot (-\frac{1}{3})\partial_b - 2\theta\,\partial_b = -2(\frac{1}{3}+\theta)\partial_b$ units. Its translation-invariant part is, by (c) and Lemma 11.25, $-\frac{2}{3}$ units times the letter count plus $\sum_a (c_{\varphi_a} - 2\theta w_{X_a})$ units, which is $-2(\frac{1}{3}+\theta)w_O$ units, by $\sum_a (c_{\varphi_a} + \frac{2}{3} w_{X_a}) = \frac{2}{3}\sum_a (2 + |\alpha_a|)$. Together they give $-(\frac{1}{3}+\theta)(2x\cdot\partial + 2w_O)$ units, which is $-(\frac{1}{3}+\theta)$ units times v. (Lemma 11.9). (ii) By Theorem 11.22 the position part of D_1, which is $-2(\frac{1}{3}+\theta)$ units times ∂_b on cohomology, is exact, and $\partial_b O$ is not exact for that cocycle, so $\theta = -\frac{1}{3}$; then Theorem **??** and Corollary **??**. (iii) Corollary 11.24 (iii). The converse reverses the two computations of (i). □

Proposition 11.29 (The arity-two equation fixes each sector coefficient)**.** *Assume the hypotheses of Theorem 11.22 and of Definition* **??***. Let $\mathfrak{F}_\perp$, $\mathfrak{F}_\lambda$, $\mathfrak{F}_V$ be families of gauge-invariant cocycles whose two-field parts have letters of the sector of $A|_{W_{\mathbb{R}}^\perp}$, of λ and of $A|_V$ alone, each containing a cocycle O for which $\{S^{(1)}(P_b), O\}$ is not exact. If (b) holds along b on $\mathfrak{F}_\perp$, then $\theta = -\frac{1}{3}$; if it holds along b on $\mathfrak{F}_\lambda$, then $\theta_\lambda = -\frac{1}{3}$; if it holds along b on $\mathfrak{F}_V$, then $\theta_V = -\frac{1}{3}$. In particular (b) on the three families gives $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$, the equality asserted in (a), and the cubic part of* (**??**) *admits the solution $\theta - \theta_V = \theta_\lambda - \theta_V = 0$.*

Proof. By Corollary 11.16, Proposition 11.10 (i) and Lemma 11.23, the position part of D_1 is twice the sum of the composite pair kernels along P_b minus the bracket with the position part of $\hbar F_1^{(1)}(\Delta)$. On the two-field part $\hbar F_1^{(1)}(\Delta)$ acts through its quadratic part; the bracket of the sector part of the sector of X has the position part $2\partial_b X$ and the other sector parts have bracket 0 with X, and the κ-directions are s-exact (Proposition 11.21 (ii)). With (b), on the cohomology of the family of the sector with coefficient $\theta_\star \in \{\theta, \theta_\lambda, \theta_V\}$,

$$\big(\text{position part of } D_1 \text{ along } b\big) \cong -\tfrac{2}{3}\,\text{units}\cdot\partial_b - 2\theta_\star\,\text{units}\cdot\partial_b = -2\big(\tfrac{1}{3}+\theta_\star\big)\,\text{units}\cdot\partial_b.$$

The left side is s-exact on cocycles (Theorem 11.22) and $\partial_b O$ is not exact, so $\theta_\star = -\frac{1}{3}$. The last statement is Proposition 11.21 (iii); the existence of the quantisation is a hypothesis of Theorem 11.22. □

Corollary 11.30 (The dilatation component without the cubic equation)**.** *Under the hypotheses of Proposition 11.29, if (b) holds on $\mathfrak{F}_\perp$, $\mathfrak{F}_\lambda$ and $\mathfrak{F}_V$, and (b) and (c) hold on $\mathfrak{F}$, then on the cohomology of $\mathfrak{F}$, D_1 induces four times the map of Theorem* **??***, with spectrum -8 units times the eigenvalues of $\sum_l H_{l,l+1}$ on the planar families of Corollary* **??** *and of Table 11.5, and $P_1 \cong 0$ in the translation-covariant scheme.*

Proof. Proposition 11.29 gives $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$, and Theorem 11.28 (i)–(iii) applies. □

Proposition 11.31 (The planar scalar sector)**.** *On the part with the outputs $A|_{W_{\mathbb{R}}^{\perp}}$ of the planar scalar sector (Proposition* **??***), (b) and (c) hold as identities modulo terms whose sum over the cyclically adjacent pairs of every trace vanishes: per adjacent pair the composite pair kernel along Δ is $\big[-\frac{1}{6}v.(X_aX_b)-\frac{2}{3}X_aX_b\big]$ units and along P_b it is $-\frac{1}{6}$ units times $\partial_b(X_aX_b)$, so that on $O=\mathrm{tr}(\iota_{w_1}A\cdots\iota_{w_L}A)$ the composite term along Δ is $-\big(\frac{1}{3}v.+\frac{2}{3}L\big)O$ units, the composite term along P_b is $-\frac{1}{3}$ units times $\partial_b O$, and the sum of the translation-invariant parts is $-\frac{4}{3}L$ units, $-\frac{2}{3}$ units times the letter count. On the bilinears these are exact identities (Example 11.18).*

Proof. By Proposition **??** (i), (e), the composite term of an adjacent pair is $-v.(X_aX_b)$ units, and the bracket with the renormalised composite two-point functional on a letter is $(\frac{5}{3}v.X-\frac{2}{3}X)$ units (Remark **??**); by Theorem 11.14 (v) their sum is the sum of the composite pair kernels, and the one-body term distributed over the two adjacent pairs of each letter gives the density per pair. For L letters without derivatives the letter count is $2L$ and the translation-invariant part of $v.$ is $2L$: $-\frac{1}{3}\cdot 2L-\frac{2}{3}L=-\frac{4}{3}L$. Along P_b: Theorem **??**, $-2+\frac{5}{3}=-\frac{1}{3}$. □

11.5 The composite kernels on the families

Throughout this section $\mathfrak{g}=\mathfrak{su}(N)$ at leading order in N and O is a single trace, with the conventions of Section 10.7. The projection to cohomology on the two-field part is the evaluation on the free equations of motion, $|p|^2=0$ on the scalar legs, the Weyl equation on the gaugino legs and transverse on-shell gauge legs, which removes the s-exact terms $s(\text{antifield}\times\text{field})$; total derivatives are kept, as descendants through ω. The letters are written in the components of Definition **??**.

Definition 11.32 (Test families)**.** Let $n_m:=\sigma_{11}^m=(0,0,1,i)$, a null vector, and $D_n:=n_m(\partial_m+[A_m,-])$, the covariant derivative $\iota_n d_{A|_V}$ along n; let $Z=\iota_w A$ with $w\in W_{\mathbb{R}}^{\perp}\otimes\mathbb{C}$, $(w,w)=0$; let $\psi:={\lambda_1}^i$ for a fixed i, the component $\alpha=1$ of $\lambda|_{S_+\otimes W}$; and let F_{11} be the component $\alpha=\beta=1$ of $(\sigma^m\bar\sigma^n)_\alpha{}^\gamma\varepsilon_{\gamma\beta}F_{mn}$, $F_{mn}=\partial_mA_n-\partial_nA_m+[A_m,A_n]$, whose two-field part is a component of $dA|_V$ and whose three-field part is a component of $[A|_V\wedge A|_V]$. The test families are the spans of the following single traces and their gauge completions:

- the scalars: $\mathrm{tr}(\iota_{w_1}A\cdots\iota_{w_L}A)$, the bilinears, and the $\mathfrak{su}(2)$ sector of Proposition **??** for $L\le 8$;
- twist two: $\mathrm{tr}\big(D_n^kZ\,D_n^{S'-k}Z\big)$, $0\le k\le S'$ (Proposition **??**);
- the $\mathfrak{su}(1|1)$ sector: single traces of the letters Z and ψ of Proposition **??**, $L\le 6$;
- the fermion $\mathfrak{sl}(2)$ sector: $\mathrm{tr}\big(D_n^k\psi\,D_n^{S'-k}\psi\big)$;
- two gaugino components: $\mathrm{tr}\big(D_n^k\psi^{(3)}\,D_n^{S'-k}\psi^{(4)}\big)$ with $\psi^{(3)}={\lambda_1}^3$, $\psi^{(4)}={\lambda_1}^4$, together with $\mathrm{tr}\big(D_n^kZ'\,D_n^{S'-k}F_{11}\big)$, Z' the scalar letter with the R-symmetry weights of $\psi^{(3)}\psi^{(4)}$, with which the family closes;
- the gluon $\mathfrak{sl}(2)$ sector: $\mathrm{tr}\big(D_n^kF_{11}\,D_n^{S'-k}F_{11}\big)$;
- the mixed gluon–gaugino family: $\mathrm{tr}\big(D_n^kF_{11}\,D_n^{S'-k}\psi\big)$.

Twist two, the fermion and the gluon $\mathfrak{sl}(2)$ sectors are families of Proposition 11.29 for the sectors of $A|_{W_{\mathbb{R}}^{\perp}}$, of λ and of $A|_V$.

The two-field part of a gauge-invariant operator with covariant derivatives receives the pairs of its two leading letters with two outputs and the pairs of a letter $A|_V$ from a covariant derivative, or from the three-field part of a field strength, with another letter, with one output (Proposition **??**). For the composite pair kernels these are the leg sets $A|_{W_{\mathbb{R}}^{\perp}}$ of $(A|_V,A|_{W_{\mathbb{R}}^{\perp}})$, λ of $(A|_V,\lambda)$ and $A|_V$ of $(A|_V,A|_V)$ of Table 11.1, all with $k=1$, with derivatives.

Proposition 11.33 (Spectra from the harmonic action)**.** *On the two-site graded cyclic states, single traces of two letters with S' derivatives D_n in total, the eigenvalues of $\sum_l H_{l,l+1}$ of [B03c, (1.1)–(1.3)], computed from the harmonic action [B03c, (4.27)–(4.29)] on the oscillator letters [B03c, (4.9)], are:*

- *(i) scalars $D_n^k Z$: $4h(S)$, $S \le S'$ even [B03c, (2.31)];*
- *(ii) gluons $D_n^k F_{11}$: $4h(S+2)$, $S \le S'$ even, so that $\mathrm{tr}(F_{11}F_{11})$ has the Konishi value 6;*
- *(iii) one gaugino component $D_n^k \psi$: $4h(S+1)$, $S \le S'$ odd;*
- *(iv) two gaugino components $\psi^{(3)}$, $\psi^{(4)}$ with $Z'F_{11}$: $0, 6$ at $S'=0$; $0,6,6,6$ at $S'=1$; $0,6,6,6,6,\frac{25}{3}$ at $S'=2$; 0, four times 6 and three times $\frac{25}{3}$ at $S'=3$; 0, four times 6, four times $\frac{25}{3}$ and $\frac{49}{5}$ at $S'=4$;*
- *(v) (F_{11},ψ): 6; $6,6$; $6,6,\frac{25}{3}$; $6,6,\frac{25}{3},\frac{25}{3}$; $6,6,\frac{25}{3},\frac{25}{3},\frac{25}{3},\frac{49}{5}$ at $S'=0,\dots,4$.*

*On the $\mathfrak{su}(1|1)$ sector the two-site Hamiltonian is one minus the graded exchange (Proposition **??**), with the cyclic spectra of Table 11.5.*

Proof. Exact evaluation of [B03c, (4.27)–(4.29)], with the harmonic numbers of the coefficients $c_{n,0,0} = h(\frac{n}{2})$, on bases of the families closed under H_{12}, with the graded exchange for H_{21} and the projection to graded cyclic states, and exact diagonalisation. The same evaluation gives [B03c, (2.31)] on twist two and the values 0, twenty times, and 6 on the scalar bilinears, which fixes the central charge per site, the signs of the fermionic oscillators and the graded cyclicity. □

(L,M)	eigenvalues on the graded cyclic states of $\mathrm{tr}(Z^{L-M}\psi^M)$
$M \le 1$	0
$(3,2)$, $(3,3)$	6
$(4,2)$, $(4,3)$	4
$(5,2)$, $(5,3)$	$5 \pm \sqrt{5}$
$(5,4)$, $(5,5)$	10
$(6,2)$	2, 6
$(6,3)$	2, 6, 8, 8
$(6,4)$	8, 8, 8
$(6,5)$	8

The eigenvalues of $\sum_l H_{l,l+1}$ on the $\mathfrak{su}(1|1)$ sector, $L \le 6$; the sectors $(2,2)$, $(4,4)$, $(6,6)$ have no graded cyclic state.

Theorem 11.34 (The composite pair kernels on the families)**.**

- *(i) (The two-site density.) For every ordered adjacent pair (X_a, X_b) of letters among $A|_{W_{\mathbb{R}}^{\perp}}$, $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$ without derivatives, on the free shell, and for two letters $A|_{W_{\mathbb{R}}^{\perp}}$ also off the shell and for all their components, the planar composite pair kernels have, on the same word X_aX_b, the translation-invariant part $-\frac{4}{3}$ units times X_aX_b along Δ and the value $-\frac{1}{6}$ units times $\partial_b(X_aX_b)$ along P_b; on the exchanged word and on the trace they vanish.*
- *(ii) On the parts with the fewest fields of the families of Definition 11.32, the two-field parts for two letters, on the free shell, the sum over the pairs of the translation-invariant parts of the composite pair kernels along Δ is the diagonal operator of Table 11.5, which is $-\frac{2}{3}$ units times the letter count, and the sum of the composite pair kernels along n is $-\frac{1}{3}$ units times ∂_n; on twist two for $S' \le 4$ the sum along every $b \in V$ is $-\frac{1}{3}$ units times ∂_b.*
- *(iii) Hence (b) along n and (c) of Definition 11.26 hold on each family in the ranges of Table 11.5, and (b) along every b holds on twist two for $S' \le 4$.*

Proof. Exact evaluation of the special-edge composite cycles by Theorem 11.11, Propositions 11.12 and 11.13, with the colour words reduced to the planar adjacent form of Lemma **??** (iv), on the leg sets of Table 11.1 with the derivatives of the families, the outputs evaluated on the free shell (Remark 11.36). On the families with derivatives the outputs depend on the output momenta through (n,p), every derivative being contracted with n and $(n,n)=0$; the sums are diagonal in the bases of the families. (iii) is (i), (ii) and the values of the letter count: $2L$ on L letters without derivatives, $4+S'$ on twist two and on two gaugino letters, $5+S'$ on $Z'F_{11}$ and on (F_{11},ψ), $6+S'$ on two gluon letters, a letter F_{11} counting 3 through its two-field part $dA|_V$. □

family	range	sum of the translation-invariant parts along Δ	spectrum of D_1 at $\theta=\theta_\lambda=\theta_V=-\frac{1}{3}$, in units
scalars	bilinears; $\mathfrak{su}(2)$, $L\le 8$	$-\frac{4}{3}L$	0, -48; -8 times the Heisenberg spectra
twist two	$S'\le 6$	$-\frac{2}{3}(4+S')$	0, -48, $-\frac{200}{3}$, $-\frac{392}{5}$: $-32h(S)$, $S\le S'$ even
$\mathfrak{su}(1\|1)$	$L\le 6$	$-\frac{4}{3}L$	-8 times the $\mathfrak{su}(1\|1)$ spectra
fermion $\mathfrak{sl}(2)$	$S'\le 5$	$-\frac{2}{3}(4+S')$	-48, $-\frac{200}{3}$, $-\frac{392}{5}$: $-32h(S+1)$, $S\le S'$ odd
$\psi^{(3)}\psi^{(4)}$ with $Z'F_{11}$	$S'\le 2$	$-\frac{2}{3}(4+S')$ on $\psi^{(3)}\psi^{(4)}$, $-\frac{2}{3}(5+S')$ on $Z'F_{11}$	$0,-48$; $0,-48,-48,-48$; 0, four times -48, $-\frac{200}{3}$
gluon $\mathfrak{sl}(2)$	$S'\le 2$	$-\frac{2}{3}(6+S')$	-48, $-\frac{200}{3}$: $-32h(S+2)$, $S\le S'$ even
(F_{11},ψ)	$S'\le 4$	$-\frac{2}{3}(5+S')$	at $S'=4$: -48, -48, $-\frac{200}{3}$, $-\frac{200}{3}$, $-\frac{392}{5}$

The composite pair kernels and the spectrum of D_1 on the families: the third column is identity (c), the fourth is -8 units times the eigenvalues of Proposition 11.33, of Proposition **??** on the $\mathfrak{su}(2)$ sector and of Table 11.5 on the $\mathfrak{su}(1|1)$ sector.

Corollary 11.35 (D_1 and P_1 on the families)**.** *Under the hypotheses of Definition* **??***, on the cohomology of the part with the fewest fields of every family of Definition 11.32, with every free direction of Definition 11.19,*

$$D_1 \cong 4\,(\textit{the map}) - \sum_a \big(\theta_{X_a}+\tfrac{1}{3}\big)\ \textit{units}\cdot v.X_a, \qquad P_1 \cong -\sum_a \big(\theta_{P,X_a}+\tfrac{1}{3}\big)\ \textit{units}\cdot \partial_b X_a,$$

with the map of Theorem **??***, the position part of D_1 and the component P_1 taken along $b=n$, and along every $b\in V$ on twist two for $S'\le 4$ and on the letters without derivatives (Theorem 11.34), where θ_{X_a} is θ, θ_λ or θ_V and θ_{P,X_a} is θ_P, $\theta_{P,\lambda}$ or $\theta_{P,V}$ according to the sector of the letter X_a, $v.X_a$ and $\partial_b X_a$ act on the letter X_a alone, and the directions $Q\int(\iota_x A|_V, c^*)$, $Q\int(\rho(x)\lambda^*,\lambda^*)$ contribute s-exact terms. Under the hypotheses of Theorem 11.22, twist two gives $\theta=-\frac{1}{3}$, the fermion $\mathfrak{sl}(2)$ sector $\theta_\lambda=-\frac{1}{3}$ and the gluon $\mathfrak{sl}(2)$ sector $\theta_V=-\frac{1}{3}$ (Proposition 11.29); then D_1 induces four times the map of Theorem* **??** *on every family, with the spectra of Table 11.5, -8 units times the eigenvalues of $\sum_l H_{l,l+1}$, which is twice the one-loop anomalous dimensions; and $P_1\cong 0$ in the translation-covariant scheme, while for equal coefficients θ_P along P_b, $P_1\cong(\theta-\theta_P)$ units times ∂_b with $\theta=-\frac{1}{3}$.*

Proof. Corollary 11.16 and Theorem 11.34: on the two-field part $\hbar F_1^{(1)}(\Delta)$ acts through its quadratic part, the bracket of the sector part of the sector of X_a with X_a being $v.X_a$ and the others vanishing. As in the proof of Theorem 11.28 (i), letter by letter, the position part of the composite pair kernels and of the bracket is $-2(\frac{1}{3}+\theta_{X_a})\partial_b$ units and the translation-invariant part is $-2(\frac{1}{3}+\theta_{X_a})w_{X_a}$ units; along P_b the composite pair kernels give $-\frac{1}{3}$ units times ∂_b and the bracket with $\hbar F_1^{(1)}(P_b)$ gives $-\theta_{P,X_a}$ units times $\partial_b X_a$. The κ-directions are s-exact (Proposition 11.21 (ii)). The rest is Proposition 11.29, Theorem 11.28 and Corollary 11.24 (iii), (iv). □

Remark 11.36 (Method of computation)**.** Each special-edge composite cycle is evaluated exactly: the Gaussian loop integral and the integrals of (11.1) over the ordered subsimplices of the region where α_σ is the smallest coordinate, the letter derivatives through Proposition 11.12, the position through the injected momentum r (Proposition 11.10), the chains with $k=1$ through Proposition 11.13, the fibre algebra in explicit realisations of V_{10} and Σ over $\mathbb{Q}(i)$, and the colour words by Lemma **??**, with its planar adjacent rule also checked in realisations of $\mathfrak{su}(2)$, $\mathfrak{su}(3)$ and $\mathfrak{gl}(3)$. In this evaluation every chain carries the sign of Definition 11.5, the factor $(-1)^{[c<\sigma]}$ included, and the families are evaluated

at generic null leg momenta. A second evaluation works in momentum space with the kernel form of Lemma 11.2 (ii), $v.$ acting on the Fourier components as an Euler operator plus constants, Feynman parameters for the edges P and the special Schwinger time integrated last, applies the sign of Definition 11.5 as written, and evaluates the families on on-shell legs with momenta proportional to one null vector $\bar{n}$, $(n,\bar{n}) \neq 0$, which determine the translation-invariant parts and the parts along n (proof of Theorem 11.34). The two evaluations give the same off-shell entries with at most two outputs and no letter derivatives, for the pairs of bosonic letters and for the pairs with a gaugino letter or output, whose values lie in $\mathbb{Q} + \mathbb{Q}\log 2 + \mathbb{Q}\log 3$ for $k \leq 2$, and the same family matrices on the scalars, on twist two for $S' \leq 4$, on the $\mathfrak{su}(1|1)$ sector, on the fermion $\mathfrak{sl}(2)$ sector for $S' \leq 5$ and on the gluon $\mathfrak{sl}(2)$ sector for $S' \leq 2$; the first evaluation extends twist two to $S' \leq 6$, and the second evaluation gives the two families with F_{11} and a gaugino letter. A third evaluation, a Taylor expansion in momentum space at generic null leg momenta, gives the sums on twist two for $S' \leq 4$ along a generic b. Every integral is a four-dimensional Schwinger integral with the cutoff ε on every edge.

11.6 The two-body kernels of Q_1, S_1 and K_1

For x in a homogeneous basis of $\mathfrak{psl}(4|4)$, λ_x is its ad_Δ eigenvalue, -1 on Σ, $+1$ on Σ^* and $+2$ for K_b, and the degree of x in the position is 0, 1 and 2 (Section 10.9). The two-body kernel of a pair of letters along x is the kernel of Proposition **??**; this section computes it along the supercharges, the special supercharges and the special conformal fields. Along Σ the vertex $S^{(1)}(\xi)$ meets the gauge-fixing operator in a new way: the commutator $[\xi., Q^{GF}]$ joins the gaugino to the ghost, and $S^{(1)}(\xi)$ sits on a cycle as a line from λ to c whose image under Q is nonzero, in contrast to the composite lines along Δ and P_b. The section shows that the component along x of every local functional is nevertheless a sum over its pairs of letters of local two-body kernels, the finite parts of the cycles with one vertex $S^{(1)}(x)$ and one special edge, plus a one-body term; writes these kernels as simplex integrals, independent of the endpoint along K_b; computes them on the scalar bilinears, where the two-body part of Q_1 on the traceless part is $-2\log\frac{3}{2}$ units times the classical transformation, and derives S_1 and K_1 there; and evaluates the pair entries of the twisted differential along a rank-one supercharge. Edges carry $P(\varepsilon,\infty)$ along Σ and Σ^* (Proposition **??**) and $P(\varepsilon,T)$ at a finite T along K_b (Definition **??**). In this section ψ denotes the finite part of the operators of Definition **??**, and $L := \log\frac{3}{2}$.

Definition 11.37 (The terms of $S^{(1)}(x)$)**.** For $\xi \in S_+ \otimes W$ the functional $S^{(1)}(\xi)$ of (**??**) is the sum of the seven terms of Table 11.6: the quadratic terms $T_1, \dots, T_4$ and the three cubic terms T_5, with $F_A = dA + \frac{1}{2}[A \wedge A]$ reduced to V and $dA|_V$, $dA|_{W_{\mathbb{R}}^\perp}$ the parts of dA in $\wedge^2 V$ and in $V \wedge W_{\mathbb{R}}^\perp$; T_3 and T_4 carry one derivative on the leg A. For $\xi \in S_- \otimes W^*$ the table holds with $\lambda|_{S_+\otimes W}$ and $\lambda|_{S_-\otimes W^*}$ exchanged and $\lambda^*|_{S_+\otimes W^*}$ and $\lambda^*|_{S_-\otimes W}$ exchanged. The quadratic part of $S^{(1)}(\xi)$ is $S_2^V(\xi) + S_2^\perp(\xi)$, with $S_2^V(\xi) := T_1 + T_3$, the part with the legs $(\lambda, A^*|_V)$ and $(A|_V, \lambda^*)$, and $S_2^\perp(\xi) := T_2 + T_4$, the part with the legs $(\lambda, A^*|_{W_{\mathbb{R}}^\perp})$ and $(A|_{W_{\mathbb{R}}^\perp}, \lambda^*)$. For $\eta \in \Sigma^*$ and an orthonormal basis e_μ of V, (**??**) reads

$$S^{(1)}(\eta) = \sum_\mu \int x^\mu s_\mu + \int s_0, \qquad \int s_\mu := S^{(1)}\big(\rho(e_\mu)\eta\big), \qquad s_0 := -\big(\rho(A|_{W_{\mathbb{R}}^\perp})\eta, \lambda^*\big); \tag{11.3}$$

for $\eta \in S_+ \otimes W^*$ the spinors $\rho(e_\mu)\eta$ lie in $S_- \otimes W^*$ and s_0 has the legs $A|_{W_{\mathbb{R}}^\perp}$, $\lambda^*|_{S_+\otimes W^*}$, and for $\eta \in S_- \otimes W$ they lie in $S_+ \otimes W$ and s_0 has the legs $A|_{W_{\mathbb{R}}^\perp}$, $\lambda^*|_{S_-\otimes W}$. For an even generator with vector field v, $S^{(1)}(x) = \int \big((v.A, A^*) - (v.\lambda, \lambda^*) - (v.c, c^*)\big)$ (Definition **??**).

term	functional, $\int$ dvol understood	legs
T_1	$-\big(\Gamma(\xi,\lambda)\vert_V, A^*\vert_V\big)$	$\lambda\vert_{S_-\otimes W^*}$, $A^*\vert_V$
T_2	$-\big(\Gamma(\xi,\lambda)\vert_{W_{\mathbb{R}}^\perp}, A^*\vert_{W_{\mathbb{R}}^\perp}\big)$	$\lambda\vert_{S_+\otimes W}$, $A^*\vert_{W_{\mathbb{R}}^\perp}$
T_3	$\frac{1}{2}\big(\rho(dA\vert_V)\xi,\lambda^*\big)$	$A\vert_V$, $\lambda^*\vert_{S_+\otimes W^*}$
T_4	$\frac{1}{2}\big(\rho(dA\vert_{W_{\mathbb{R}}^\perp})\xi,\lambda^*\big)$	$A\vert_{W_{\mathbb{R}}^\perp}$, $\lambda^*\vert_{S_-\otimes W}$
T_5	$\frac{1}{2}\big(\rho(\frac{1}{2}[A\wedge A])\xi,\lambda^*\big)$	$(A\vert_V, A\vert_V, \lambda^*\vert_{S_+\otimes W^*})$, $(A\vert_V, A\vert_{W_{\mathbb{R}}^\perp}, \lambda^*\vert_{S_-\otimes W})$, $(A\vert_{W_{\mathbb{R}}^\perp}, A\vert_{W_{\mathbb{R}}^\perp}, \lambda^*\vert_{S_+\otimes W^*})$

The terms of $S^{(1)}(\xi)$ for $\xi\in S_+\otimes W$ and their legs; their two-component forms are in Remark 11.60.

Lemma 11.38 (The generators on the regulator and their lines)**.** *For an odd generator x, $x.$ is the odd operator on $\mathscr{E}$ that induces the bracket with the quadratic part of $S^{(1)}(x)$ (Proposition **??**); for an even one it is v. of Definition **??**. Let $g(H):=\int_\varepsilon^T e^{-tH}dt$, so that $P\star = Q^{GF}g(H)$ and $QP\star + P\star Q = e^{-\varepsilon H}-e^{-TH} =: E(H)$ (Proposition **??** (iii)).*

- *(i) For $\xi\in\Sigma$, $\xi.$ maps the slot of λ to the slot of A by $\lambda\mapsto+\Gamma(\xi,\lambda)$, the slot of A to the slot of λ by $A\mapsto+\frac{1}{2}\rho(dA)\xi$, the slots of A^* and of λ^* to each other by the transposed arrows, and vanishes on the slots of c and c^*. In particular*

$$\Big\{S^{(1)}(\xi),\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)\Big\}=\int\omega\,\Big[\big(\iota_{w_1}A,(\rho(w_2)\xi,\lambda)\big)+\big(\iota_{w_2}A,(\rho(w_1)\xi,\lambda)\big)\Big].$$

- *(ii) $[\xi.,Q]=0$, $[\xi.,H]=0$, and $C_\xi:=[\xi.,Q^{GF}]$ is an even operator of degree -1 with constant coefficients and of order one, with four components, each nonzero for $\xi\neq0$: $\lambda\mapsto-d^*\big(\Gamma(\xi,\lambda)\vert_V\big)$ in the slot of c, on a plane wave $e^{i(p,x)}$ the map $\lambda\mapsto+i\big(p,\Gamma(\xi,\lambda)\vert_V\big)$; a component from $A^*\vert_V$ to λ, a nonzero multiple of $A^*\vert_V\mapsto(p,A^*\vert_V)\,\xi$; and their transposes from λ^* to $A\vert_V$ and from c^* to λ^*. For $\xi\in S_+\otimes W$ these are $\lambda\vert_{S_-\otimes W^*}\to c$, $A^*\vert_V\to\lambda\vert_{S_+\otimes W}$, $\lambda^*\vert_{S_+\otimes W^*}\to A\vert_V$ and $c^*\to\lambda^*\vert_{S_-\otimes W}$. The variation kernel $(\xi.\otimes1+1\otimes\xi.)P(\varepsilon,T)=(C_\xi\,g(H)\otimes1)K_0$ has its components in $\lambda\otimes A\vert_V$, $c\otimes\lambda^*$ and their transposes.*

- *(iii) The term T_1 with both legs on edges P, its leg λ on the component $\lambda\otimes\lambda$ and its leg $A^*\vert_V$ on the component $A^*\vert_V\otimes c$, is the composite line along ξ, with the kernel $2(P\star\xi.P\star\otimes1)K_0$, whose components lie in $c\otimes\lambda$ and $\lambda\otimes c$. For $\xi\in S_+\otimes W$ and $T=\infty$ it is $-2\,\xi\,e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $c\otimes\lambda\vert_{S_+\otimes W}$ and $+2\,\xi\,e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $\lambda\vert_{S_+\otimes W}\otimes c$, and*

$$P\star\,\xi.\,P\star=C_\xi\,Q^{GF}g(H)^2-\xi.\,(Q^{GF})^2g(H)^2,\qquad\big[Q,P\star\,\xi.\,P\star\big]=E(H)\,C_\xi\,g(H)\neq0.$$

  The terms of $S^{(1)}(\xi)$ with a leg $A^*\vert_{W_{\mathbb{R}}^\perp}$ or λ^* enter a cycle with that leg on the special edge, and a term T_5 also with its leg λ^* as an output.

- *(iv) For $\eta\in\Sigma^*$, with $\xi_\mu:=\rho(e_\mu)\eta$,*

$$\eta.=\sum_\mu x^\mu\,\xi_\mu.+\eta^0.+Y_\eta,\qquad[\eta.,Q]=0,\qquad[\eta.,H]=2\sum_\mu\xi_\mu.\,\partial_\mu,$$

  where $\eta^0.$ is the constant operator of order zero induced by s_0, with the arrows $A\vert_{W_{\mathbb{R}}^\perp}\to\lambda$ and $\lambda^*\to A^*\vert_{W_{\mathbb{R}}^\perp}$, and Y_η is the constant operator of order zero from λ^* to A^* produced when the derivative of T_3, T_4 in $\sum_\mu\int x^\mu\big(S_2^V(\xi_\mu)+S_2^\perp(\xi_\mu)\big)$ is moved off A, $\partial_\nu(x^\mu\,\cdot\,)=x^\mu\partial_\nu+\delta^\mu_\nu$. On a cycle $S^{(1)}(\eta)$ forms the line of (iii) at the spinor $\rho(x)\eta$, linear in the position, and s_0 enters with its leg λ^* on the special edge.

- *(v) For K_b, with $v(x) = (x,x)b - 2(b,x)x$ and $\operatorname{div} v = -8(b,x)$, on a slot of Weyl weight w with the rotation generators $\Sigma^{\rho\nu}$, and with $\sigma^\rho(D) := [D, x^\rho]$ for a constant-coefficient operator D,*

$$[v., Q^{GF}] = 4(b,x)\, Q^{GF} + C_K, \qquad C_K := 2\,\sigma^\rho(Q^{GF})\big(w\, b_\rho - b_\nu \Sigma^{\rho\nu}\big),$$
$$[v., H] = 4(b,x)\, H - 4(w-1)(b,\partial) + 4 b_\nu \Sigma^{\rho\nu}\partial_\rho,$$

*C_K of order zero with constant coefficients, zero on $A^*|_{W_{\mathbb{R}}^\perp}$, with $C_K A|_V = -4(b, A|_V)$ in the slot of c. $S^{(1)}(K_b)$ forms on a cycle the composite line from c to $A|_V$ with the kernel $P\star\, v.\, P\star = v.(P\star)^2 + [P\star, v.]P\star$, quadratic in the position, at a finite T (Definition **??**).*

*In each case every letter, $A^*|_{W_{\mathbb{R}}^\perp}$, λ^* and c^* included, is active (Lemma **??**).*

Proof. (i) The terms $-(\Gamma(\xi,\lambda), A^*)$ and $\frac{1}{2}(\rho(dA)\xi, \lambda^*)$ of (**??**) pair the slots (λ, A^*) and (A, λ^*); the bracket of a quadratic functional with a coordinate functional is the coordinate functional composed with the corresponding arrow (proof of Lemma **??** (i)), and each pairing term gives one arrow between field slots and the transposed arrow between antifield slots (proof of Proposition **??**). In the reading of Proposition **??** the term (dc, A^*) of (**??**) gives $c \mapsto -dc$; the term $-(\Gamma(\xi,\lambda), A^*)$, whose letter λ is odd like c, gives $\lambda \mapsto +\Gamma(\xi,\lambda)$, and $[\xi., Q] = 0$ on the slot of A, the part of polynomial degree two of $\{S, S^{(1)}(\xi)\} = 0$, fixes $A \mapsto +\frac{1}{2}\rho(dA)\xi$. With $\iota_w \Gamma(\xi,\lambda) = (\rho(w)\xi, \lambda)$ [ESW, (28)] the classical component on the bilinear is the first functional of Corollary **??**.

(ii) $[\xi., Q] = 0$ and $[\xi., H] = 0$ are Proposition **??**. Q^{GF} maps $A^* \mapsto -2A^*$ into the slot of A, $A \mapsto -d^*(A|_V)$ into the slot of c, $c^* \mapsto dc^*$ into the slot of $A^*|_V$ and $\lambda^* \mapsto -\not{d}\lambda^*$ into the slot of λ, and vanishes on c and λ (Chapter 3). Hence $\xi.Q^{GF}$ has the arrows $A^* \to A \to \lambda$, $c^* \to A^*|_V \to \lambda^*$, $\lambda^* \to \lambda \to A$, and $Q^{GF}\xi.$ the arrows $\lambda \to A \to c$, $A^* \to \lambda^* \to \lambda$, $\lambda^* \to A^* \to A$; each composite has order one. On λ, $C_\xi \lambda = Q^{GF}\Gamma(\xi,\lambda) = -d^*\big(\Gamma(\xi,\lambda)|_V\big)$, and $d^* = -\iota_\partial$ is $-i\iota_p$ on plane waves. On $A^* = a\, e^{i(p,x)}$ the two composites to λ are opposite multiples of $\rho(p)\rho(a)\xi$ plus, through $\rho(q(p\wedge a)) = \rho(p)\rho(a) - (p,a)$ [ESW, (29)] in the first, a multiple of $(p,a)\xi$; for $a \in W_{\mathbb{R}}^\perp$ the two cancel. The components from λ^* and c^* are the transposes, since $\xi.$ preserves the pairing and Q^{GF} has the adjointness of Proposition **??**, so $(C_\xi \otimes 1)K_0$ is graded symmetric. $\Gamma(\xi, -)|_V$ is nonzero on one chirality of λ for $\xi \neq 0$, and the component from A^* to λ is nonzero by Proposition **??**. The variation kernel is the third item of Proposition **??** with T in place of ∞.

(iii) As in Proposition **??** (i), with the signs of Lemma **??** for the odd $\xi.$, both orders of the two legs give $(P\star\, \xi.\, P\star \otimes 1)K_0$. Since $[\xi., g(H)] = 0$, $[\xi., P\star] = C_\xi g(H)$, which gives the first identity; with $[\xi., Q] = 0$,

$$Q\,(P\star\, \xi.P\star) = E\,\xi.P\star - P\star\, Q\,\xi.P\star = E\,\xi.P\star + P\star\,\xi.E - P\star\,\xi.P\star\, Q,$$

so $[Q, P\star\,\xi.P\star] = E\,(\xi.P\star + P\star\,\xi.) = E\,C_\xi g$. The components follow the arrows $\lambda^* \to \lambda \to A \to c$ and $c^* \to A^*|_V \to \lambda^* \to \lambda$. On a plane wave the composite on c^* is a multiple of $\rho(p)\rho(p)\xi = |p|^2\xi$ times $g(H)^2 = e^{-2\varepsilon|p|^2}|p|^{-4}$ at $T = \infty$, a scalar line of momentum degree -2; the multiple, -2 on $c \otimes \lambda$, is the value of both evaluations of Remark 11.58. A leg $A^*|_{W_{\mathbb{R}}^\perp}$ or λ^* has no component of P (Proposition **??**).

(iv) By (11.3) the field arrows of $\int x^\mu S_2(\xi_\mu)$, S_2 the quadratic part, are x^μ times those of $\xi_\mu.$; its antifield arrows, the transposes, are obtained by integrating by parts the derivative of T_3, T_4 against $x^\mu\lambda^*$, which gives x^μ times the antifield arrows of $\xi_\mu.$ plus Y_η. For the odd Q, $[x^\mu\xi_\mu., Q] = x^\mu[\xi_\mu., Q] + [Q, x^\mu]\xi_\mu.$, and the vanishing of $[\eta., Q]$ is the part of polynomial degree two of $\{S, S^{(1)}(\eta)\} = 0$ (Lemma **??**); it requires Y_η. H is $-\partial^2$ on every component, $[x^\mu, -\partial^2] = 2\partial_\mu$, and $\eta^0.$, Y_η and $\xi_\mu.$ have constant coefficients.

(v) For a conformal Killing field, $\partial_\mu v_\nu = \frac{1}{4}(\operatorname{div} v)\delta_{\mu\nu} + \omega_{\mu\nu}$ with ω antisymmetric and $v. = v^\mu\partial_\mu + \frac{1}{2}\omega_{\mu\nu}\Sigma^{\mu\nu} + \frac{w}{4}\operatorname{div} v$; the transport, spin and weight terms and the rotation covariance of a constant-

coefficient operator D of weight -2 give

$$[v., D] = -\tfrac{1}{2}(\operatorname{div} v)D - \tfrac{w}{4}\sigma^{\rho}(D)\,\partial_{\rho}(\operatorname{div} v) - \tfrac{1}{2}\sigma^{\rho}(D)(\partial_{\rho}\omega_{\mu\nu})\Sigma^{\mu\nu},$$
$$[v., H] = -\tfrac{1}{2}(\operatorname{div} v)H + \tfrac{w-1}{2}(\partial_{\rho}\operatorname{div} v)\partial_{\rho} + (\partial_{\rho}\omega_{\mu\nu})\Sigma^{\mu\nu}\partial_{\rho},$$

with $\partial^2 v_{\mu} = -\frac{1}{2}\partial_{\mu}\operatorname{div} v$. For K_b, $\omega_{\mu\nu} = 2(x_{\mu}b_{\nu} - x_{\nu}b_{\mu})$ and $\partial_{\rho}\omega_{\mu\nu} = 2(\delta_{\rho\mu}b_{\nu} - \delta_{\rho\nu}b_{\mu})$. On $A^*|_{W_{\mathbb{R}}^{\perp}}$, $Q^{GF} = -2$ has order zero; on $A|_V$, $\sigma^{\rho}(Q^{GF})Y = Y_{\rho}$, $w = 1$ and $\sigma^{\rho}(Q^{GF})\Sigma^{\rho\nu}A = 3A_{\nu}$, so $C_K A|_V = 2(b, A|_V) - 6(b, A|_V)$. The line is Lemma 11.2 (iii), (iv) with this v. □

Definition 11.39 (Defect cycles and two-body kernels)**.** Let x be a generator, X_a, X_b two letters of any slots and $0 < \varepsilon < T \leq \infty$.

- (i) A defect cycle of (X_a, X_b) along x is a chain $X_a - e_0 - V_1 - e_1 - \cdots - V_n - e_n - X_b$ of Definition **??** whose vertices are vertices of I and exactly one vertex V_s given by a term of $S^{(1)}(x)$, entered and left through two of its legs, the third leg of a cubic term being an output, and whose edges are components of $P(\varepsilon, T)$ and exactly one special edge $e_{\sigma} = K_{\varepsilon} - K_0$, with the components of K_0. The vertex V_s is placed on a line, with both legs on edges P; at the special edge, with one leg on e_{σ}; or cubic, a term T_5 with two legs on the cycle.

- (ii) $\mathcal{D}_x(X_a, X_b; \varepsilon, T)$ is the sum of the defect cycles of the pair, amputated, with the outputs left open as in Definition **??**: the δ-linear graphs of arity one along x of the flow of the defect vertex of Lemma **??**, applied to $e^{(J+\delta M(\varepsilon))/\hbar}$ with $M = \int \omega\, X_a X_b$, whose cycle is a defect cycle of the pair. A chain read from X_a carries, beyond the signs of Lemma **??**, the factor

  $$-(-1)^{n(<\sigma)+|S^{(1)}(x)|\,(n(<s)+[s<\sigma])},$$

  with $n(< j)$ the number of odd outputs at the vertices before the position j, $[s < \sigma] = 1$ if V_s precedes e_{σ} and 0 otherwise, and $|S^{(1)}(x)|$ the total parity of $S^{(1)}(x)$, 0 for odd x and 1 for even x; the leading sign is that of the special edge (Proposition 11.4), and for even x the factor is that of Definition 11.5 times this sign.

- (iii) Let $\mathrm{ct}_m(M)$ and $\mathrm{lc}(M)$ be the coefficients of ε^{-m} and of $\log\varepsilon$ in $\partial_{\delta} I^{CT}[\varepsilon](M)$, and $\mathrm{ct}^{(1)}_{x,m}(M)$ and $\mathrm{lc}^{(1)}_{x}(M)$ those of the arity-one counterterm of M along x, the $\hbar\,\mathrm{Sing}_{\varepsilon}$ of the cycles through two letters of M with one vertex $S^{(1)}(x)$ on the cycle and no special edge (Definition 11.3 for $x \in \{\Delta, P_b\}$). The finite special vertex of the pair is

  $$\mathcal{F}_x(X_a, X_b) := -\sum_{m\geq 1} \frac{(-1)^m}{m!}\Big(\big\{I, \mathrm{ct}^{(1)}_{x,m}(X_a X_b)\big\}_{H^m} + \big\{S^{(1)}(x), \mathrm{ct}_m(X_a X_b)\big\}_{H^m}\Big),$$

  with $\{-,-\}_{H^m}$ the bracket of (**??**).

- (iv) The two-body kernel of (X_a, X_b) along x is

  $$\Upsilon_x(X_a, X_b) := [\varepsilon^0]\,\mathcal{D}_x(X_a, X_b; \varepsilon, T) + \mathcal{F}_x(X_a, X_b),$$

  with $[\varepsilon^0]$ the coefficient of ε^0 in the expansion of Theorem 11.41: a local density of the outputs at the point of the pair, with values in the tensor product of the fibres and colour spaces of the two letters.

For $x \in \{\Delta, P_b\}$ the defect cycles with V_s on a line are the special-edge composite cycles of Definition 11.5.

Theorem 11.40 (Two-body form of the components along every generator)**.** *Let x be a generator and $O = \int \omega\, X_1 \cdots X_n$ a monomial local functional, gauge-variant ones included. Let $J = I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] - \hbar F_1$ with $\hbar F_1$ a one-loop quantisation through arity one along x, that is, (??) modulo $\hbar^2$ in arity zero and in arity one along x, and let $\psi = 0$. The endpoint is $T = \infty$ along Σ, Σ^*, Δ, P_b and the rotations, and any $T \in (0,\infty)$ along K_b.*

- *(i) The δ-linear graphs of arity one along x and of order $\hbar$ of the flow of the defect vertex of Lemma ??, applied to $e^{(J+\delta O(\varepsilon))/\hbar}$, form the classes: (F) the one-loop graphs of the vertex $\delta\{S^{(1)}(x), O\}$ and of the vertex $\delta(Q + \{I, -\})O$ with a separate vertex $S^{(1)}(x)$; (C2) the defect cycles of the pairs of letters of O, with trees of I on the outputs; (C0) the cycles through two letters of O with the special edge or a contraction through K_ε, with $S^{(1)}(x)$ on a tree; (Br) the graphs with the special edge on a tree; (Ac) the graphs whose cycle passes through vertices of the action alone; (Ct) the trees with one counterterm vertex of order $\hbar$.*

- *(ii) (F) is the one-loop part of the arity-one flow of sO, and (C0) with the arity-zero counterterm vertices of (Ct) tends to the trees of $\hbar L_1(O)^{(0)}$ with one vertex $S^{(1)}(x)$, both on the left side of (??); (Br) tends to 0; (Ac) with the defect vertices of the action tends to 0 by the master equation in arity zero and in arity one along x.*

- *(iii) With the Koszul signs of (??),*

$$\hbar L_1(O)^{(1)}(x) = \sum_{1\le a<b\le n} (-1)^{|X_b|(|X_{a+1}|+\cdots+|X_{b-1}|)} \int \omega\, X_1 \cdots X_{a-1}\, \Upsilon_x(X_a, X_b)\, X_{a+1} \cdots \widehat{X_b} \cdots X_n - \big\{\hbar F_1^{(1)}(x), O\big\}, \tag{11.4}$$

  the two-body kernels of Definition 11.39 summed over the unordered pairs of letters, and a one-body term, the bracket with the arity-one finite counterterm along x.

- *(iv) The coefficients of $\log\varepsilon$ and of ε^{-m} cancel. The defect cycles carry no $\log\varepsilon$ (Theorem 11.41), and the first cancellation is the logarithmic covariance*

$$\mathrm{lc}\big(\{S^{(1)}(x), O\}\big) - \big\{S^{(1)}(x), \mathrm{lc}(O)\big\} = s\,\mathrm{lc}^{(1)}_x(O) - \mathrm{lc}^{(1)}_x(sO) + \big\{L^{(1)}(x), O\big\}, \tag{11.5}$$

  with $L^{(1)}(x)$ the coefficient of $\log\varepsilon$ in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$, which has no term ε^{-m} along Σ, Σ^*, Δ, P_b and K_b. The second is the pole identity, for $m \ge 1$,

$$\sum_{a<b} \pm\Big([\varepsilon^{-m}]\, \mathcal{D}_x(X_a, X_b) - \sum_{j\ge 1} \tfrac{(-1)^j}{j!} \Big(\big\{I, \mathrm{ct}^{(1)}_{x,m+j}(X_aX_b)\big\}_{H^j} + \big\{S^{(1)}(x), \mathrm{ct}_{m+j}(X_aX_b)\big\}_{H^j} \Big) \Big) = s\,\mathrm{ct}^{(1)}_{x,m}(O) + \big\{S^{(1)}(x), \mathrm{ct}_m(O)\big\} - \mathrm{ct}_m\big(\{S^{(1)}(x), O\}\big) - \mathrm{ct}^{(1)}_{x,m}(sO), \tag{11.6}$$

  whose right side, for a single pair $O = \int \omega\, X_aX_b$, contains the vacuum poles of the pairs of the cubic monomials of $\{S^{(1)}(x), O\}$ and of sO times their third letter.

- *(v)* (11.4) *holds along K_b at every $T \in (0,\infty)$ with the same right side. Given the master equation through arity one along x, the local representative of $L(O)$ exists through arity one along x for every local functional O: (iii) of Definition ?? holds along every generator.*

Proof. (i) At order $\hbar$ a graph of the flow has one cycle and vertices of order $\hbar^0$, or no cycle and one vertex of order $\hbar$ (Proposition ??); the part of arity one along x and linear in δ places x on one vertex and δ on one vertex. Expand the defect vertex in δ, in the arity and in $\hbar$, with $I_{\mathfrak{psl}(4|4)} = I + S^{(1)} + S^{(2)}$ and $O(\varepsilon) = O - \partial_\delta I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon](O)$. The classical master equation (??) leaves at order $\hbar^0$ the vertices

$\delta\{S^{(1)}(x),O\}$ and $\delta(Q+\{I,-\})O$ and the special-edge parts of $\{S^{(1)}(x),O\}_\varepsilon$, $\{I,O\}_\varepsilon$, $\{I,S^{(1)}(x)\}_\varepsilon$, $\frac12\{I,I\}_\varepsilon$, together with $\delta\hbar\Delta_\varepsilon O$; $\hbar\Delta_\varepsilon S^{(1)}(x)$ vanishes, its quadratic terms pairing a field with the antifield of another field and its cubic terms carrying λ^* and no λ. A cycle through a vertex of the first two kinds is in (F). A cycle with a special edge passes through O along two letters (Theorem **??** (i)) or avoids O, and the positions of $S^{(1)}(x)$ and of the special edge sort it into (C2), (C0), (Br) or (Ac).

(ii) By Theorem **??** (iv) the left side of (**??**) in arity one along x is $\big(L(O)^{(1)}(x)\big)[T]+\big(L(O)^{(0)}\big)[T]^{(1)}(x)$ with $L(O)=sO+\hbar L_1(O)$; the one-loop part of $(sO)[T]^{(1)}(x)$ consists of the graphs (F) and their counterterm trees. For (C0), the arity-zero analysis of Theorem **??**, (**??**) and Proposition **??** applies with a tree carrying $S^{(1)}(x)$ attached at generic momenta. (Br): a special edge on a tree is $e^{-\varepsilon|p|^2}-1$ at a generic momentum times a renormalised subgraph (Theorem **??** (ii), Proposition **??**). (Ac): the cycle and its counterterm vertices form the flow of the defect of the action in arity zero or in arity one along x, whose limit is 0 by Definition **??** and the master equation.

(iii) What remains is (C2) with the arity-one vertices of (Ct), against the tree of $\hbar L_1(O)^{(1)}(x)$ and the counterterm trees of $-\partial_\delta I^{CT}[\varepsilon](\{S^{(1)}(x),O\})$ and $-\mathrm{ct}_x^{(1)}(sO)$ on the left. The vertex $-\{\hbar F_1^{(1)}(x),O\}_\varepsilon$ gives the one-body term, its special-edge part tending to 0. With $\{f,g\}_\varepsilon-\{f,g\}=\sum_{j\geq1}\frac{(-\varepsilon)^j}{j!}\{f,g\}_{H^j}$, the part of order ε^0 of the special-edge part of $-\varepsilon^{-m}\{f,g_m\}_\varepsilon$ is $-\frac{(-1)^m}{m!}\{f,g_m\}_{H^m}$, which for the vertices $-\{I,\mathrm{ct}_x^{(1)}(O)\}_\varepsilon$ and $-\{S^{(1)}(x),\partial_\delta I^{CT}[\varepsilon](O)\}_\varepsilon$ is the finite special vertex; the terms with $j<m$ are poles, those with $j>m$ tend to 0, and $\log\varepsilon$ times a special edge tends to 0. Each defect cycle is a sum of poles, $[\varepsilon^0]\mathcal{D}_x$ and a vanishing remainder (Theorem 11.41); the trees on its outputs converge, and Theorem **??** (i) identifies the limit with the tree flow of (11.4). The sum over the pairs with the Koszul signs is that of Theorem **??** (iii).

(iv) The left side has a limit; collecting the coefficients of $\log\varepsilon$ and of ε^{-m} of (C2), of the special-edge parts of the (Ct) brackets shifted by j, of the brackets without special edge and of the counterterm trees moved from the left side gives (11.5) and (11.6). A coefficient of ε^{-m} in $I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon]^{(1)}(x)$ is a colour-singlet local functional of degree -1 and total Weyl weight $4-\frac12\lambda_x-2m$, the position at weight -1 and of degree p at most that of x (Proposition **??**); a colour singlet of degree -1 contains an antifield of weight at least $\frac52$ and a second adjoint field of weight at least 1, or c^* and c, so its density has weight at least $\frac72$ and the total weight is at least $\frac72-p$. For $m\geq1$ the total weight is at most $\frac52$, $\frac32$, 2, 3, 1 along Σ, Σ^*, Δ, P_b, K_b, below $\frac72$, $\frac52$, $\frac52$, $\frac72$, $\frac32$. Read for one pair, the cubic terms of $\{S^{(1)}(x),O\}$ and of sO have pairs with the empty leg set of Theorem **??** (iv), whose poles stand on the right with the third letter.

(v) The argument uses sO through (F), which cancels between the two sides, and through the counterterm trees. Along K_b at a fixed finite T the edges $P(\varepsilon,T)$ are bounded uniformly in T and decay as $e^{-|y|^2/4T}$ (Definition **??**); the graphs of (Br) and (Ac) carry at fixed ε the term $\frac{3}{8\pi^2}\hbar\log T$ times a tree with the factor $e^{-\varepsilon|p|^2}-1$ (remark after Theorem **??**), which tends to 0; Υ_{K_b} is independent of T by Theorem 11.41 (iii), and Theorem **??** gives the independence of the local representative. The classes (C2) and (Ct) have local limits, so Theorem **??** (i) gives the local representative. □

Theorem 11.41 (Defect cycles as simplex integrals)**.** *For a defect cycle with n vertices and the edges $e_0,\dots,e_n$, let q_j, the centre, the momentum form and the m-th simplex integrand F_m be those of Definition **??** with n in place of k, the numerator containing the function of the term of $S^{(1)}(x)$ at V_s (Table 11.6, Definition **??**) and the special-edge factor $-|l+q_\sigma|^2$ times the fibre factor of K_0; along η and K_b a momentum r is injected at V_s and carried by $e_s,\dots,e_n$ to X_b and ω. Let D be the superficial degree of Proposition 11.43.*

- *(i) At generic output momenta and $T=\infty$, for $x\in\Sigma\cup\Sigma^*\cup\{\Delta,P_b\}\cup\mathfrak{so}(V)$, the value of the*

cycle is $\sum_{m\geq 1}\varepsilon^{-m}a_m + a_0 + o(1)$ *with*

$$\begin{aligned} 16\pi^2 a_m &= \frac{1}{m}\int_{\alpha_\sigma \text{ smallest}} \Big[\big(\min_{f\neq\sigma}\alpha_f\big)^m - \alpha_\sigma^m\Big] F_m \, d\alpha \qquad (m\geq 1), \\ 16\pi^2 a_0 &= \int_{\alpha_\sigma \text{ smallest}} F_0 \, \log\frac{\min_{f\neq\sigma}\alpha_f}{\alpha_\sigma}\, d\alpha, \end{aligned} \tag{11.7}$$

and no term in $\log\varepsilon$*;* a_m *is a polynomial of degree* $D-2m$ *in the output momenta and in* r*, and* $a_0 = 0$ *for* $D<0$*.*

- *(ii) Along* $\eta\in\Sigma^*$*, at the point* x_O *of the pair,* $\Upsilon_\eta = \sum_\mu x_O^\mu\,\Upsilon_{\rho(e_\mu)\eta} + \Upsilon^0_\eta$*: the position part along* b *is the kernel along* $\rho(b)\eta\in\Sigma$*, and the translation-invariant part* Υ^0_η *is* $-i\sum_\mu \partial_{r_\mu}$ *at* $r=0$ *of* (11.7) *with* $S^{(1)}(\rho(e_\mu)\eta)$ *at* V_s*, plus* (11.7) *with* s_0 *at* V_s*.*

- *(iii) Along* K_b*, with* $z = x - x_O$*,*

$$v(x) = v(x_O) + \big(-2(b,x_O)z + \omega_{x_O}z\big) + v(z), \qquad \omega_{x_O}z := 2\big((x_O,z)b - (b,z)x_O\big),$$

and correspondingly

$$\Upsilon_{K_b} = v(x_O)^\mu\,\Upsilon_{P_{e_\mu}} - (b,x_O)\,\Upsilon^0_\Delta + \Upsilon^0_{M_{\omega_{x_O}}} + \Upsilon^0_{K_b}, \tag{11.8}$$

with Υ^0_y *the translation-invariant part along the generator* y *centred at* x_O*, the rotation* $M_{\omega_{x_O}}$ *having the vector field* $\omega_{x_O}z$*; the monomials* z^μ *and* $z^\mu z^\nu$ *of the Killing data act as* $-i\partial_{r_\mu}$ *and* $-\partial_{r_\mu}\partial_{r_\nu}$ *at* $r=0$*. At every finite* T *the coefficients of the expansion in* ε *are given by* (11.7)*, and* $T\partial_T$ *of every defect cycle is* $O(\varepsilon)$*:* Υ_{K_b} *is independent of* T*.*

- *(iv) For a cycle with* $n_t = n+1$ *edges and a constant integrand* F_0 *(*$D=0$*),* $16\pi^2 a_0 = F_0\log\frac{n_t}{n_t-1}/(n_t-1)!$*, so that* $16\pi^2 a_0 = \frac12 F_0\log\frac32$ *for* $n_t=3$ *and* $\frac16 F_0\log\frac43$ *for* $n_t=4$*; for a constant* F_1 *(*$D=2$*),* $16\pi^2 a_1 = F_1/\big(n_t^2(n_t-1)(n_t-1)!\big)$*. For* $n=1$*, the vertex* V_s *at the special edge, with* $\alpha := \alpha_\sigma\in[0,\frac12]$*,*

$$16\pi^2 a_0 = \int_0^{1/2} F_0(\alpha)\log\frac{1-\alpha}{\alpha}\,d\alpha, \qquad 16\pi^2 a_m = \frac{1}{m}\int_0^{1/2}\big((1-\alpha)^m - \alpha^m\big)F_m(\alpha)\,d\alpha,$$

in $\mathbb{Q}+\mathbb{Q}\log 2$ *for polynomial numerators; for* $n=2$ *with* V_s *on a line between a letter or a vertex of* I *and the special edge, Proposition 11.13 applies, with values in* $\mathbb{Q}+\mathbb{Q}\log 2+\mathbb{Q}\log 3$*.*

Proof. (i) The domain $\{t_\sigma\leq\varepsilon,\ t_f\geq\varepsilon\ (f\neq\sigma)\}$ is the difference of the domains with the cutoff shapes $1/\min_{f\neq\sigma}\alpha_f$ and $1/\min_f\alpha_f$, which differ on the region where α_σ is the smallest coordinate; Proposition **??** gives for each shape the poles, the same coefficient of $\log\varepsilon$ and the finite part, and the difference is (11.7), as in Lemma **??**. The region $\tau\geq 1$ converges at $T=\infty$: the edges carry distinct generic momenta (proof of Theorem **??** (ii)), the composite line along ξ is of order $|p|^{-2}$ at small momentum (Lemma 11.38 (iii)), the composite kernels along Δ, P_b and $\mathfrak{so}(V)$ of order $|p|^{-3}$ (Proposition **??** (iv)), and along η the line carries the spinor $\rho(b)\eta$ in the position part and one derivative in r in the translation-invariant part; the measure of $\{\alpha_\sigma\leq\varepsilon/\tau\}$ is of order ε. The coefficient of τ^{-1-m} after the Gaussian integral is homogeneous of degree $D-2m$. (ii) By (11.3), insert $x^\mu = x_O^\mu + (x-x_O)^\mu$ and argue as in Proposition 11.10 (i): on plane waves the factor $(x-x_O)^\mu$ is $-i\partial/\partial r_\mu$ of the momentum injected at V_s and absorbed by ω, which commutes with the Gaussian integral and the simplex integration; the same holds for the special edges, the counterterm vertices and the finite special vertices. (iii) Expand $(x_O+z,x_O+z)b - 2(b,x_O+z)(x_O+z)$; $S^{(1)}$ of Definition **??** is linear in v with its weight and spin terms, and $\operatorname{div} v(x) = -8(b,x_O) - 8(b,z)$, $\operatorname{div}(2z)=8$, $\operatorname{div}(\omega_{x_O}z)=0$. At a finite T the region $\tau\geq\tau_0$

is compact in τ, so the expansion in ε is that of the region of small τ. $T\partial_T$ replaces one edge e by TL_T, of scalar factor $Te^{-T|l+q_e|^2}$, which confines the loop momentum, while the special edge carries $e^{-\varepsilon|l+q_\sigma|^2}-1=O\big(\varepsilon(1+|l|)^2\big)$; the polynomial dependence on the position enters through derivatives in r, which preserve these bounds, and the finite special vertices are built from the coefficients ct_m, $\mathrm{ct}^{(1)}_{x,m}$, independent of T (Proposition **??**). (iv) For a constant integrand the integrals of (11.7) over the region where α_σ is the smallest coordinate are elementary; for $n=1$ the simplex is $[0,1]$, the region is $\alpha_\sigma\le\frac12$ and $\min_{f\neq\sigma}\alpha_f=1-\alpha_\sigma$, and $\int_0^{1/2}\alpha^a(1-\alpha)^b\log\frac{1-\alpha}{\alpha}\,d\alpha\in\mathbb{Q}+\mathbb{Q}\log 2$. Proposition 11.13 uses that the two segments adjacent to V_s carry the same momentum at $r=0$ and that the special edge is the third edge. □

Example 11.42 (The cubic term at the special edge)**.** The defect cycles with $n=1$ and an output have a cubic term of $S^{(1)}(\xi)$ at the special edge. For $\xi\in S_+\otimes W$ and the pair $(A|_{W^\perp_{\mathbb{R}}},\lambda|_{S_+\otimes W})$, the cycle

$$A|_{W^\perp_{\mathbb{R}}}\xrightarrow{P}T_5\big(A|_{W^\perp_{\mathbb{R}}},A|_{W^\perp_{\mathbb{R}}},\lambda^*|_{S_+\otimes W^*}\big)\xrightarrow{K_\varepsilon-K_0}\lambda|_{S_+\otimes W},$$

with the output $A|_{W^\perp_{\mathbb{R}}}$ of momentum q, has by Theorem 11.41 (iv) and Lemma **??** the value

$$\frac{\hbar}{16\pi^2}\Big(-\frac{1}{2\varepsilon}-\frac{3}{8}|q|^2\Big)$$

times the value of T_5 on the fibre elements and colour factors of the letters and of the output, with the fibre factor $2(-,-)^{-1}$ of the edge $A\otimes A$ and the inverse fibre pairing of (λ,λ^*) on the special edge. Its contribution to $\Upsilon_\xi\big(A|_{W^\perp_{\mathbb{R}}},\lambda|_{S_+\otimes W}\big)$ on the leg set $A|_{W^\perp_{\mathbb{R}}}$ is $\frac38\frac{\hbar}{16\pi^2}$ times ∂^2 of the output times that factor, and its pole enters (11.6) against ct_1 of the pair $(\lambda|_{S_+\otimes W},\lambda|_{S_+\otimes W})$ of the classical transformation $\{S^{(1)}(\xi),A|_{W^\perp_{\mathbb{R}}}\}=\Gamma(\xi,\lambda)|_{W^\perp_{\mathbb{R}}}$, whose coefficient of ε^{-1} has the leg set $A|_{W^\perp_{\mathbb{R}}}$ (Table **??**).

Proposition 11.43 (Output bounds along Σ, Σ^* and K_b)**.** *A defect cycle of (X_a,X_b) along x whose outputs have total Weyl weight w, fields and derivatives, has the superficial degree*

$$D=w_{X_a}+w_{X_b}+\delta-w,$$

with $\delta=\frac12$ along $\xi\in\Sigma$ and for the position part along η, $\delta=-\frac12$ for Υ^0_η, and $\delta=1,0,-1$ for the parts of Υ_{K_b} multiplying $v(x_O)$, the first derivatives of v at x_O and its second derivatives. Hence:

- *(i) along $\xi\in\Sigma$: $\Upsilon_\xi(X_a,X_b)$ is translation invariant, of Weyl weight $w_{X_a}+w_{X_b}+\frac12$, of the pair degree and of parity $|X_a|+|X_b|+1$, with at most $w_{X_a}+w_{X_b}+\frac12$ plus the pair degree output fields, as for the second-order operator on pairs of letters of [BK, (3.7)];*
- *(ii) along $\eta\in\Sigma^*$: $\Upsilon_\eta=\sum_\mu x^\mu\,\Upsilon_{\rho(e_\mu)\eta}+\Upsilon^0_\eta$, of parity $|X_a|+|X_b|+1$, with Υ^0_η translation invariant of Weyl weight $w_{X_a}+w_{X_b}-\frac12$ and at most $w_{X_a}+w_{X_b}-\frac12$ plus the pair degree output fields;*
- *(iii) along K_b: Υ_{K_b} is a polynomial of degree at most two in the position, of Weyl weight $w_{X_a}+w_{X_b}-1$ with the position at weight -1, of the pair degree and of parity $|X_a|+|X_b|$, with at most $w_{X_a}+w_{X_b}+1$ plus the pair degree output fields; $\Upsilon^0_{K_b}$ has Weyl weight $w_{X_a}+w_{X_b}-1$.*

*The same bounds hold for the counterterm cycles, whose outputs carry the pair degree minus one, and for the finite special vertices, which are supported on the leg sets with $D\ge2$. These are the bounds of Proposition **??** with $\lambda_x=-1,+1,+2$, valid without (iii) of Definition **??** by Theorem 11.40 (v).*

Proof. Lemma **??** with one vertex of weight $4+\delta$: every vertex of I has weight 4, the term of $S^{(1)}(x)$ has weight $4+\delta$ on its fields and derivatives once the Killing data carry the weights of Proposition

?? ($-\frac{1}{2}$ for a constant spinor, -1, 0, $+1$ for the value and the first and second derivatives of a vector field, the position at weight -1), every edge P has numerator degree $w_1 + w_2 - 2$, and the special edge, $-|l+q|^2$ over a Schwinger interval $[0, \varepsilon]$, has degree $0 = w + (4 - w) - 4$. The degree of the coefficient of ε^0 is Theorem 11.41 (i). $S^{(1)}(x)$ has degree -1 and total parity 0 for odd x and 1 for even x, the special edge degree $+1$ and parity 1. Weyl weight plus degree is at least 1 for every field and antifield, with equality for c and A (Proposition **??**), which bounds the number of outputs. The position parts are Theorem 11.41 (ii), (iii). On a leg set of degree D the term of $\mathcal{F}_x$ with ct_m or $\mathrm{ct}^{(1)}_{x,m}$ contains a coefficient of ε^{-m}, of degree $D - 2m \geq 0$ in the momenta, and $m \geq 1$. □

Proposition 11.44 (Census of the defect cycles). • *(i) Every defect cycle conserves the degree and the centre $\mathbb{Z}/4$ of $SL(4,\mathbb{C})$, $W \mapsto iW$, once x carries its charge, $+1$ on $S_+ \otimes W$ and on $S_- \otimes W$, -1 on $S_- \otimes W^*$ and on $S_+ \otimes W^*$, 0 on the even part; it shifts the total parity by that of $S^{(1)}(x)$ plus one.*

- *(ii) Along every generator, defect cycles exist for 45 of the 55 unordered pairs of the ten slots; the other ten are the pairs of two letters among $A^*|_{W^\perp_{\mathbb{R}}}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$, c^*, each of which needs the special edge.*

- *(iii) A defect cycle without vertices of I has V_s at the special edge, letter–P–V_s–special edge–letter; a cycle without outputs is homogeneous in ε of degree $-\frac{D}{2}$ (Lemma **??**), so its coefficient of ε^0 is nonzero at $D = 0$ alone.*

- *(iv) Along $\xi \in S_+ \otimes W$ the leg sets through the pairs of field and ghost letters without derivatives are those of Table 11.6; the numbers of leg sets with $n \leq 3$ derivatives on the two letters, along $\xi \in S_+ \otimes W$ and for the translation-invariant parts along $\eta \in S_+ \otimes W^*$ and along K_b, are those of Table 11.6. Along $S_- \otimes W^*$ and $S_- \otimes W$ the tables hold with the chiralities exchanged, and for the position parts with δ of Proposition 11.43.*

- *(v) Along $\xi \in S_+ \otimes W$ the pair (c, c) has the empty set of leg sets at the pair weight without derivatives, and the pairs of c with $A|_V$, $A|_{W^\perp_{\mathbb{R}}}$, $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$ have leg sets with one ghost output.*

Proof. (i) Each vertex of I, each term of $S^{(1)}(x)$ and each component of P and of K_0 carries these charges (Table 11.6, Definition **??**, Proposition **??**). (ii)–(v) Exhaustive enumeration of the chains with up to six vertices on the cycle, exactly one term of $S^{(1)}(x)$ in each of its placements and exactly one special edge, with output weights up to $w_{X_a} + w_{X_b} + 3 + \delta$; every chain is checked for degree, parity and the centre, and its superficial degree against the count of Lemma **??**. A letter $A^*|_{W^\perp_{\mathbb{R}}}$, λ^* or c^* has no component of P and lies on the special edge, of which a defect cycle has one. □

pair	output weight at most	leg sets along $\xi \in S_+ \otimes W$: fewest vertices of I; number of chains
$(A\|_V, A\|_V)$	5/2	$\lambda\|_{S_-\otimes W^*}$ (1; 2), $A\|_V\,\lambda\|_{S_-\otimes W^*}$ (1; 8), $A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_+\otimes W}$ (1; 7), $c\,\lambda^*\|_{S_+\otimes W^*}$ (1; 4)
$(A\|_V, A\|_{W^\perp_{\mathbb{R}}})$	5/2	$\lambda\|_{S_+\otimes W}$ (1; 4), $A\|_V\,\lambda\|_{S_+\otimes W}$ (1; 15), $A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_-\otimes W^*}$ (1; 15), $c\,\lambda^*\|_{S_-\otimes W}$ (1; 6)
$(A\|_V, \lambda\|_{S_+\otimes W})$	3	$\emptyset$ (0; 2), $A\|_V$ (0; 6), $A\|_V^2$ (1; 13), $(A\|_{W^\perp_{\mathbb{R}}})^2$ (1; 11), $A\|_V^3$ (1; 25), $A\|_V\,(A\|_{W^\perp_{\mathbb{R}}})^2$ (1; 55), $c\,A^*\|_V$ (2; 9), $\lambda\|_{S_+\otimes W}\,\lambda\|_{S_-\otimes W^*}$ (2; 15)
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	3	$A\|_{W^\perp_{\mathbb{R}}}$ (0; 5), $A\|_V\,A\|_{W^\perp_{\mathbb{R}}}$ (1; 19), $A\|_V^2\,A\|_{W^\perp_{\mathbb{R}}}$ (1; 51), $(A\|_{W^\perp_{\mathbb{R}}})^3$ (1; 19), $c\,A^*\|_{W^\perp_{\mathbb{R}}}$ (2; 2), $(\lambda\|_{S_-\otimes W^*})^2$ (2; 7), $(\lambda\|_{S_+\otimes W})^2$ (2; 6)
$(A\|_V, c)$	3/2	$\lambda\|_{S_-\otimes W^*}\,c$ (2; 10)
$(A\|_{W^\perp_{\mathbb{R}}}, A\|_{W^\perp_{\mathbb{R}}})$	5/2	$\lambda\|_{S_-\otimes W^*}$ (1; 2), $A\|_V\,\lambda\|_{S_-\otimes W^*}$ (1; 7), $A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_+\otimes W}$ (1; 8), $c\,\lambda^*\|_{S_+\otimes W^*}$ (1; 2)
$(A\|_{W^\perp_{\mathbb{R}}}, \lambda\|_{S_+\otimes W})$	3	$A\|_{W^\perp_{\mathbb{R}}}$ (0; 6), $A\|_V\,A\|_{W^\perp_{\mathbb{R}}}$ (1; 22), $A\|_V^2\,A\|_{W^\perp_{\mathbb{R}}}$ (1; 59), $(A\|_{W^\perp_{\mathbb{R}}})^3$ (1; 21), $c\,A^*\|_{W^\perp_{\mathbb{R}}}$ (2; 7), $(\lambda\|_{S_-\otimes W^*})^2$ (2; 7), $(\lambda\|_{S_+\otimes W})^2$ (2; 8)
$(A\|_{W^\perp_{\mathbb{R}}}, \lambda\|_{S_-\otimes W^*})$	3	$\emptyset$ (0; 2), $A\|_V$ (0; 5), $A\|_V^2$ (1; 10), $(A\|_{W^\perp_{\mathbb{R}}})^2$ (1; 11), $A\|_V^3$ (1; 18), $A\|_V\,(A\|_{W^\perp_{\mathbb{R}}})^2$ (1; 52), $\lambda\|_{S_+\otimes W}\,\lambda\|_{S_-\otimes W^*}$ (2; 13)
$(A\|_{W^\perp_{\mathbb{R}}}, c)$	3/2	$\lambda\|_{S_+\otimes W}\,c$ (2; 9)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	7/2	$\lambda\|_{S_+\otimes W}$ (1; 3), $A\|_V\,\lambda\|_{S_+\otimes W}$ (1; 11), $A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_-\otimes W^*}$ (1; 9), $c\,\lambda^*\|_{S_-\otimes W}$ (2; 7), $A\|_V^2\,\lambda\|_{S_+\otimes W}$ (2; 27), $A\|_V\,A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_-\otimes W^*}$ (2; 42), $A\|_V\,c\,\lambda^*\|_{S_-\otimes W}$ (2; 30), $(A\|_{W^\perp_{\mathbb{R}}})^2\,\lambda\|_{S_+\otimes W}$ (2; 21), $A\|_{W^\perp_{\mathbb{R}}}\,c\,\lambda^*\|_{S_+\otimes W^*}$ (3; 9)
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	7/2	$\lambda\|_{S_-\otimes W^*}$ (1; 5), $A\|_V\,\lambda\|_{S_-\otimes W^*}$ (1; 18), $A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_+\otimes W}$ (1; 17), $c\,\lambda^*\|_{S_+\otimes W^*}$ (2; 7), $A\|_V^2\,\lambda\|_{S_-\otimes W^*}$ (2; 44), $A\|_V\,A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_+\otimes W}$ (2; 79), $A\|_V\,c\,\lambda^*\|_{S_+\otimes W^*}$ (2; 30), $(A\|_{W^\perp_{\mathbb{R}}})^2\,\lambda\|_{S_-\otimes W^*}$ (2; 40), $A\|_{W^\perp_{\mathbb{R}}}\,c\,\lambda^*\|_{S_-\otimes W}$ (2; 21)
$(\lambda\|_{S_+\otimes W}, c)$	2	c (1; 5), $A\|_V\,c$ (1; 17), $A\|_V^2\,c$ (2; 39), $(A\|_{W^\perp_{\mathbb{R}}})^2\,c$ (2; 18)
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	7/2	$\lambda\|_{S_+\otimes W}$ (1; 2), $A\|_V\,\lambda\|_{S_+\otimes W}$ (1; 7), $A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_-\otimes W^*}$ (1; 8), $A\|_V^2\,\lambda\|_{S_+\otimes W}$ (2; 17), $A\|_V\,A\|_{W^\perp_{\mathbb{R}}}\,\lambda\|_{S_-\otimes W^*}$ (2; 37), $(A\|_{W^\perp_{\mathbb{R}}})^2\,\lambda\|_{S_+\otimes W}$ (2; 19), $A\|_{W^\perp_{\mathbb{R}}}\,c\,\lambda^*\|_{S_+\otimes W^*}$ (2; 12)
$(\lambda\|_{S_-\otimes W^*}, c)$	2	$A\|_{W^\perp_{\mathbb{R}}}\,c$ (1; 10), $A\|_V\,A\|_{W^\perp_{\mathbb{R}}}\,c$ (2; 44)
(c, c)	1/2	–

The leg sets of the defect cycles along $\xi \in S_+ \otimes W$ through the pairs of field and ghost letters without derivatives, with the fewest vertices of I and the number of chains; along $\xi \in S_- \otimes W^*$ the chiralities of λ and λ^* are exchanged. A leg set enters Υ_ξ when its output weight is at most the bound of the second column, and the empty leg set $\emptyset$ enters the poles.

pair	ξ, $n=0$	1	2	3	Υ^0_η, $n=0$	1	2	3	$\Upsilon^0_{K_b}$, $n=0$	1	2	3
$(A\|_V, A\|_V)$	4	9	20	43	1	4	9	20	2	4	8	19
$(A\|_V, A\|_{W_{\mathbb{R}}^\perp})$	4	9	20	43	1	4	9	20	1	2	7	16
$(A\|_V, \lambda\|_{S_+\otimes W})$	8	19	35	68	2	7	16	33	1	4	9	20
$(A\|_V, \lambda\|_{S_-\otimes W^*})$	7	16	33	61	4	8	19	35	1	4	9	20
$(A\|_V, c)$	1	4	9	20	0	1	4	9	1	2	4	8
$(A\|_V, A^*\|_V)$	8	22	49	97	3	8	22	49	1	6	15	35
$(A\|_V, A^*\|_{W_{\mathbb{R}}^\perp})$	8	22	49	97	3	8	22	49	1	5	13	29
$(A\|_V, \lambda^*\|_{S_+\otimes W^*})$	3	11	26	62	1	6	15	35	1	3	8	21
$(A\|_V, \lambda^*\|_{S_-\otimes W})$	6	15	35	73	0	3	11	26	1	3	8	21
$(A\|_V, c^*)$	3	13	34	80	0	3	13	34	1	3	10	25
$(A\|_{W_{\mathbb{R}}^\perp}, A\|_{W_{\mathbb{R}}^\perp})$	4	9	19	39	1	4	9	19	2	4	7	14
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_+\otimes W})$	7	16	33	61	4	7	16	30	1	3	7	16
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda\|_{S_-\otimes W^*})$	7	16	30	60	2	7	16	33	1	3	7	16
$(A\|_{W_{\mathbb{R}}^\perp}, c)$	1	4	9	20	0	1	4	9	0	1	2	7
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_V)$	8	22	49	97	3	8	22	49	1	5	13	29
$(A\|_{W_{\mathbb{R}}^\perp}, A^*\|_{W_{\mathbb{R}}^\perp})$	7	18	40	81	3	7	18	40	0	1	5	16
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda^*\|_{S_+\otimes W^*})$	3	10	26	59	1	5	14	31	0	1	4	14
$(A\|_{W_{\mathbb{R}}^\perp}, \lambda^*\|_{S_-\otimes W})$	5	14	31	70	0	3	10	26	0	1	4	14
$(A\|_{W_{\mathbb{R}}^\perp}, c^*)$	3	13	34	80	0	3	13	34	0	1	4	19
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_+\otimes W})$	9	20	42	79	3	7	16	35	2	6	12	24
$(\lambda\|_{S_+\otimes W}, \lambda\|_{S_-\otimes W^*})$	9	20	43	82	4	9	20	43	4	7	15	27
$(\lambda\|_{S_+\otimes W}, c)$	4	8	19	35	1	2	7	16	0	1	4	9
$(\lambda\|_{S_+\otimes W}, A^*\|_V)$	15	35	73	143	5	14	31	70	3	8	21	46
$(\lambda\|_{S_+\otimes W}, A^*\|_{W_{\mathbb{R}}^\perp})$	14	31	70	135	3	10	26	59	1	4	14	35
$(\lambda\|_{S_+\otimes W}, \lambda^*\|_{S_+\otimes W^*})$	8	21	47	88	3	8	21	47	0	2	7	19
$(\lambda\|_{S_+\otimes W}, \lambda^*\|_{S_-\otimes W})$	8	21	46	86	1	3	11	29	0	1	5	14
$(\lambda\|_{S_+\otimes W}, c^*)$	10	25	62	116	2	5	19	49	0	2	10	29
$(\lambda\|_{S_-\otimes W^*}, \lambda\|_{S_-\otimes W^*})$	7	16	35	68	4	9	20	42	2	6	12	24
$(\lambda\|_{S_-\otimes W^*}, c)$	2	7	16	33	2	4	8	19	0	1	4	9
$(\lambda\|_{S_-\otimes W^*}, A^*\|_V)$	14	31	70	135	6	15	35	73	3	8	21	46
$(\lambda\|_{S_-\otimes W^*}, A^*\|_{W_{\mathbb{R}}^\perp})$	10	26	59	123	5	14	31	70	1	4	14	35
$(\lambda\|_{S_-\otimes W^*}, \lambda^*\|_{S_+\otimes W^*})$	3	11	29	60	3	8	21	46	0	1	5	14
$(\lambda\|_{S_-\otimes W^*}, \lambda^*\|_{S_-\otimes W})$	8	21	47	88	3	8	21	47	0	2	7	19
$(\lambda\|_{S_-\otimes W^*}, c^*)$	5	19	49	101	3	10	25	62	0	2	10	29
(c, c)	0	1	4	9	0	0	1	4	0	1	2	4
$(c, A^*\|_V)$	9	20	43	82	4	9	20	43	4	8	19	35
$(c, A^*\|_{W_{\mathbb{R}}^\perp})$	9	20	43	82	4	9	20	43	2	7	16	32
$(c, \lambda^*\|_{S_+\otimes W^*})$	6	14	30	53	4	8	19	35	1	4	9	20
$(c, \lambda^*\|_{S_-\otimes W})$	8	19	35	64	2	6	14	30	1	4	9	20
(c, c^*)	8	22	49	91	3	8	22	49	1	6	15	35
$(A^*\|_V, A^*\|_V)$	13	34	80	169	3	13	34	80	3	10	25	61
$(A^*\|_V, A^*\|_{W_{\mathbb{R}}^\perp})$	13	34	80	169	3	13	34	80	1	4	19	49
$(A^*\|_V, \lambda^*\|_{S_+\otimes W^*})$	5	19	49	101	3	10	25	62	0	2	10	29
$(A^*\|_V, \lambda^*\|_{S_-\otimes W})$	10	25	62	116	2	5	19	49	0	2	10	29
$(A^*\|_V, c^*)$	4	15	46	101	1	4	15	46	0	1	7	24

The number of leg sets of the defect cycles entering Υ_ξ, $\xi \in S_+ \otimes W$, the translation-invariant part Υ^0_η, $\eta \in S_+ \otimes W^*$, and the translation-invariant part $\Upsilon^0_{K_b}$, for n derivatives on the two letters and up to six vertices on the cycle, on the 45 pairs of slots with defect cycles; the other ten pairs, of two letters among $A^*|_{W^\perp_\mathbb{R}}$, $\lambda^*|_{S_+\otimes W^*}$, $\lambda^*|_{S_-\otimes W}$, c^*, are those of Proposition 11.44 (ii).

Proposition 11.45 (The quadratic free directions along Σ and Σ^*).

- *(i) The quadratic local functionals of degree -1, of Weyl weight $\frac{9}{2}$ in the fields and derivatives, linear in $\xi \in S_+ \otimes W$, translation invariant and covariant under* $\mathrm{Spin}(4) \times \mathrm{Spin}(6) \times \mathfrak{g}$*, are spanned by T_1, T_2, T_3, T_4 and $\int \big((d^*A|_V)\xi, \lambda^*\big)$; their Q-closed combinations are spanned by $S_2^V(\xi)$ and $S_2^\perp(\xi)$, and the Q-exact ones are* 0.

- *(ii) The arity-one equation along ξ, the arity-one part along ξ of $s(\hbar F_1) = \hbar\,\mathrm{Ob}$ (Theorem* **??** *(ii)), is*

$$(Q + \{I, -\})\,\hbar F_1^{(1)}(\xi) + \big\{S^{(1)}(\xi), \hbar F_1^{(0)}\big\} = \hbar\,\mathrm{Ob}^{(1)}(\xi). \tag{11.9}$$

  At fixed $\hbar F_1^{(0)}$ two solutions differ, in the quadratic sector, by θ_V^ξ units times $S_2^V(\xi)$ plus $\theta_\perp^\xi$ units times $S_2^\perp(\xi)$, and the cubic part of (11.9) *then contains $\theta_V^\xi\{I_3, S_2^V(\xi)\} + \theta_\perp^\xi\{I_3, S_2^\perp(\xi)\}$ units, I_3 the cubic part of I. For $\theta_V^\xi = \theta_\perp^\xi =: \theta^\xi$ the direction extends to θ^ξ units times $S^{(1)}(\xi)$, the arity-one value of G_B (Definition* **??**) *and of the field redefinition by the R-rotation $S^{(1)}(R_\xi)$, $[R_\xi, \xi] = \xi$ (Theorem* **??**). *The cubic part fixes $\theta_V^\xi - \theta_\perp^\xi$ when the class of $\{I_3, S_2^V(\xi)\}$ modulo the images under Q of the cubic local functionals of degree -1 is nonzero, as for the sector coefficients along Δ (Proposition 11.21 (iii)).*

- *(iii) The one-body kernel of a letter X along ξ is $-\{\hbar F_1^{(1)}(\xi), X\}$; the free directions add $-\theta_V^\xi$ units times the arrows of $S_2^V(\xi)$ and $-\theta_\perp^\xi$ units times those of $S_2^\perp(\xi)$, which are $\{S_2^V(\xi), A|_V\} = \Gamma(\xi,\lambda)|_V$ and $\{S_2^\perp(\xi), A|_{W^\perp_\mathbb{R}}\} = \Gamma(\xi,\lambda)|_{W^\perp_\mathbb{R}}$ on the gauge field and the scalars and $\frac{1}{2}\rho(dA|_V)\xi$ and $\frac{1}{2}\rho(dA|_{W^\perp_\mathbb{R}})\xi$ on λ (Lemma 11.38 (i)).*

- *(iv) Along Σ^*, with the position included at weight -1 and η at weight $\frac{1}{2}$, the Q-closed quadratic directions of degree -1 and Weyl weight* 4*, linear in η, are spanned by $\sum_\mu \int x^\mu S_2^V(\rho(e_\mu)\eta)$ and $\sum_\mu \int x^\mu S_2^\perp(\rho(e_\mu)\eta) + \int s_0$, whose sum is the quadratic part of $S^{(1)}(\eta)$.*

The same holds along $S_- \otimes W^*$ with the chiralities exchanged.

Proof. (i) The field contents of degree -1 and field weight $\frac{9}{2}$ linear in a constant spinor are (λ, A^*) without derivatives and (A, λ^*) with one derivative, since c and c^* would need a further weight $\frac{1}{2}$; the covariant pairings are $\Gamma(\xi,\lambda)$ with A^* in its parts along V and $W^\perp_\mathbb{R}$, and $\rho(\partial A)\xi$ with λ^*, ∂A being $dA|_V$, $d^*A|_V$ or the mixed part $dA|_{W^\perp_\mathbb{R}}$, the volume element acting by a sign on each chirality. On functionals Q substitutes $A \mapsto -dc$, $A^* \mapsto -\frac{1}{2}d^*dA$, $\lambda^* \mapsto \not{d}\lambda$, $c^* \mapsto d^*(A^*|_V)$ and annihilates c and λ (Proposition **??**). $Q\int((d^*A|_V)\xi, \lambda^*)$ contains $-\int((d^*dc)\xi, \lambda^*)$, produced by no other structure, so its coefficient vanishes in a closed combination; Q preserves the split into the parts along V and along $W^\perp_\mathbb{R}$, and $S_2^V(\xi)$, $S_2^\perp(\xi)$ are closed, being the two parts of the quadratic part of $\{S, S^{(1)}(\xi)\} = 0$. A Q-exact functional is Q of a functional of degree -2 and field weight $\frac{9}{2}$, and two antifields have weight at least 5, while c^* needs a further field of weight $\frac{1}{2}$. Both evaluations of Remark 11.58 give the same ranks on plane-wave normal forms. (ii) Theorem **??** (iii) restricted to arity one along ξ at fixed arity-zero part. The cubic part of $s\big(\theta_V^\xi S_2^V + \theta_\perp^\xi S_2^\perp\big)$ is the bracket with I_3, and $\{I_3, S_2^V(\xi) + S_2^\perp(\xi)\} = -QS_3(\xi)$ with $S_3(\xi)$ the cubic part of $S^{(1)}(\xi)$, the cubic part of $\{S, S^{(1)}(\xi)\} = 0$; $G_B^{(1)}(\xi) = S^{(1)}(B\xi) = S^{(1)}(\xi)$ for $\xi \in S_+ \otimes W$. (iii) Theorem 11.40 (iii) and Lemma 11.38 (i). (iv) The same enumeration with $\int x^\mu$ times the structures of (i) at $\rho(e_\mu)\eta$, $\int s_0$ and $\int(\rho(A|_V)\eta, \lambda^*)$, and the exact ranks of Q on plane-wave normal forms; s_0 enters the closed direction together with the part along $W^\perp_\mathbb{R}$. □

Proposition 11.46 (Renormalisation freedom of Q_1, S_1 and K_1)**.** *Under the changes of Proposition* **??** *the components change as follows;* $\{S^{(1)}(x),-\}$ *is the classical transformation along* x*.*

- *(i) A field redefinition* $F_{\mathfrak{psl}(4|4)} \mapsto F_{\mathfrak{psl}(4|4)}+\hbar s\phi$ *changes the component along* x *by* $-\{(s\phi)^{(1)}(x),-\}$*, a derivation of the letters: it changes the one-body kernels and leaves the two-body kernels unchanged. For the field countings* $\phi^{(0)} = \sum_f y_f \int(f,f^*)$ *the change of* Q_1 *multiplies each arrow* $f \to f'$ *of the classical transformation by* $y_{f'} - y_f$*; for* $\phi^{(0)} = t\,S^{(1)}(\Delta)$ *it is a multiple of the classical transformation proportional to* λ_x*.*

- *(ii) The coupling direction leaves* Q_1*,* S_1*,* K_1 *unchanged. The outer direction* G_B *changes* Q_1 *along* $S_+ \otimes W$ *by* $-a$ *and along* $S_- \otimes W^*$ *by* $+a$ *times the classical transformation,* S_1 *along* $S_+ \otimes W^*$ *by* $+a$ *and along* $S_- \otimes W$ *by* $-a$ *times it, and leaves* K_1*,* D_1*,* P_1*,* M_1*,* R_1 *unchanged.*

- *(iii) The free directions of* $\hbar F_1^{(1)}(x)$ *change the one-body kernels: along* Σ *by Proposition 11.45 (iii); along* Δ *and* P_b *through the sector coefficients of Definition 11.19, which enter* D_1*,* P_1 *and, through* P_1*, the relations of Proposition 11.51 and Corollary 11.52; along* K_b *through the s-closed directions of degree* -1 *quadratic in the Killing data.* $\hbar F_1^{(0)}$ *enters the one-body kernels through* (11.9) *and through the trees with* $S^{(1)}(x)$ *(Remark* **??** *(a), (b)).*

- *(iv) A finite part* ψ' *of the operators changes the component along* x *by* $[\{S^{(1)}(x),-\},\psi']$*, on* O *by* $\{S^{(1)}(x),\psi'(O)\} - \psi'(\{S^{(1)}(x),O\})$*. For* $\psi' = \frac{a}{2}\{S^{(1)}(\Delta),-\}$ *this is* $\frac{a}{2}\lambda_x\{S^{(1)}(x),-\}$*:* $+a$*,* $-\frac{a}{2}$*,* $+\frac{a}{2}$ *times the classical transformation along* K_b*,* Σ*,* Σ^* *(Proposition* **??***), a change of the one-body kernels. A finite part* ψ' *that vanishes on* O *changes the component of* O *along* x *by* $-\psi'(\{S^{(1)}(x),O\})$*: the finite parts of the operators of the Weyl weight of* $\{S^{(1)}(x),O\}$ *enter the two-body kernels, as the finite part of Lemma 11.47 on the trace bilinear and the shift of the Konishi entry in Remark 11.55.*

- *(v) The cutoff shape enters the finite parts of the defect cycles (Proposition* **??***, Lemma* **??***), of the special-edge cycles of* $L_1^{(0)}$ *and of the finite special vertices; since the counterterms are unique up to finite local terms [C07, Thm B], a change of the shape is a change of* $\hbar F_1$ *and of* ψ*, a combination of (i)–(iv).*

- *(vi) Under the hypotheses of Theorem 11.22, in the translation-covariant scheme at* $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$*,* $P_1 \cong 0$ *on the families of Definition 11.32 (Corollary 11.35), and the condition* $P_1 = 0$ *of Definition* **??** *holds there with* $\psi' = 0$*: the components along* K_b*,* Σ *and* Σ^* *are those of the scheme. For equal coefficients* $\theta_P \neq \theta$ *along* P_b*, the finite part with* $a = (\theta - \theta_P)$ *units gives* $P_1 \cong 0$ *and changes the components along* K_b*,* Σ*,* Σ^* *by* a*,* $-\frac{a}{2}$*,* $\frac{a}{2}$ *times the classical transformation (Corollary 11.24 (iv)).*

- *(vii) Under (i), (iv) and the coupling direction the representation of* $\mathfrak{psl}(4|4)$ *on the cohomology of* $\Lambda^{(0)}$ *changes by an isomorphism, under* G_B *by the automorphism* $e^{-a\hbar B}$*, and the spectrum of* D_1 *in a fixed classical Weyl weight is invariant (Theorem* **??***); the class* $[Q_1^\xi]$ *on the cohomology of* Q_0^ξ*, its kernel, image and rank are invariant under (i), (ii) and (iv) (Theorem* **??***).*

Proof. (i), (ii) Proposition **??** (a), (c); the bracket with a local functional is a derivation in its second argument, so it acts letter by letter; for $\phi^{(0)} = t\,S^{(1)}(\Delta)$, $\{S^{(1)}(x),S^{(1)}(\Delta)\}$ is $S^{(1)}$ of $\pm[x,\Delta] = \mp\lambda_x x$ by Lemma **??**, with $S^{(2)}(x,\Delta) = 0$ (Definition **??**). (iii) Proposition 11.45, Definition 11.19, Remark **??**. (iv) Proposition **??** (b) and Proposition **??**, with $[\{S^{(1)}(x),-\},\psi'](O) = \{S^{(1)}(x),\psi'(O)\} - \psi'(\{S^{(1)}(x),O\})$. (v) Remark **??** (d); the finite counterterm and the finite parts of the operators absorb the change. (vi) Corollaries 11.35 and 11.24 (iv). (vii) Theorems **??** and **??**. □

Lemma 11.47 (The trace bilinear in a cocycle scheme)**.** *Let $O = \int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$, $\xi \in S_+ \otimes W$ or $\xi \in S_- \otimes W^*$, and let $\mathcal{B}_1, \dots, \mathcal{B}_4$ be the four functionals of Corollary* **??***,*

$$\mathcal{B}_1 := \{S^{(1)}(\xi), O\}, \qquad \mathcal{B}_2 := (w_1, w_2) \int \omega\,\big(\rho(A|_{W_{\mathbb{R}}^\perp})\xi, \lambda\big),$$
$$\mathcal{B}_3 := (w_1, w_2) \int \omega\,\big(\rho(A|_V)\xi, \lambda\big), \qquad \mathcal{B}_4 := (w_1, w_2) \int \omega\,\big(c, (\xi, \lambda^*)\big),$$

$\mathcal{B}_1$ being the first functional listed there (Lemma 11.38 (i)). Write the two-body part of $Q_1(O)$ as $a_1\mathcal{B}_1 + a_2\mathcal{B}_2 + a_3\mathcal{B}_3 + a_4\mathcal{B}_4$.

- *(i) If $L_1^{(0)} = 0$ on O and on $\mathcal{B}_1$, then $a_3 = a_4 = 0$, and the one-body terms of $Q_1(O)$ are multiples of $\mathcal{B}_1$.*
- *(ii) If $L_1^{(0)}(O) = 2\kappa_O(w_1, w_2)\int \omega\,(dc, A|_V)$ and the part of $L_1^{(0)}(\mathcal{B}_1)$ with the two letters c, λ is $-\kappa_{\mathcal{B}}\, s\mathcal{B}_3$, the finite part $\psi(O) = \kappa_O(w_1, w_2)\int \omega\,(A|_V, A|_V)$, $\psi(\mathcal{B}_1) = \kappa_{\mathcal{B}}\mathcal{B}_3$ removes both and changes $Q_1(O)$ by $(2\kappa_O - \kappa_{\mathcal{B}})\mathcal{B}_3$. At $\psi = 0$ the part with the letters c, λ of (R1) of Proposition 11.53 is the identity $a_3 + 2\kappa_O - \kappa_{\mathcal{B}} = 0$.*

Proof. (i) By (R1), $s\,Q_1(O) = 0$. $s\mathcal{B}_1 = 0$, since $[s, \{S^{(1)}(\xi), -\}] = \{\{S, S^{(1)}(\xi)\}, -\} = 0$ at tree level and $sO = 0$; $s\mathcal{B}_2 = 0$, since Q annihilates the coordinate functionals of $A|_{W_{\mathbb{R}}^\perp}$ and λ and the cubic parts of s on them are the gauge transformation with parameter c, which leaves the invariant pairing fixed. In the sector with the two letters c, λ, $s\mathcal{B}_3 = -(w_1, w_2)\int \omega\,(\rho(dc)\xi, \lambda)$ and $s\mathcal{B}_4 = (w_1, w_2)\int \omega\,(c, (\xi, \not{d}\lambda))$, whose densities carry the derivative on different letters and are linearly independent for arbitrary ω; hence $a_3 = a_4 = 0$. The one-body terms are $-\{\hbar F_1^{(1)}(\xi), O\}$; a term of $\hbar F_1^{(1)}(\xi)$ acting on the letters $A|_{W_{\mathbb{R}}^\perp}$ of O has the leg $A^*|_{W_{\mathbb{R}}^\perp}$, and by Proposition 11.45 (i) it is a multiple of T_2, which gives a multiple of $\mathcal{B}_1$. (ii) $Q\int \omega\,(A|_V, A|_V) = -2\int \omega\,(dc, A|_V)$ (Remark **??**), and the cubic part of $s\int \omega\,(A|_V, A|_V)$ vanishes by the invariance of $(-,-)$; L_1 changes by $[s, \psi]$ (Lemma **??**), the component along ξ by $[\{S^{(1)}(\xi), -\}, \psi]$ (Proposition **??** (b)), and $\{S^{(1)}(\xi), \int \omega\,(A|_V, A|_V)\} = 2\int \omega\,(\Gamma(\xi, \lambda)|_V, A|_V)$ with $(\Gamma(\xi, \lambda)|_V, a) = (\rho(a)\xi, \lambda)$ [ESW, (28)]. In the shifted scheme (i) gives $a_3 + 2\kappa_O - \kappa_{\mathcal{B}} = 0$; at $\psi = 0$ the part with the letters c, λ of (R1), $s\,Q_1(O) + L_1^{(0)}(\mathcal{B}_1) + \{S^{(1)}(\xi), L_1^{(0)}(O)\} = 0$, is the same equation, since there $s\,Q_1(O) = a_3\, s\mathcal{B}_3$, $L_1^{(0)}(\mathcal{B}_1) = -\kappa_{\mathcal{B}}\, s\mathcal{B}_3$ and $\{S^{(1)}(\xi), L_1^{(0)}(O)\} = 2\kappa_O\, s\mathcal{B}_3$. □

Theorem 11.48 (Q_1 on the scalar bilinears)**.** *Let $O = \int \omega\,(\iota_{w_1}A, \iota_{w_2}A)$, $\xi \in S_+ \otimes W$ or $\xi \in S_- \otimes W^*$, and $\mathcal{B}_1, \dots, \mathcal{B}_4$ as in Lemma 11.47. At $\psi = 0$:*

- *(i) the two-body part of $Q_1(O)$ is*

$$-2L\,\mathcal{B}_1 + 4L\,\mathcal{B}_2 + 4L\,\mathcal{B}_3 + 0\cdot\mathcal{B}_4 \text{ units}, \qquad L = \log\tfrac{3}{2}, \tag{11.10}$$

  without poles and the same for both chiralities; its trace part is $4L$ units times $(w_1, w_2)\int \omega\,(\rho(A)\xi, \lambda)$, with the whole connection $A = A|_V + A|_{W_{\mathbb{R}}^\perp}$;

- *(ii) on the traceless part, $(w_1, w_2) = 0$,*

$$Q_1(O) = c_Q\,\{S^{(1)}(\xi), O\}, \qquad c_Q = -2\log\tfrac{3}{2} - t_\perp \text{ units},$$

  with $t_\perp$ units the coefficient of T_2 in $\hbar F_1^{(1)}(\xi)$, which (11.9) fixes up to the free direction $\theta_\perp^\xi$ of Proposition 11.45; c_Q moves by $-\theta_\perp^\xi$ under that direction, by $\mp a$ under G_B along $S_+ \otimes W$ and $S_- \otimes W^*$, by $-\frac{a}{2}$ under $\psi' = \frac{a}{2}\{S^{(1)}(\Delta), -\}$, and under the field countings as in Proposition 11.46 (i);

- *(iii)* $L_1^{(0)}(O) = -(2+4L)$ *units times* $(w_1,w_2)\int\omega\,(dc, A|_V)$*, and the part of* $L_1^{(0)}(\mathcal{B}_1)$ *with the letters* c*,* λ *is* 2 *units times* $s\mathcal{B}_3$*; Lemma 11.47 (ii) applies with* $\kappa_O = -(1+2L)$ *units and* $\kappa_{\mathcal{B}} = -2$ *units, and in the cocycle scheme*

$$\psi(O) = -(1+2L)\,\textit{units}\cdot(w_1,w_2)\int\omega\,(A|_V, A|_V), \qquad \psi(\mathcal{B}_1) = -2\,\textit{units}\cdot\mathcal{B}_3,$$

the two-body part of $Q_1(O)$ *is* $-2L\,\mathcal{B}_1 + 4L\,\mathcal{B}_2$ *units,* $a_3 = 4L + 2\kappa_O - \kappa_{\mathcal{B}} = 0$*;*

- *(iv) at* $\psi = 0$*, for* $\xi \in S_+\otimes W$*, the part with the letters* c*,* $\lambda|_{S_-\otimes W^*}$ *of (R1) is the identity*

$$\begin{aligned} &s\,Q_1(O) + L_1^{(0)}\big(\{S^{(1)}(\xi), O\}\big) + \big\{S^{(1)}(\xi), L_1^{(0)}(O)\big\} \\ &\qquad = \big(-4L - 2 + (2+4L)\big)\,\textit{units}\cdot(w_1,w_2)\int\omega\,\big(\rho(dc)\xi, \lambda\big) = 0. \end{aligned}$$

Proof. (i) By Theorem 11.40 (iii) the two-body part is Υ_ξ on the single pair $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ of O, whose leg sets are those of Table 11.6: $\lambda|_{S_-\otimes W^*}$, $A|_V\,\lambda|_{S_-\otimes W^*}$, $A|_{W_{\mathbb{R}}^\perp}\lambda|_{S_+\otimes W}$ and $c\,\lambda^*|_{S_+\otimes W^*}$ for $\xi\in S_+\otimes W$. The single output Y has the colour factor $(1\otimes \mathrm{ad}_Y)(-,-)^{-1}$, which the colour pairing of O contracts to $\mathrm{tr}_{\mathfrak{g}}\,\mathrm{ad}_Y = 0$. The other three leg sets have $D = 0$, so they carry no poles and no finite special vertex (Proposition 11.43), and they give $\mathcal{B}_1$, $\mathcal{B}_2$ (outputs $A|_{W_{\mathbb{R}}^\perp}\lambda$), $\mathcal{B}_3$ (outputs $A|_V\lambda$) and $\mathcal{B}_4$ (outputs $c\,\lambda^*$). Their chains have three edges and, at $D = 0$, constant integrands, so each chain gives a rational multiple of $\frac{1}{2}\log\frac{3}{2}$ (Theorem 11.41 (iv)); the exact evaluation of the chains in all placements, in both ways of Remark 11.58, gives (11.10), with the same coefficient for $\mathcal{B}_2$ and $\mathcal{B}_3$. (ii) By Corollary **??** the functionals at $(w_1,w_2) = 0$ are the multiples of $\mathcal{B}_1$; the one-body term is $-t_\perp\mathcal{B}_1$ units (proof of Lemma 11.47 (i)); the changes are Propositions 11.45 and 11.46. (iii) $L_1^{(0)}$ is given by the arity-zero two-body kernels, the cycles with one special edge of the proof of Theorem **??**, of the pairs $(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ of O and $(A|_{W_{\mathbb{R}}^\perp}, \lambda)$ of $\mathcal{B}_1$, evaluated in the same way; then Lemma 11.47 (ii). (iv) With $s\mathcal{B}_3 = -(w_1,w_2)\int\omega\,(\rho(dc)\xi,\lambda)$ in this sector, the three terms are $4L\,s\mathcal{B}_3$, $2\,s\mathcal{B}_3$ and $2\kappa_O\,s\mathcal{B}_3 = -(2+4L)\,s\mathcal{B}_3$ units (proof of Lemma 11.47 (ii)). □

Proposition 11.49 (S_1 and K_1 on the scalar bilinears)**.** *Let* $O = \int\omega\,(\iota_{w_1}A, \iota_{w_2}A)$ *and* $\psi = 0$*.*

- *(i)* $\Upsilon^0_\eta(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ *has the single leg set* λ*, which vanishes on the colour singlet, and* s_0 *has zero bracket with a letter* $A|_{W_{\mathbb{R}}^\perp}$*. Hence the two-body part of* $S_1(O)$ *at the point* x *of the density is the two-body part of* $Q_1(O)$ *along the Killing spinor* $\rho(x)\eta$*, given by* (11.10) *along* $S_-\otimes W^*$ *for* $\eta\in S_+\otimes W^*$ *and along* $S_+\otimes W$ *for* $\eta\in S_-\otimes W$*; the term linear in* $\not{d}(\rho(x)\eta) = 4\eta$ *is absent (Corollary* **??***).*

- *(ii)* $\Upsilon^0_{K_b}(A|_{W_{\mathbb{R}}^\perp}, A|_{W_{\mathbb{R}}^\perp})$ *has the leg sets* $\emptyset$*, of superficial degree* 1*, and* $A|_V$*, both zero on the colour singlet. In a scheme in which* $\hbar F_1^{(1)}(K_b)$ *decomposes as* $S^{(1)}(K_b)$ *in Theorem 11.41 (iii), on the part with two legs* $A|_{W_{\mathbb{R}}^\perp}$*,*

$$K_1(O) = v(x)^\mu\,P_1(O)\big|_{P_{e_\mu}} - (b,x)\,D_1^0(O),$$

with D_1^0 *the translation-invariant part of* D_1 *(Definition 11.8); the rotation part and the intrinsic part vanish. With Theorems* **??** *and* **??** *at* $\theta_P = \theta$*,*

$$\begin{aligned} &K_1(O) = -\big(\tfrac{1}{3}+\theta\big)\,\textit{units}\cdot\{S^{(1)}(K_b), O\} + \begin{cases} 0 & \textit{(traceless part)},\\ 48\,\textit{units}\cdot(b,x)\,O & \textit{(trace)}, \end{cases} \\ &\{S^{(1)}(K_b), O\} = v^\mu\partial_\mu O - 4(b,x)\,O, \end{aligned}$$

so that at $\theta = -\frac{1}{3}$, $K_1(O) = -(b,x)D_1(O)$*: 0 on the traceless bilinears and* 48 *units times* $(b,x)\,O$ *on the trace, the relation of Proposition* **??** *with* $\kappa = -24$ *units.*

Proof. (i) Table 11.6 and the evaluation of the leg set: a single output Y gives $\operatorname{tr}_{\mathfrak{g}} \operatorname{ad}_Y = 0$ on the colour singlet; s_0 has the legs $A|_{W_{\mathbb{R}}^{\perp}}$, λ^*, and its bracket with a letter needs the leg $A^*|_{W_{\mathbb{R}}^{\perp}}$. Theorem 11.41 (ii) gives the position part. (ii) The leg set $A|_V$ has a single output; the empty leg set is homogeneous in ε of degree $-\frac{1}{2}$ (Lemma **??**). Theorem 11.41 (iii) for the two-body parts, with the one-body term decomposed in the same way; by Corollary **??** the term in the second derivatives of v and the term in the antisymmetric part of dv vanish on a scalar bilinear. By (**??**), $P_1(O)|_{P_{e_\mu}} = -(\frac{1}{3} + \theta_P)$ units times $\partial_\mu O$; by (**??**), with $\{S^{(1)}(\Delta), O\} = 2x^\mu \partial_\mu O + 4O$, $D_1^0(O)$ at the point of the density is $-4(\frac{1}{3} + \theta)$ units times O, plus -48 units times O on the trace; and $\{S^{(1)}(K_b), O\} = v^\mu \partial_\mu O + \frac{2}{4}(\operatorname{div} v)O$ (Definition **??**). Proposition **??**: $K_1(O) = -2(b,x)\kappa\, O$ for $D_1(O) = 2\kappa\, O$. □

Corollary 11.50 (The dilatation and the translations by the defect cycles)**.** *On every local functional* O *with* $sO = 0$*, with the Koszul signs of* (**??**)*,*

$$\sum_{a<b} \pm \Upsilon_\Delta(X_a, X_b) = \sum_{a<b} \pm \big(4 \operatorname{lc}(X_a X_b) + \textit{composite pair kernel along } \Delta\big),$$
$$\sum_{a<b} \pm \Upsilon_{P_b}(X_a, X_b) = \sum_{a<b} \pm \big(\textit{composite pair kernel along } P_b\big),$$

and the one-body terms of the two forms agree. On the part with two legs $A|_{W_{\mathbb{R}}^{\perp}}$ *of the scalar bilinears, at the point of the density,* Υ^0_Δ *is* $\frac{40}{3}$ *units times* O *on the traceless part and* $-\frac{104}{3}$ *units times* O *on the trace, and* Υ_{P_b} *is* $-\frac{1}{3}$ *units times* $\partial_b O$*; with the position part* $2x^\mu \Upsilon_{P_{e_\mu}}$ *this is* $\frac{44}{3}O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ *and* $-\frac{100}{3}O - \frac{1}{3}\{S^{(1)}(\Delta), O\}$ *units, the first row of the table of Theorem* **??***, and the value* $-\frac{1}{3}$ *of Theorem* **??***.*

Proof. Both forms compute the unique local representative (Theorem **??** (i)): for Δ this is Proposition 11.17, whose route of the special edges is (11.4), and for P_b the same argument applies with Corollary 11.16 and the one-body term $-\{\hbar F_1^{(1)}(P_b), -\}$ in both forms. On the bilinears, the defect cycles with $S^{(1)}(\Delta)$ and $S^{(1)}(P_b)$ in all their placements, evaluated by Theorem 11.41, give the stated values, with $\{S^{(1)}(\Delta), O\} = 2x^\mu \partial_\mu O + 4O$ at the point of the density: $\frac{40}{3} + \frac{4}{3} = \frac{44}{3}$ and $-\frac{104}{3} + \frac{4}{3} = -\frac{100}{3}$. □

Proposition 11.51 (The special supercharges and the supercharges)**.** *Let* $\eta \in \Sigma^*$ *and* $b \in V$*, so that* $\rho(b)\eta \in \Sigma$*.*

- *(i) Graph by graph, the position part along* b *of the two-body kernel of* (X_a, X_b) *along* η *is the two-body kernel of* (X_a, X_b) *along* $\rho(b)\eta$*; its translation-invariant part is computed with* $S^{(1)}(\eta)$ *at the spinor* $\rho(x - x_O)\eta$*,* x_O *the point of the density, together with the term* $-\int(\rho(A|_{W_{\mathbb{R}}^{\perp}})\eta, \lambda^*)$*.*
- *(ii) Under the hypotheses of Proposition* **??** *and Corollary* **??***, on cohomology, the position part along* b *of* S_1 *along* η *is* Q_1 *along* $\rho(b)\eta$ *minus the commutator of* $\{S^{(1)}(\eta), -\}$ *with* P_1*.*
- *(iii) On a family on which* $P_1 \cong 0$*, in particular, under the hypotheses of Theorem 11.22, on the families of Definition 11.32 in the translation-covariant scheme at* $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$*, the position part of* S_1 *along* η *is* Q_1 *along* $\rho(b)\eta$ *on cohomology, and* S_1 *along* η *is fixed by* Q_1 *along* Σ *up to the term of Proposition* **??** *linear in* $\not{d}(\rho(x)\eta) = 4\eta$*. On the scalar bilinears this term is absent (Corollary* **??***), and* S_1 *along* η *is, on cohomology, the combination of the functionals of Corollary* **??** *with the coefficients of* Q_1 *along* Σ *in Corollary* **??***, given by Theorem 11.48 and Proposition 11.49 (i).*

- *(iv) In a reflection-compatible scheme, on the family formed by a traceless scalar bilinear O at the origin and its descendants $\{S^{(1)}(\xi), O\}$, on which $Q_1 = c_Q\{S^{(1)}(\xi), -\}$ along the unit generator Q^i_α (Theorem 11.48 (ii)) and $S_1 = c_S\{S^{(1)}(\eta), -\}$ along $S^\alpha{}_i$, Corollary **??** reads*

$$c_S = \overline{c_Q} + g_{5/2} - g_2,$$

with g_w the eigenvalue of $\mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1$ in Weyl weight w on the family; the outer direction changes c_Q by $-a$ and c_S by $+a$ and preserves the relation for $\bar{a} = -a$.

Proof. (i) By (11.3), $S^{(1)}(\eta) = \int x^\mu s_\mu + \int s_0$, where $\int s_\mu$ is $S^{(1)}$ along the constant spinor $\rho(e_\mu)\eta \in \Sigma$. Insert $x^\mu = x^\mu_O + (x - x_O)^\mu$ and argue as in the proof of Proposition 11.10 (i): on plane waves the factor x^μ is $-i\partial/\partial r^\mu$ of the momentum injected at the vertex, and its action on the momentum delta function is $-i$ times the graph with $\int s_\mu$; the same holds for the special edges, the vertices $S^{(1)}(\eta)$ on the cycle and the counterterm vertices, which are the defect insertions of Proposition **??** (Theorem 11.41 (ii)). (ii) Corollary **??** (ii) at (η, P_b): by Proposition **??**, $[P_b, \eta] = \rho(b)\eta$, so $[\eta, P_b] = -\rho(b)\eta$, and $S^{(2)}(\eta, P_b) = 0$ (Definition **??**). At order $\hbar$ the identity states that the commutator of S_1 along η with $\{S^{(1)}(P_b), -\}$, plus the commutator of $\{S^{(1)}(\eta), -\}$ with P_1, minus Q_1 along $\rho(b)\eta$, is a graded commutator with $\Lambda^{(0)}$, which induces 0 on cohomology; at order $\hbar^0$ the same identity is the statement of (i) for the classical transformation, which fixes the signs. By Proposition **??**, S_1 along η is at most linear in the position, and Lemma 11.9 identifies the first commutator with its position part. (iii) By Proposition **??** the representative of S_1 along η at a point x_0 is a term linear in the value $\rho(x_0)\eta$ plus a term linear in 4η; the first has the position part given by its value at $\rho(b)\eta$, and as η and b vary, $\rho(b)\eta$ spans Σ, so by (ii) it is the representative of Q_1 evaluated at $\rho(x_0)\eta$ on cohomology. (iv) Write $X_0 + \hbar X_1$ for the matrix of $-\Lambda^{(1)}(Q^i_\alpha)$ and $Y_0 + \hbar Y_1$ for that of $-\Lambda^{(1)}(-S^\alpha{}_i)$ (Theorem **??** (i)): $\hbar X_1 = c_Q X_0$ and $\hbar Y_1 = c_S Y_0$, and Y_0 maps Weyl weight $\frac{5}{2}$ to Weyl weight 2. Corollary **??** gives $\hbar Y_1 = \overline{c_Q}\, X_0^{\mathrm{adj}} + \hbar[Y_0, \mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1] = \big(\overline{c_Q} + g_{5/2} - g_2\big)Y_0$ on the descendants, with $Y_0 = X_0^{\mathrm{adj}}$. The outer direction: Proposition 11.46 (ii) and Corollary **??**. □

Corollary 11.52 (K_1 in the translation-covariant scheme)**.** *Under the hypotheses of Theorems **??** and 11.22, on a family of Definition 11.32 at the origin of one length, in the translation-covariant scheme at $\theta = \theta_\lambda = \theta_V = -\frac{1}{3}$, K_1 along $K_{\bar{b}}$ equals $\hbar\big[\{S^{(1)}(K_{\bar{b}}), -\}, \mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1\big]$, the statement $K = P^\dagger$ of Corollary **??**. For equal coefficients θ_P along P_b, K_1 along $K_{\bar{b}}$ is $(\theta - \theta_P)$ units times $\{S^{(1)}(K_{\bar{b}}), -\}$ plus the same commutator, with the radial Gram matrix of Definition **??** in that scheme. On a scalar bilinear at the origin both sides vanish: the defect cycles give $K_1(O) = 0$ there (Proposition 11.49 (ii)), and the commutator vanishes on a primary at the origin on which $\mathrm{Gram}^{-1}_{\mathrm{tree}}\mathrm{Gram}_1$ is diagonal. On its descendants $\partial_\mu O$ the relation equates the intrinsic kernels $\Upsilon^0_{K_b}$ of the descendant pairs with the difference of the two-point normalisations in Weyl weights $w_O + 1$ and w_O times the classical transformation.*

Proof. Corollary 11.35 gives $P_1 \cong 0$, resp. $P_1 \cong (\theta - \theta_P)$ units times $\{S^{(1)}(P_b), -\}$ with a real coefficient; Corollary **??** gives the two relations. At the origin $v(0) = 0$ and $(b, 0) = 0$, so the formula of Proposition 11.49 (ii) vanishes; $\{S^{(1)}(K_{\bar{b}}), -\}$ lowers the Weyl weight by one and annihilates a primary at the origin. □

Proposition 11.53 (The relations at order $\hbar$)**.** *Write $\Lambda^{(1)}(x) = \Lambda^{(1)}_0(x) + \hbar\Lambda^{(1)}_1(x)$ with $\Lambda^{(1)}_0(x) = \pm\{S^{(1)}(x), -\}$ (Theorem **??** (iv)), so that $\hbar\Lambda^{(1)}_1(x)O$ is the component of O along x, and $\Lambda^{(0)} = s + \hbar L^{(0)}_1$. Let $\mathfrak{F}$ be a family of colour-singlet local functionals of degree 0, stable under Λ and under the classical transformations, with $L^{(0)}_1 = 0$ on $\mathfrak{F}$ for a finite part ψ as in Definition **??** (ii), and let $O \in \mathfrak{F}$ with $sO = 0$.*

- *(R1) $s\big(\hbar\Lambda^{(1)}_1(x)O\big) = 0$ for every generator x; along ξ, $s\,Q_1(O) = 0$, an identity of local functionals.*

- *(R2) For $\xi,\xi'\in\Sigma$ with $b=\Gamma(\xi,\xi')|_V$, $[\Lambda_1^{(1)}(\xi),\Lambda_0^{(1)}(\xi')]O+[\Lambda_0^{(1)}(\xi),\Lambda_1^{(1)}(\xi')]O+\Lambda_1^{(1)}(P_b)O=\Lambda^{(0)}(\chi)$, with χ local of degree -1, built from the arity-two component of O along (ξ,ξ'); for $\xi,\xi'\in S_+\otimes W$, $b=0$.*

- *(R3) For $\xi\in\Sigma$ and $\eta\in\Sigma^*$ with $[\xi,\eta]=\frac12(\xi,\eta)\Delta+M_{\xi\eta}+R_{\xi\eta}$ (Proposition **??**), $M_{\xi\eta}\in\mathfrak{so}(V)$, $R_{\xi\eta}\in\mathfrak{so}(W_{\mathbb{R}}^{\perp})$,*

$$[\Lambda_1^{(1)}(\xi),\Lambda_0^{(1)}(\eta)]O+[\Lambda_0^{(1)}(\xi),\Lambda_1^{(1)}(\eta)]O+\tfrac12(\xi,\eta)\,D_1(O)+M_1(O)+R_1(O)=\Lambda^{(0)}(\chi'),$$

  the components M_1, R_1 taken along $M_{\xi\eta}$, $R_{\xi\eta}$.

- *(R4) For a rank-one ξ, $Q_0^\xi Q_1^\xi+Q_1^\xi Q_0^\xi=0$ on every local functional (Theorem **??**).*

- *(R5) With $[K_b,P_{b'}]=(b,b')\Delta+M_{bb'}$ (Remark **??**), $[\Lambda_1^{(1)}(K_b),\Lambda_0^{(1)}(P_{b'})]+[\Lambda_0^{(1)}(K_b),\Lambda_1^{(1)}(P_{b'})]+(b,b')\hbar\Lambda_1^{(1)}(\Delta)+\hbar\Lambda_1^{(1)}(M_{bb'})\cong 0$ on cohomology; with Theorem 11.41 (iii) it states that the parts of K_1 multiplying the position are those of* (11.8).

- *(R6) For $\eta_1,\eta_2\in\Sigma^*$ with $[\eta_1,\eta_2]=-K_b$, $b=\Gamma(\eta_1,\eta_2)|_V$, at a finite T, K_1 along K_b is, on cohomology, $[\Lambda_1^{(1)}(\eta_1),\Lambda_0^{(1)}(\eta_2)]+[\Lambda_0^{(1)}(\eta_1),\Lambda_1^{(1)}(\eta_2)]$.*

*In (R2), (R3), (R5) and (R6) the right sides vanish on cohomology. For O without letters λ and c, such as the scalar bilinears and the families of scalars, the parts of order $\hbar^0$ of χ, χ' vanish, the right sides are s-exact, and their parts with two field letters and no antifield vanish on free on-shell plane waves, $|p|^2=0$ on the scalars and the gauge field, $\rho(p)\lambda=0$ and $(p,a)=0$ on the polarisations a of $A|_V$. On the scalar bilinears at $\theta=-\frac13$, (R3) contains $D_1(O)=0$ on the traceless part and -48 units times O on the trace (Corollary **??**), the Konishi relation between Q_1, S_1 and the anomalous dimension, and (R2) with $\xi\in S_+\otimes W$, $\xi'\in S_-\otimes W^*$ contains $P_1(O)=-(\frac13+\theta_P)$ units times $\partial_b O$ (Theorem **??**).*

Proof. Corollary **??** (i) at order $\hbar$ reads $[s,\Lambda_1^{(1)}(x)]+[L_1^{(0)},\Lambda_0^{(1)}(x)]=0$; on O with $sO=0$, $L_1^{(0)}O=0$ and $L_1^{(0)}\{S^{(1)}(x),O\}=0$ (stability of $\mathfrak{F}$) this is (R1). Corollary **??** (ii) at order $\hbar$ for the pairs (ξ,ξ'), (ξ,η), $(K_b,P_{b'})$, (η_1,η_2), with the brackets of Proposition **??** and Remark **??**, has the right side $[\Lambda^{(0)},H]$ with H built from $\Lambda^{(2)}$, which on O with $\Lambda^{(0)}O=0$ is $\Lambda^{(0)}(HO)$; the part of order $\hbar^0$ of H is $\pm\{S^{(2)}(x_1,x_2),-\}$, and $S^{(2)}$ has the legs (λ^*,λ^*) and (A,c^*) (Definition **??**), whose bracket with a functional of the fields A alone vanishes. (R4) is Theorem **??**. For the vanishing on shell: the part of $s\chi$ with two field letters is $Q\chi_2+\{I_3,\chi_1\}$, with χ_k the part of χ with k letters; χ_1 is a single adjoint letter contracted with a constant tensor, zero for a colour singlet; χ_2 of degree -1 has one antifield and one field, or c^* and c, and Q substitutes $A^*\mapsto-\frac12 d^*dA$, $\lambda^*\mapsto\not{d}\lambda$, $c^*\mapsto d^*(A^*|_V)$, $A\mapsto -dc$ (Proposition **??**), so the part of $Q\chi_2$ without antifields and ghosts is a free field equation times a field. The values on the bilinears are Corollary **??** and Theorem **??**, with $R_1=0$ on them (Remark **??**). □

Proposition 11.54 (The chiral-primary bilinears along a rank-one supercharge)**.** *Let $\xi\in S_+\otimes W$ be of rank one, ${\xi_\alpha}^i=u_\alpha z^i$, and let $w_1,w_2\in W_{\mathbb{R}}^{\perp}\otimes\mathbb{C}$ satisfy $\rho(w_1)\xi=\rho(w_2)\xi=0$. Then $(w_1,w_2)=0$, and under the hypotheses of Proposition **??** the component along ξ of $O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$ vanishes at every order at which they hold: its classical part $\{S^{(1)}(\xi),O\}$ and $Q_1(O)$ are 0, in every scheme in which the counterterms are equivariant for translations, $\mathfrak{so}(V)$, $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$ and $\mathfrak{g}$.*

Proof. $\rho(w_1)\rho(w_2)+\rho(w_2)\rho(w_1)=2(w_1,w_2)$ applied to $\xi\neq0$ gives $(w_1,w_2)=0$. By Corollary **??**, for $(w_1,w_2)=0$ the component is a multiple of $\int\omega\,\big[(\iota_{w_1}A,(\rho(w_2)\xi,\lambda))+(\iota_{w_2}A,(\rho(w_1)\xi,\lambda))\big]$, which vanishes; the classical component is the same functional (proof of Corollary **??**). In the evaluation of Theorem 11.48 the two-body part on these bilinears, $-2L\,\mathcal{B}_1+4L(\mathcal{B}_2+\mathcal{B}_3)$ units, vanishes term by term, with $\mathcal{B}_1=\{S^{(1)}(\xi),O\}=0$ and $(w_1,w_2)=0$, and the one-body term is a multiple of $\mathcal{B}_1$. □

Remark 11.55 (The pair entries of the holomorphic twist)**.** The twisted one-loop differential is $\hbar Q_1^\xi(O) = \hbar L_1(O)^{(0)} + Q_1(O) - \frac{1}{2}\hbar L_1(O)^{(2)}(\xi,\xi) - \cdots$ (Definition **??**), so its pair rule [BK, (3.7)–(3.8)] corresponds to the two-body kernels of Q_1 along ξ and of the arity-two component along (ξ,ξ). By Proposition **??**, applied once for each argument, a component of arity k along ξ of an operator of Weyl weight w and degree d has representatives of Weyl weight $w + \frac{k}{2}$ and degree $d + 1 - k$. [BK, (3.51)] relates the holomorphic letters to the fields up to field redefinitions; with ${\xi_\alpha}^i = u_\alpha z^i$, $u' \in S_+$ with $u'^\alpha u_\alpha = 1$, and the representatives $\gamma \leftrightarrow A_{ij}z^j$ (Weyl weight 1, degree 0), $\beta \leftrightarrow u'^\alpha {\lambda_\alpha}^i$ modulo z^i $(\frac{3}{2},0)$, $\partial_{\dot\alpha} c \leftrightarrow \bar\lambda_{\dot\alpha i} z^i$ $(\frac{3}{2},0)$, $c \leftrightarrow c$ $(0,1)$, $b \leftrightarrow u'^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u'_\beta$ $(2,0)$ or a component of λ^* $(\frac{5}{2},-1)$, and $\partial_{\dot\alpha} \leftrightarrow u'^\alpha \sigma^m_{\alpha\dot\alpha} D_m$ $(1,0)$, the pair entries of [BK, (3.64)–(3.69)] lie in the following arities:

entry	Weyl weight and degree of the pair	of the output	arity
$b\,c$ [BK, (3.64)]	$(2,1)$	$(3,0)$	2
$b\,b$ [BK, (3.65)]	$(4,0)$	$(\frac{9}{2},0)$	1
$\beta\,\gamma$ [BK, (3.66)]	$(\frac{5}{2},0)$	$(3,0)$	1
$\beta\,\beta$ [BK, (3.67)]	$(3,0)$	$(\frac{7}{2},0)$	1
$b\,\gamma$ [BK, (3.68)]	$(3,0)$	$(\frac{7}{2},0)$	1
$\beta\,b$ [BK, (3.69)]	$(\frac{7}{2},0)$	$(4,0)$	1

The arities of the one-loop pair entries of the holomorphic twist, with b represented by the self-dual field strength; with b represented by a component of λ^* the entry $b\,c$ has arity 1.

The entry $\beta\gamma$, the Konishi entry, is the two-body kernel of Q_1 along ξ on the pair $(\lambda|_{S_+\otimes W}, A|_{W_\mathbb{R}^\perp})$ with two outputs $\lambda|_{S_-\otimes W^*}$: the coefficient of $\int\omega\,(\bar\lambda_{\dot\alpha i}z^i, \bar\lambda^{\dot\alpha}{}_j z^j)$ in Q_1 of $\int\omega\,(u'^\alpha{\lambda_\alpha}^i, A_{ij}z^j)$. The classical component of this pair along ξ is a number κ times a cubic functional of the scalars, and a finite part of the operators of Weyl weight 3 that adds $y\int\omega\,(\bar\lambda_{\dot\alpha i}z^i, \bar\lambda^{\dot\alpha}{}_j z^j)$ to that cubic functional changes the entry by $-\kappa y$ (Proposition 11.46 (iv)), the two functionals having the same Weyl weight and quantum numbers. The pair entries are therefore compared through the class $[Q_1^\xi]$ on the cohomology of Q_0^ξ (Theorem **??**), transported to the holomorphic letters through [ESW, Thm 10.9] (remark after Theorem **??**). Proposition 11.54 is the exact untwisted form, for two letters, of [BK, (3.77)], $Q_1(\gamma^I\gamma^J) = 0$: for $\rho(w_a)\xi = 0$, $\iota_{w_a}A$ is a combination of the components $A_{ij}z^j$, the letters γ. Theorem 11.56 gives the arity-one entries at $\psi = 0$, and Theorem 11.57 states what their comparison with [BK] tests.

Theorem 11.56 (The pair entries along a rank-one supercharge)**.** *Let $\xi = u\otimes z \in S_+\otimes W$ be of rank one, with u', the twisted letters and their representatives of Remark 11.55, and let $\psi = 0$. Normalise the letters as the components along the weight vectors of a Cartan subalgebra of $\mathfrak{so}(V)\oplus\mathfrak{so}(W_\mathbb{R}^\perp)$ for which u, u' and z are weight vectors, read the outputs on the dual weight vectors, take b as the component $F(d_1,d_2)$ of the field strength on the two null vectors d_1, d_2 of $V\otimes\mathbb{C}$ along which $\partial_{\dot\alpha}$ differentiates, and expand the output momenta in the null coordinates; this fixes each entry up to one nonzero factor for each letter and each output. Reduce modulo $\operatorname{Im} Q_0^\xi$, total derivatives kept: the derivatives $u^\alpha\sigma^m_{\alpha\dot\alpha}\partial_m$ are Q_0^ξ-exact on the letters, and so is, up to terms with antifields, the holomorphic Dirac component of a derivative on an output $\partial_{\dot\alpha}c$. Then the arity-one pair entries of Q_1 are those of Table 11.6, and:*

- *(i) the entry $\beta\gamma$, the kernel of $(\lambda|_{S_+\otimes W}, A|_{W_\mathbb{R}^\perp})$ on the leg set $(\lambda|_{S_-\otimes W^*})^2$, vanishes for every $\xi \in S_+\otimes W$, the placements on a line and at the special edge cancelling chain by chain; the one-body terms and $L_1^{(0)}$ have outputs of other Weyl weights or with a spectator letter, so the Konishi entry of Remark 11.55 is 0 at $\psi = 0$; the entry bc with b represented by (ξ,λ^*) and the entry $\gamma\gamma$ vanish as well;*
- *(ii) the entries $\beta\beta$ and $\beta b \to \gamma\gamma$ carry the colour and derivative structure of [BK, (3.67)] and of the last term of [BK, (3.69)]: the output γ^K, resp. the outputs γ^J, γ^K, with the indices complementary to those of the letters β, the same value in the two colour orders, and, for $\beta b \to \gamma\gamma$, the antisymmetric combination of the two holomorphic derivatives;*

- *(iii) the entry $b\gamma$ has different values in the two colour orders, each with a term in $\log\frac{3}{2}$, and the entry $\beta b \to \beta\,\partial c$ is nonzero in one colour order;*
- *(iv) every derivative in the entries is holomorphic.*

Statements (i)–(iv) are independent of the normalisation factors; the values of the individual entries depend on the scheme (Proposition 11.46 (iv), Remark 11.55).

entry	pair → outputs	order (Y_1, Y_2)	order (Y_2, Y_1)	Dirac components
$\beta\gamma$ [BK, (3.66)]	$(\beta_I, \gamma^J) \to \partial c\,\partial c$	0	0	0, 0
bc [BK, (3.64)], $b \leftrightarrow (\xi, \lambda^*)$	$(c, b) \to \partial c\,\partial c$	0	0	0, 0
$\beta\beta$ [BK, (3.67)]	$(\beta_I, \beta_J) \to \gamma^K\,\partial c$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{3} - 2L$, $\frac{1}{3} - 2L$
$b\gamma$ [BK, (3.68)]	$(b, \gamma^I) \to \gamma^I\,\partial c$	$\frac{64}{3} - 128L$	$32 + 64L$	$-\frac{64}{3} - 128L$, $-\frac{32}{3} + 64L$
$\beta b \to \beta\,\partial c$ [BK, (3.69)]	$(\beta_I, b) \to \beta_I\,\partial c$	$-\frac{64}{3}$	0	0, 0
$\beta b \to \gamma\gamma$ [BK, (3.69)]	$(\beta_I, b) \to \gamma^J\gamma^K$	$\frac{8}{3}\big(\partial_2\gamma^J\partial_1\gamma^K - \partial_1\gamma^J\partial_2\gamma^K\big)$	the same	
$\gamma\gamma$ [BK, (3.77)]	$(\gamma^I, \gamma^J) \to$ twisted letters	0	0	

The arity-one pair entries of Q_1 along $\xi = u \otimes z$ at $\psi = 0$, reduced modulo $\operatorname{Im} Q_0^\xi$, in the normalisation of Theorem 11.56: coefficients of $\frac{\hbar}{16\pi^2}$ times the colour factor $(1 \otimes \mathrm{ad}_{Y_2}\mathrm{ad}_{Y_1})(-,-)^{-1}$ of the colour order (Y_1, Y_2), Y_1 the first output listed, with $L = \log\frac{3}{2}$. The third and fourth columns give the coefficient of the holomorphic derivative on the output other than ∂c, and for $\beta b \to \gamma\gamma$ the derivative structure, ∂_1 and ∂_2 being the derivatives along d_1 and d_2; the last column gives, in the two orders, the coefficient of the holomorphic Dirac component on the output ∂c, which is Q_0^ξ-exact up to terms with antifields; K is complementary to I, J, and J, K to I.

Proof. The entries are the kernels on the leg sets $(\lambda|_{S_-\otimes W^*})^2$ of $(\lambda|_{S_+\otimes W}, A|_{W_{\mathbb{R}}^\perp})$; $A|_{W_{\mathbb{R}}^\perp}\lambda|_{S_-\otimes W^*}$ of $(\lambda|_{S_+\otimes W}, \lambda|_{S_+\otimes W})$ and of $(A|_V, A|_{W_{\mathbb{R}}^\perp})$ with one derivative on $A|_V$; $\lambda|_{S_+\otimes W}\lambda|_{S_-\otimes W^*}$ and $(A|_{W_{\mathbb{R}}^\perp})^2$ of $(\lambda|_{S_+\otimes W}, A|_V)$ with one derivative on $A|_V$; and $(\lambda|_{S_-\otimes W^*})^2$ of $(c, \lambda^*|_{S_+\otimes W^*})$ (Tables 11.6 and 11.6); all have superficial degree 0, so they carry no poles and no finite special vertex (Proposition 11.43), and Theorem 11.41 evaluates them exactly, with the letter vectors contracted exactly in the weight basis (Remark 11.58). The colour factors are those of Lemma **??** with the output of a term T_5 read as an output of a cubic vertex. (i) The vanishing of $\beta\gamma$ for general ξ and of bc holds in both evaluations. The outputs $(\lambda|_{S_-\otimes W^*})^2$ have Weyl weight 3, while $L_1^{(0)}$ preserves the Weyl weight $\frac{5}{2}$ of the Konishi descendant, a one-body term keeps the other letter as a spectator, and the arity-two kernels have outputs of degree -1; hence the coefficient of $\int \omega\,(\bar\lambda_{\dot\alpha i} z^i, \bar\lambda^{\dot\alpha}{}_j z^j)$ in $\hbar Q_1^\xi$ of the Konishi descendant is the entry $\beta\gamma$. $\gamma\gamma$ is Proposition 11.54. (ii)–(iv) are read off the table. Rescaling a weight vector rescales its letter by the inverse factor and its output by the factor, which multiplies an entry by one product of factors, the same for the two colour orders, for the reduced and the Dirac coefficients, and for the rational part and the part in $\log\frac{3}{2}$; the vanishing, the equality of the two orders, the indices and the derivative structure are therefore independent of the normalisation. □

Theorem 11.57 (What the comparison with the holomorphic twist tests)**.** *Let $\xi = u \otimes z$ be of rank one, let Φ be the quasi-isomorphism from the cohomology of Q_0^ξ on local operators to that of the free holomorphic differential on the letters c, γ, β, b, $\partial_{\dot\alpha}$ of [BK], transported through [ESW, Thm 10.9] (remark after Theorem **??**), and let D be the difference, read through Φ, of the one-loop quantisation $r_\xi(\hbar F_1)$ of the twisted theory restricted from the equivariant one and the holomorphic quantisation of [BK, (2.42), footnote 22] and [W18, Thm 3.4] (B. R. Williams, Renormalization for holomorphic field theories, arXiv:1809.02661). If $D \in \mathbb{C}\,S^\xi + \operatorname{Im} Q_0^\xi$, then $\Phi \circ [Q_1^\xi] \circ \Phi^{-1}$ is the class of the one-loop differential of [BK, (3.7), (3.64)–(3.69)], and:*

- *(a) the one-body terms of Q_1^ξ on the twisted letters are $a\,Q_0^\xi + [Q_0^\xi, N]$ for a number a and a derivation N of the letters, zero on cohomology;*

- *(b) on the pairs from $\{c,\gamma\}$ and on (β,c) the two-body kernels lie, on the twisted letters, in* $\operatorname{Im} Q_0^\xi$ *plus terms cancelled on cohomology by the one-body terms, as in [BK, (3.11)]; on (γ,γ) they vanish (Proposition 11.54);*

- *(c) on the pairs $\beta\gamma$, $\beta\beta$, $b\gamma$, βb, bb and bc the kernels, reduced modulo* $\operatorname{Im} Q_0^\xi$ *as in Theorem 11.56, have the colour factors $(1\otimes \mathrm{ad}_{Y_2}\mathrm{ad}_{Y_1})(-,-)^{-1}$, which are $\frac{1}{2}f_{ACD}f_{BCE}$ of [BK] up to normalisation, and holomorphic derivatives $\partial_{\dot\alpha}=u'^{\alpha}\sigma^m_{\alpha\dot\alpha}\partial_m$ on the outputs;*

- *(d) their coefficients satisfy, modulo the finite parts of Proposition 11.46 (iv), the relations [BK, (3.59)–(3.63)], which are the order-$\hbar$ part of Corollary* **??** *(ii) at (ξ,R) for the elements R of the tree-level commutant $\mathfrak{psl}(3|3)$ of ξ, restricted by r_ξ;*

- *(e) every statement of the holomorphic twist about $[Q_1]$ on classes of several letters is the image under Φ of $[Q_1^\xi]$, computed by the pair rule from the kernels of (c), with the normalisation of $\hbar$ relative to the unit fixed by Φ.*

For a general D the two class maps differ by $[\{D,-\}]$, and (a)–(e) compute that class.

Proof. Two quantisations of the twisted theory differing by a Q_0^ξ-closed D of degree 0 have class maps differing by $[\{D,-\}]$, which vanishes for $D\in\mathbb{C}\,S^\xi+\operatorname{Im}Q_0^\xi$, since $\{S^\xi,-\}=Q_0^\xi$ and $\{Q_0^\xi\chi,-\}=\pm[Q_0^\xi,\{\chi,-\}]$ (Theorem **??** and the remark after it). (a) The one-body terms are brackets with local functionals (Theorem 11.40 (iii)), and the condition on D makes them aQ_0^ξ plus commutators with Q_0^ξ on the twisted letters. (b), (c) The pair rule [BK, (3.7)–(3.11)], the colour factors of Lemma **??** with two outputs, and the exactness of the antiholomorphic derivatives, $\{Q,\bar Q^1_{\dot\alpha}\}=P_{-\dot\alpha}$ [BK, (2.5)]. (d) Corollary **??** (ii) at (ξ,R) with $[\xi,R]=0$, at order $\hbar$, mapped by r_ξ (Lemma **??**). (e) $[Q_1^\xi]$ is determined by its values on representatives, polynomials in the letters, through the pair rule. □

Remark 11.58 (Method of computation)**.** Each defect cycle is evaluated exactly as in Remark 11.36: the Gaussian loop integral and the integrals of (11.7) over the ordered subsimplices of the region where α_σ is the smallest coordinate, the letter derivatives through Proposition 11.12 with $n+1$ edges, the position along η and K_b through the injected momentum r, the fibre algebra in an explicit realisation of V_{10} and Σ over $\mathbb{Q}(i)$ with the vertices $S^{(1)}(x)$ of Chapter 2 and the sign of Definition 11.39 (ii), letter vectors with fractional components contracted exactly, and the colour words by Lemma **??**; the twisted letters are built from the weights of a Cartan subalgebra. A second evaluation works in momentum space with an independently written realisation of $S^{(1)}(x)$ in the components of Definition **??**, the special edge as the factor $e^{-\varepsilon|p|^2}-1$, Feynman parameters for the other edges and numerical integration at several values of ε with thirty digits, extrapolated to ε^0 with the poles fitted; along K_b it evaluates the cycles at the endpoints $T=1,4,16$. The two evaluations give the same operators C_ξ, Y_η, C_K and the same arrows of Lemma 11.38, the same composite line along ξ, the same free directions of Proposition 11.45, the same kernels Υ_ξ on every leg set with at most two outputs of the pairs of field and ghost letters without derivatives and with one letter derivative, the same coefficients of Theorem 11.48 for both chiralities, the same entries $\beta\gamma$, $\beta\beta$ and bc, and the same translation-invariant kernels along η and K_b where both are evaluated; both reproduce Example 11.42, Proposition 11.54 and the value along P_b of Corollary 11.50. The first evaluation gives the values along Δ of Corollary 11.50, extends Υ_ξ to three letter derivatives without the finite special vertices, and gives the entries of Table 11.6 and the identity (iv) of Theorem 11.48; the second gives the closed forms of Theorem 11.41 (iv) for $n_t=2,\dots,6$, the finite special vertices on the pairs of gaugino letters, the pole identity (11.6) on these pairs and on the pair of Example 11.42 with the vacuum-pole terms, the position identity of Theorem 11.41 (ii), and, along K_b, $T\partial_T$ of the defect cycles of the intrinsic vertex $S^{(1)}(K_b)$ on $(A|_V,A|_V)$ with the output $A|_V$, with the vertex on a line and at the special edge, decreasing at least as ε at the three

endpoints, as Theorem 11.41 (iii) states. Every integral is a four-dimensional Schwinger integral with the cutoff ε on every edge.

Remark 11.59 (The data that complete the one-loop components)**.** The one-body kernels of Q_1, S_1 and K_1 are given by the action side of the arity-one equations. Along ξ it is (11.9): the quadratic part of $\hbar\,\mathrm{Ob}^{(1)}(\xi)$ in the sectors $(\lambda, A|_V)$, $(\lambda, A|_{W_{\mathbb{R}}^{\perp}})$ and (c, λ^*), the bracket $\{S^{(1)}(\xi), \hbar F_1^{(0)}\}$ with the finite counterterm of Theorem **??**, a particular solution, whose coefficient of T_2 gives $t_\perp$ and with it c_Q of Theorem 11.48, and the cubic part, which fixes $\theta_V^\xi - \theta_\perp^\xi$; the same computation along η and K_b gives the remaining one-body kernels. The next computations of the two-body kernels are the finite special vertices on the leg sets with $D \geq 2$, with their vacuum-pole terms; the arity-two kernels along (ξ, ξ), which give the entry bc with b represented by the self-dual field strength; and the entry bb on the pair $(A|_V, A|_V)$ with two derivatives. With the one-body kernels, the relations (R2)–(R6) of Proposition 11.53 become finite computations on the scalar bilinears, with the cocycle finite part of Theorem 11.48 (iii) extended beyond the two-field sector; (R3) on the trace bilinear is the Konishi relation between Q_1, S_1 and $D_1 = -48$ units. The class-level comparison of Theorem 11.57 uses the transport map Φ built from [ESW, Thm 10.9], the condition on D and the relations [BK, (3.59)–(3.63)] applied to the entries of Table 11.6. The inner-product relations of Proposition 11.51 (iv) and Corollary 11.52 use Gram_1 from the two-point graphs of Definition **??**, and $K_1 \cong -(b,x)D_1$ on the families uses the rotation component M_1.

Remark 11.60 (Two-component forms)**.** With the components of Definition **??**:

- Letters. $A|_V$, $A|_{W_{\mathbb{R}}^{\perp}}$, $\lambda|_{S_+\otimes W}$, $\lambda|_{S_-\otimes W^*}$ are A_m, A_{ij} (the $\mathbf{6}$ of $SU(4)_R$), $\lambda_\alpha{}^i$ (undotted, $\mathbf{4}$), $\bar\lambda_{\dot\alpha i}$ (dotted, $\bar{\mathbf{4}}$), with the antifields A_m^*, A_{ij}^*, $\lambda^{*\alpha}{}_i$, $\bar\lambda^{*\dot\alpha i}$, c^* and the degrees $1,0,0,0,0,-1,-1,-1,-1,-2$ for $c, A_m, A_{ij}, \lambda_\alpha{}^i, \bar\lambda_{\dot\alpha i}, A_m^*, A_{ij}^*, \lambda^{*\alpha}{}_i, \bar\lambda^{*\dot\alpha i}, c^*$; $(\lambda,\lambda^*) = (\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})$ and $(u,w) = \frac{1}{4}u^{ij}w_{ij}$ on $W_{\mathbb{R}}^{\perp}$.

- Composite kernels. The internal vertex along Δ is $\int\big((2E+2)A_m, A_m^*\big) = 2\int x_n(\partial_n A_m, A_m^*) + 2\int(A_m, A_m^*)$ with $E = x_n\partial_n$, along P_b it is $\int(b_n\partial_n A_m, A_m^*)$; it sits between $\langle A_m A_n\rangle = 2(-,-)^{-1}\delta_{mn}$ and $\langle A_m^*\, c\rangle = -(-,-)^{-1} i p_m$, each times $e^{-\varepsilon|p|^2}|p|^{-2}$ at $T=\infty$, and the insertion factors are 2, $2i(l+q_c)_n$ and $ib_n(l+q_c)_n$. The special edge carries $e^{-\varepsilon|p|^2}-1$ times $(-,-)^{-1}$ and the inverse fibre pairing of (A_m, A_m^*), $\frac{1}{4}(A^{ij}, A_{ij}^*)$, (λ,λ^*) and (c,c^*): on $(\lambda_\alpha{}^i, \lambda^{*\beta}{}_j)$ the factor $-|l+q_\sigma|^2\delta_\alpha^\beta\delta_j^i(-,-)^{-1}$, on $(\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\beta j})$ the opposite sign.

- Bilinears. For $O = \frac{1}{16}\int\omega\,(w_1^{ij}A_{ij}, w_2^{kl}A_{kl})$ the composite pair kernel along Δ is

$$\Big[-\tfrac{1}{3}\big((2E+2)A_{ij}\big)A_{kl} - \tfrac{1}{3}A_{ij}\big((2E+2)A_{kl}\big) - \tfrac{4}{3}A_{ij}A_{kl}\Big]\ \text{units},$$

  and along P_b it is $-\frac{1}{3}$ units times $b_m\partial_m(A_{ij}A_{kl})$, with the same contraction.

- Two-site density. On adjacent pairs (A_{ij}, A_{kl}), (Z,ψ), (ψ, Z), (ψ,ψ) of a single trace the composite pair kernels give $-\frac{4}{3}$ units (translation-invariant part along Δ) and $-\frac{1}{6}$ units times $b_m\partial_m$ (along P_b) on the same word, and 0 on the exchanged word.

- Free directions. The sector parts of $S^{(1)}(\Delta)$ are $\frac{1}{4}\int\big((2E+2)A^{ij}, A_{ij}^*\big)$, $-\int\big[\big((2E+3)\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i\big) - \big((2E+3)\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i}\big)\big]$ and $\int\big((2E+2)A_m, A_m^*\big) - \int\big(2E(c), c^*\big)$; those of $S^{(1)}(P_b)$ are $\frac{1}{4}\int(b_m\partial_m A^{ij}, A_{ij}^*)$, $-\int\big[(b_m\partial_m\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) - (b_m\partial_m\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\big]$ and $\int(b_m\partial_m A_n, A_n^*) - \int(b_m\partial_m c, c^*)$. The exact directions are $Q\int x_m(A_m, c^*)$ and $Q\int(\rho(x)\lambda^*, \lambda^*) = -2i\,Q\int x_m\big(\lambda^{*\alpha}{}_i, \sigma^m_{\alpha\dot\alpha}\bar\lambda^{*\dot\alpha i}\big)$. The further class of Proposition 11.20 without the reflection is a multiple of $\int\big[(\lambda_\alpha{}^i, \lambda^{*\alpha}{}_i) + (\bar\lambda_{\dot\alpha i}, \bar\lambda^{*\dot\alpha i})\big]$, the rotation $\lambda_\alpha{}^i \mapsto e^{i\varphi}\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i} \mapsto e^{-i\varphi}\bar\lambda_{\dot\alpha i}$ of the free gaugino.

- Letter count. It counts 2 for each letter A_m, A_{ij}, $\lambda_\alpha{}^i$, $\bar\lambda_{\dot\alpha i}$ and 1 for each derivative ∂_m on it; (b) reads: the composite pair kernels along P_b sum to $-\frac13$ units times $b_m\partial_m$ on cohomology.

- Families and spectra. $n_m=\sigma^m_{11}$, $D_n=n_m(\partial_m+[A_m,-])$, $\psi=\lambda_1{}^i$, and F_{11} the self-dual field strength with both undotted indices 1. In Srednicki's normalisation, $A_m=-igA^a_mT^a$, $A_I=-igA^a_IT^a$, $\lambda=-\frac{ig}{\sqrt2}\lambda^aT^a$, $\mathrm{tr}(T^aT^b)=\frac12\delta^{ab}$, the unit is $-\frac{g^2N}{32\pi^2}$ and $-\frac13$ units is $\frac{g^2N}{96\pi^2}$. On the families $D_1=-8$ units times the eigenvalues E of $\sum_l H_{l,l+1}$, which is $\frac{g^2N}{4\pi^2}E=2\gamma$ with the one-loop anomalous dimension $\gamma=\frac{g^2N}{8\pi^2}E$: $\gamma=\frac{g^2N}{2\pi^2}h(S)$ on twist two, $\gamma=\frac{3g^2N}{4\pi^2}$ for $\mathrm{tr}(F_{11}F_{11})$ and the trace bilinear, the Konishi multiplet.

- Odd generators. $Q_1(O)=\xi_\alpha{}^i\,Q_1(O)^\alpha{}_i+\bar\xi_{\dot\alpha i}\,\bar Q_1(O)^{\dot\alpha i}$ and $S_1(O)=\eta^\alpha{}_i\,S_1(O)_\alpha{}^i+\bar\eta^{\dot\alpha i}\,\bar S_1(O)_{\dot\alpha i}$ (Definition **??**); the Killing spinor of η has the components $(\rho(x)\eta)_\alpha{}^i=-i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$ and $(\rho(x)\eta)_{\dot\alpha i}=i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$ (**??**), so on a family with $P_1\cong0$ the position part along b of S_1 along η is, on cohomology, Q_1 along $\xi_\alpha{}^i=-i\,b_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\xi_{\dot\alpha i}=i\,b_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$. For $\xi_\alpha{}^i=u_\alpha z^i$, $(\rho(w)\xi)^\alpha{}_j=\varepsilon^{\alpha\beta}w_{ji}u_\beta z^i$, so $\rho(w)\xi=0$ exactly when $w_{ij}z^j=0$, and then $\iota_wA=\frac14w^{ij}A_{ij}$ is a combination of the components $A_{ij}z^j$: Proposition 11.54 is the vanishing of Q_1 on $\int\omega\,(m_1^iA_{ij}z^j,m_2^kA_{kl}z^l)$, $m_1,m_2\in W$.

- Terms of $S^{(1)}(\xi)$. For $\xi=(\xi_\alpha{}^i,0)$, with $(\xi^j\lambda^i):=\xi^{\alpha j}\lambda_\alpha{}^i$ and $F^0_{mn}=\partial_mA_n-\partial_nA_m$: $T_1=-i\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i,A^*_m\big)$, $T_2=\frac12\varepsilon_{jikl}\big((\xi^j\lambda^i),A^{*kl}\big)$, $T_3=-\frac14\big(F^0_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta\xi_\beta{}^i,\lambda^{*\alpha}{}_i\big)$, $T_4=-\frac i2\big(\xi^{\alpha j}\sigma^m_{\alpha\dot\alpha}\partial_mA_{ij},\bar\lambda^{*\dot\alpha i}\big)$ and T_5 the three terms $-\frac14\big([A_m,A_n](\sigma^m\bar\sigma^n)_\alpha{}^\beta\xi_\beta{}^i,\lambda^{*\alpha}{}_i\big)$, $-\frac i2\big(\xi^{\alpha j}\sigma^m_{\alpha\dot\alpha}[A_m,A_{ij}],\bar\lambda^{*\dot\alpha i}\big)$, $-\frac14\big([A^{ik},A_{kj}]\xi_\alpha{}^j,\lambda^{*\alpha}{}_i\big)$; for $\xi_\alpha{}^i=u_\alpha z^i$ these are the terms of the remark after Theorem **??**. The terms with $\bar\xi_{\dot\alpha i}$ follow by $\lambda_\alpha{}^i\leftrightarrow\bar\lambda_{\dot\alpha i}$, $\lambda^{*\alpha}{}_i\leftrightarrow\bar\lambda^{*\dot\alpha i}$, $\sigma\leftrightarrow\bar\sigma$. The intrinsic term along η is $s_0=\big(A^{kj}\varepsilon_{\alpha\beta}\eta^\beta{}_j,\lambda^{*\alpha}{}_k\big)-\big(A_{kj}\varepsilon_{\dot\beta\dot\alpha}\bar\eta^{\dot\alpha j},\bar\lambda^{*\dot\beta k}\big)$, and $S^{(1)}(K_b)=\int\big\{(v.A_m,A^*_m)+\frac14(v.A^{ij},A^*_{ij})-(v.\lambda_\alpha{}^i,\lambda^{*\alpha}{}_i)+(v.\bar\lambda_{\dot\alpha i},\bar\lambda^{*\dot\alpha i})-(v.c,c^*)\big\}$ with $v^m=x^2b^m-2(b\cdot x)x^m$, $v.A_m=v^n\partial_nA_m+A_n\partial_mv^n$, $v.A_{ij}=v^n\partial_nA_{ij}-2(b\cdot x)A_{ij}$, $v.\lambda_\alpha{}^i=v^n\partial_n\lambda_\alpha{}^i-\frac12(x_mb_n-x_nb_m)(\sigma^m\bar\sigma^n)_\alpha{}^\beta\lambda_\beta{}^i-3(b\cdot x)\lambda_\alpha{}^i$, $v.c=v^n\partial_nc$.

- Arrows and lines. $\xi.$ maps $A_{kl}\mapsto\Gamma_{kl}(\xi,\lambda)=-2\varepsilon_{jikl}(\xi^j\lambda^i)$ and $A_m\mapsto\Gamma^m(\xi,\lambda)=i\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i+\bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big)$. On a plane wave C_ξ maps λ to c by $\lambda\mapsto-p_m\big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i+\bar\xi_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i\big)$. The composite line along $\xi_\alpha{}^i$ joins a leg $\lambda_\alpha{}^i$ to a leg c through $\langle\lambda_\alpha{}^i\bar\lambda_{\dot\beta j}\rangle$, the term T_1 and $\langle A^*_m\,c\rangle$; with $\sigma^k_{\alpha\dot\beta}\bar\sigma^{m\dot\beta\gamma}p_kp_m=-|p|^2\delta^\gamma_\alpha$ it is $-2\,\xi_\alpha{}^i\,e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $c\otimes\lambda_\alpha{}^i$ and $+2\,\xi_\alpha{}^i\,e^{-2\varepsilon|p|^2}|p|^{-2}(-,-)^{-1}$ on $\lambda_\alpha{}^i\otimes c$. Along η, $(\rho(e_m)\eta)_\alpha{}^i=-i\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$ and $(\rho(e_m)\eta)_{\dot\alpha i}=i\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$; along K_b, $\mathrm{div}\,v=-8\,b\cdot x$, $\Sigma^{\mu\nu}$ acts on A_m as $(\Sigma^{\mu\nu}A)_m=\delta_{m\mu}A_\nu-\delta_{m\nu}A_\mu$ and on $\lambda_\alpha{}^i$ through $-\frac14(\sigma^\mu\bar\sigma^\nu-\sigma^\nu\bar\sigma^\mu)$, and C_K maps A_m to $-4b_mA_m$ in the slot of c.

- Bilinears along ξ. For $O=\frac{1}{16}\int\omega\,(w_1^{ij}A_{ij},w_2^{kl}A_{kl})$,

$$\mathcal{B}_1=\tfrac14\int\omega\,\big[(w_1^{kl}A_{kl},w_{2\,ji}\xi^{\alpha i}\lambda_\alpha{}^j)+(1\leftrightarrow2)\big],\qquad\mathcal{B}_2=\tfrac14w_1^{ij}w_{2\,ij}\int\omega\,(A_{ji},\xi^{\alpha i}\lambda_\alpha{}^j),$$

  $\mathcal{B}_3=\frac14w_1^{ij}w_{2\,ij}\int\omega\,i\big(\bar\lambda_{\dot\alpha i},A_m\bar\sigma^{m\dot\alpha\alpha}\xi_\alpha{}^i\big)$ and $\mathcal{B}_4=\frac14w_1^{ij}w_{2\,ij}\int\omega\,(c,\xi_\alpha{}^i\lambda^{*\alpha}{}_i)$; Theorem 11.48 reads, in units $-\frac{g^2N}{32\pi^2}$,

$$Q_1(O)\big|_{\text{two-body}}=-2\log\tfrac32\,\mathcal{B}_1+4\log\tfrac32\,(\mathcal{B}_2+\mathcal{B}_3),$$
$$L_1^{(0)}(O)=-\big(2+4\log\tfrac32\big)\tfrac14w_1^{ij}w_{2\,ij}\int\omega\,(\partial_mc,A_m),$$

  and the cocycle finite part is $\psi(O)=-\big(1+2\log\frac32\big)\frac14w_1^{ij}w_{2\,ij}\int\omega\,(A_m,A_m)$ units, $\psi(\mathcal{B}_1)=-2\,\mathcal{B}_3$ units. On the twenty traceless bilinears the transformation along Q^i_α is $1+c_Q$ times the classical one, with c_Q of Theorem 11.48 (ii).

- Special supercharges and special conformal fields on the bilinears. The two-body part of $S_1(O)$ along $\eta^\alpha{}_i$ is that of $\bar{Q}_1(O)$ along $\bar{\xi}_{\dot\alpha i} = i\,x_m\eta^\alpha{}_i\sigma^m_{\alpha\dot\alpha}$, and along $\bar\eta^{\dot\alpha i}$ that of $Q_1(O)$ along $\xi_\alpha{}^i = -i\,x_m\sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$. On the part with two scalar legs, $K_1(O) = -\big(\frac{1}{3}+\theta\big)\big(v^m\partial_m O - 4(b\cdot x)O\big)$ units on the traceless bilinears and the same plus $48\,(b\cdot x)\,O$ units on the trace $\frac{1}{4}\int\omega\,(A^{ij},A_{ij})$.

- Twist. With $z = e_4$: $\gamma^I = A_{I4}$, $\beta_I = u'^\alpha\lambda_\alpha{}^I$ $(I=1,2,3)$, $\partial_{\dot\alpha}c = \bar\lambda_{\dot\alpha 4}$, $b = u'^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u'_\beta$ or $(\xi,\lambda^*) = u_\alpha\lambda^{*\alpha}{}_4$, and $\partial_{\dot\alpha} = u'^\alpha\sigma^m_{\alpha\dot\alpha}D_m$, so that $\partial_2\gamma^J\partial_1\gamma^K - \partial_1\gamma^J\partial_2\gamma^K$ of Table 11.6 is a multiple of $\partial_{\dot\alpha}\gamma^J\partial^{\dot\alpha}\gamma^K$. The Konishi entry is the coefficient of $\int\omega\,(\bar\lambda_{\dot\alpha 4},\bar\lambda^{\dot\alpha}{}_4)$ in $Q_1\big(\int\omega\,(u'^\alpha\lambda_\alpha{}^i, A_{i4})\big)$, 0 at $\psi=0$; [BK, (3.66)] is $Q_1\big((\beta_I)^A(\gamma^J)^B\big) = \frac{1}{2}\delta^J_I f_{ACD}f_{BCE}\,\partial_{\dot\alpha}c^D\partial^{\dot\alpha}c^E$, and the entries $\beta\beta$, $b\gamma$, βb of [BK, (3.67)–(3.69)] have the same factor $\frac{1}{2}f_{ACD}f_{BCE}$ in both terms.

- Weinberg. On the Dirac column $(\lambda_\alpha{}^i,\bar\lambda^{\dot\alpha}{}_i)$, $\rho(v) = \gamma^\mu_W v_\mu$ gives Weinberg's form of each line, as in Remark **??**.